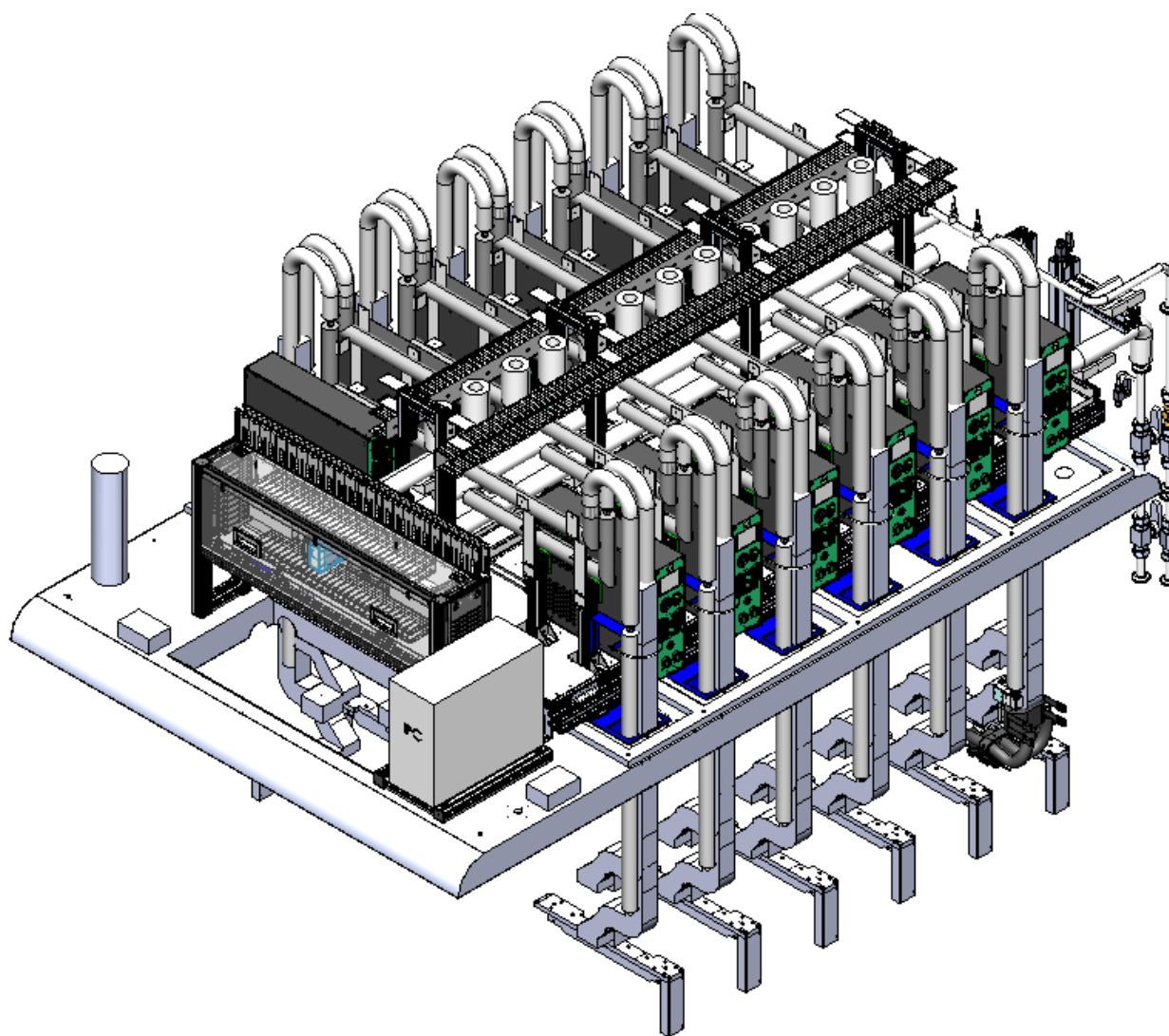




AEM 1kW Thermal Retrofit System

Operation Manual

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 - c) that has not been properly maintained in accordance with the requirements and at the frequency specified in this manual,
 - d) without properly observing the safety precautions, guidelines and instructions listed in this manual, and/or
 - e) which has been modified from the original equipment specifications (whether by AEM's customer or otherwise), including where the equipment contains parts or components not compliant with the original equipment specifications.
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6. If you have any doubt on the installation and use of the equipment, or any information contained in this manual, please contact your local AEM representative for the site for further clarifications prior to installing and using the equipment or using the information in this manual.

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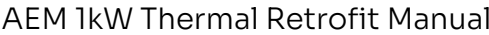
Revision Record

Manual Version	Issue Date	Description
0	02 Dec 2022	Initial Release



Abbreviations & Acronyms

Term	Definition
ATC	Active Thermal Controller
DE	Die Engage
DUT	Device under Test
DTF	DUT Temperature Following
EGW	Ethylene glycol water
GUI	Graphical User Interface
HTF	Heater Temperature Following
LOTO	Lock Out / Tag Out
MSD	Musculoskeletal Disorders
PD	DUT Power
PID	A transfer function consists of Proportional, Integral, Differentiator
PPE	Personal Protective Equipment
PSET	Set Power
PWM	Pulse Width Modulation
QD	Quick Disconnect
RTD	Resistance Temperature Detector
SSR	Solid State Relay
SW	Software
TCO	Thermal CutOff
TCS	Temperature Control System
TD	DUT Temperature
TSET	Set Temperature
TSP	Temperature Set Point
TTV	Thermal Testing Vehicle
UI	User Interface

[illegible]



1.Introduction



1.1 General

The **AEM 1kW Thermal Retrofit System** is a system designed to evaluate chip resistance in various temperatures which integrated with piping, chiller, and thermal graphics interface.

1.2 Characteristic of Equipment

1.2.1 User Friendly Graphical Interface

The AEM 1kW Thermal Retrofit System features a computerize control system with a touch screen that has a user friendly graphical interface. Through the graphical interface, the user will have full control of the tool and functionality (depending on the “user-level” granted). Testing condition of the tool will be visually displayed on the interface screen to assist the user to navigate through the tool. Monitoring of the process only requires a quick glance on the friendly graphical flowchart display menu. Errors will shows in the graphical interface screen.

1.2.2 Windows Program

Using the Windows program interface, customer can use the program that easy to understand familiar.

1.2.3 Tri-Temperature Testing

The AEM 1kW Thermal Retrofit System supports SLT testing in tri-temperature test scenarios (cold test, room test, hot test) without compromising the physical integrity of the device under test.

1.2.4 Quick Kit Change Design

The AEM 1kW Thermal Retrofit System provides better accessibility design to allow user to exchange the kit more efficiently.

1.3 Equipment Configuration

This section shows the AEM 1kW Thermal Retrofit System overview.

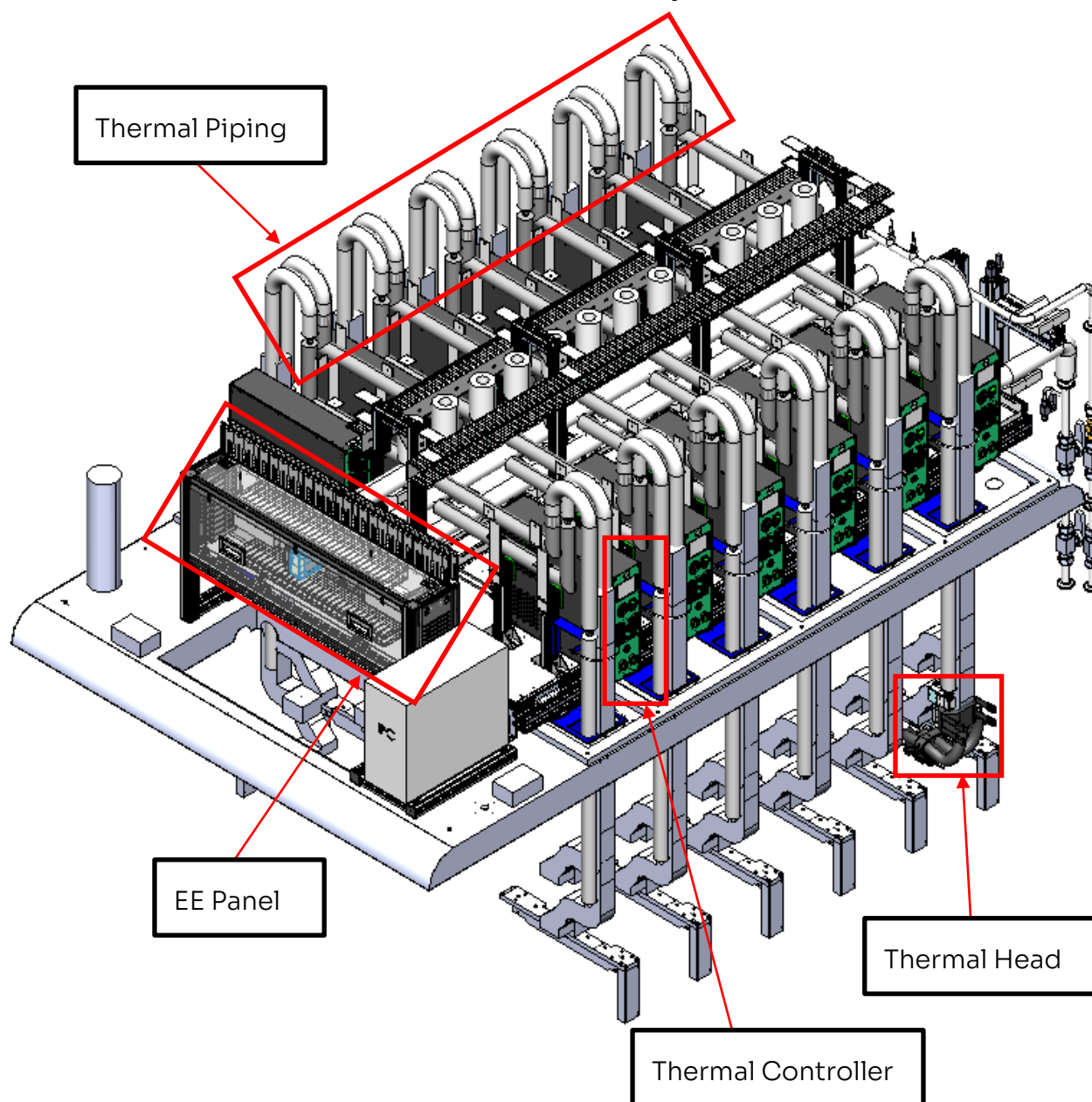


Figure 1-1 AEM 1kW Thermal Retrofit System front ISO view



1.4 Transport, Handling & Storage

Handling

The equipment will be handled by AEM and AEM authorized personnel from build to packing and crating into shipment package. Shipment and forwarding of the equipment will be handled by customer authorized and appointed forwarder. Customer, customer authorized personnel, and AEM staff onsite or local representative will handle the equipment from uncrating/unpacking, installation, operation, and maintenance.

Transportation

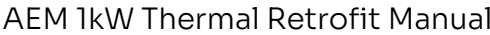
Equipment will be transported either via land transport, air transport, or via cargo container ship for sea transportation. The transport mode depends on the customer's site location, delivery time as well as the price. Customer appointed and authorized cargo forwarder will handle the shipment and transportation based on customer's request and direction. During transportation, altitude depends on the mode of transportation and all necessary requirements must be adhered by the forwarder. Equipment should be kept in a controlled environment (temperature and humidity) during transportation. Refer to **Table 3-6 Mechanical specification** for relative humidity, machine storage humidity, and machine storage temperature.

Storage

Equipment will be stored at AEM facility until it is picked up by the customer appointed forwarder. Forwarder will store the equipment at their warehouse once picked up from AEM facility until the equipment is shipped out. Customer will store at their own designated warehouse according to their onsite location until the equipment is uncrated and moved into the production floor. Equipment should be kept in a controlled environment (temperature and humidity) during storage, with altitude not exceeding 1000 m above sea level. Refer to **Table 3-6 Mechanical specification** for relative humidity, machine storage humidity, and machine storage temperature.

General Recommendations

Equipment should not be subjected to vibrations. Ensure tip and fall shock indicators are not triggered due to vibrations or mishandling. If any shock indicators are triggered, contact AEM Singapore representative immediately for further support.

[illegible]



2. Safety Precautions



2.1 Safety

2.1.1 General Safety Information

The AEM 1kW Thermal Retrofit System consists of various safety features to prevent the operators and maintenance personnel from injuring themselves or others and damaging the machine. Mechanical and electrical safety hazards may occur for any automated machines. The safety features for the AEM 1kW Thermal Retrofit System consists of circuit breakers and other devices that allow safe operations and maintenance of the equipment. These features will be further highlighted in the following chapters.

Purpose

To establish safe work practices with potential hazards.

Scope

Safety applies to all workers who perform tasks in a potentially hazardous environment during installation, operations, modifications, maintenance, and repair of machinery and equipment. Operations and maintenance of any automated equipment exposes personnel to potential hazards. Properly following the procedures set forth in this section will ensure safe operations and maintenance of the equipment.

Re-training of operator and technical personnel shall be required when equipment system calls for any changes.

2.2 Safety Alert Boxes

While using this manual, be sure to observe the alert boxes and symbols shown below.

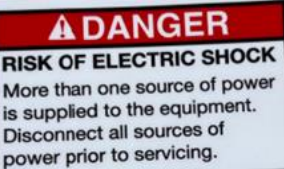
2.2.1 Safety Label Precaution

All hazardous labels are to alert personnel accessing to the area / cabinet to perform work. It does not isolate potential hazards from the personnel as such; service personnel must take the necessary safety procedures (e.g., power off) to ensure the system is fully isolated from the facility's energy supply. Failure to heed precautions can result in injury to people or damage to machine.

Always heed the information provided by any labels!

The following conventions are used to indicate and classify precautions in this manual.

Table 2-1 Safety label description

   Hazardous voltage/current. Contact may cause electric shock or burn. Turn off and lock out system power before servicing   RISK OF ELECTRIC SHOCK More than one source of power is supplied to the equipment. Disconnect all sources of power prior to servicing.	Indicates an imminently hazardous situation, which, if not avoided, will result in death or serious injury. (e.g., moving parts, high voltage electric parts) Danger alert boxes are Red.
   Crush Hazard. Keep hands clear while operating. Lock out power before servicing.   HOT SURFACE INSIDE. Contact may cause burn. Do not touch, or wear protective gear before servicing internal parts.	Indicates a potentially hazardous situation, which, if not avoided, could result in death or serious personal injury. Warning alert boxes are Orange.

  	Indicates a potentially hazardous situation, which, if not avoided, may result in minor or moderate personal injury and / or damage to the equipment. Caution alert boxes are Yellow.
---	--

Training will be given to operators and service personnel on the procedures to be followed during operations of the machine. Additionally, safety precautions will be highlighted to the operators and service personnel during the installation, operation, and maintenance. These safety precautions will be further mentioned in the later chapters.



2.2.2 General Safety Guidelines

Lock Out Tag Out (LOTO) is a critical to worker safety to isolate hazardous energies and limit service personnel exposure. LOTO can be applied to the whole tool, isolating energies at their inlet to the tool from plant facilities or applied to individual modules which allow the other modules of the tool not being serviced to continue normal operation.

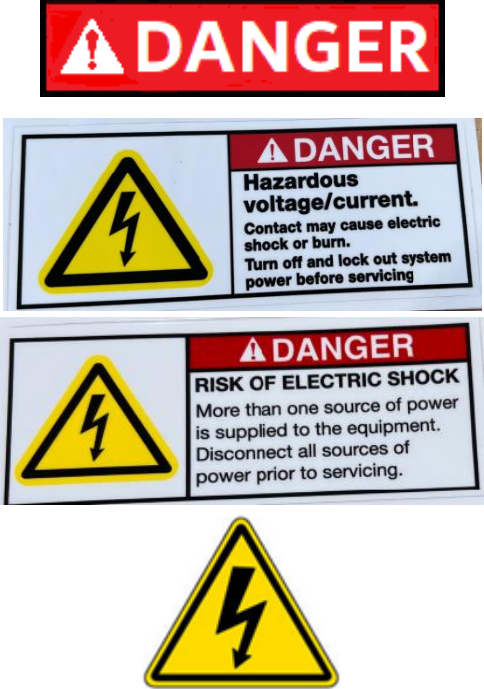
Performing LOTO on the whole tool requires that the tool be properly shut down completely. Service personnel should evaluate the scope of the work to be performed to determine which modules should have LOTO implemented or if the whole tool should have LOTO applied. If there is a possibility that the scope of work will expose service personnel to hazardous energies from any module, that module must have LOTO applied.

Reference:

LOTO Procedure ([Section 6.2.4 Lock Out/Tag Out \(LOTO\) Procedure](#))

The following are some of the hazard(s) that a worker may face while working on the tool.

Table 2-2 Type of hazards

	Electrical Hazards: Hazardous voltages may exist and can cause electric shock or burns. Possible injury or death can occur. Only qualified personnel should service this equipment. Perform equipment or module power down and power off before servicing. Before beginning work, always evaluate and perform power off on any module with hazardous energies that service personnel will be exposed to for the scope of work. Be aware that hazardous voltages will still be present on the line side of any isolator switch.
	Mechanical Hazards: These hazards exist in cases where a component can move in such a way as to impact or otherwise injure personnel performing maintenance functions on the equipment. To avoid these hazards, always operate the equipment with the safety covers in place and door interlocks enabled. When performing maintenance functions with the covers open, be sure to avoid those areas where impact with the moving parts of the equipment is possible. When removing components, it may be necessary to isolate potential mechanical energies by restraining the mechanism using cable ties or blocking.

	Compressed Hazards: OFA is supplied to the equipment for many functions. Even though the equipment is electrically powered down, pneumatic pressure is still present. When performing maintenance functions on pneumatic components, always isolate the OFA by closing the manual valve and safely bleed the pressure out of the system.
	Thermal Hazards: Actuation Pan & Bottom Pan can also operate at elevated temperatures and capable of causing burns. Be sure that the components to be worked on are not at elevated temperatures before performing maintenance functions.
	Usage of Material Handling Equipment: When operating or working near Material Handling Equipment (hoists, pallet jacks, forklifts, and overhead cranes) exercise caution to avoid injury. Safety boots (steel toes) are mandatory. Be aware of nearby personnel and equipment and avoid contact. Work with site personnel for proper preparation and reporting prior to employing Material Handling Equipment if required.
	Hand and Portable Power Tools: Exercise caution when using power tools. Safety glasses are mandatory. Hearing protection may be needed depending on the tools used and noise level. If the noise level is above 80dB, earplugs are mandatory. Always power down the equipment and perform before using power tools on the equipment. Use a vacuum system to reduce the spread of particles or debris. Work with site personnel for proper preparation and reporting prior to employing power tools if required.
	Fluid Hazards: When working on EGW lines, isolate the fluids from the section being worked on by closing valves or disconnecting QD's. Drain the section if needed.

	House Keeping: When possible, perform maintenance without blocking aisle ways. If necessary, erect warning signs and / or barricades to warn / block the necessary space to safely perform the required maintenance. Do not block emergency exits. Work with site personnel for proper preparation and reporting prior to work requiring blocking aisles way if required.
	Ergonomics: Avoid Musculoskeletal Disorders (MSD). This range of disorders may be avoided by using the correct tools, proper postures, stretching and avoiding repetitive activities or sitting for prolonged periods. Know your limits and seek assistance when needed. The symptoms of MSD include tingling, numbness, burning sensations, weakness, swelling, soreness, stiffness, spasms, pain, reduced range of motion, aches, and cramping. At the onset of symptoms, notify a supervisor. Evaluation by medical professional may be required.
	PPE: Standard Personal Protective Equipment (PPE) include shoes (closed toe, closed heel, and non-porous), gloves (latex or nitrile), and eye wear (safety, side protection). Other PPE may be required for specific locations or activities.
	Control of Hazardous Energy: Failure to follow safety protocols and reporting requirements are punishable by disciplinary action up to and including termination.



2.2.3 Personal Safety Equipment

Installation or de-commissioning of AEM 1kW Thermal Retrofit System requires full preparation in advance. Installation team must fully understand the safety precautions for machine handling as well as the factory health and safety regulations.

1. PPE

Standard Personal Protective Equipment is required as listed below:

- Safety goggles (non-conductive frame)
- Safety boots (steel toe)
- Smock / Jump suit
- Hard top helmet / bump cap
- Safety glove
- Rubber / leather / cotton glove

2. Tools and Equipment

Other than basic hand tools, the team needs to prepare the following equipment for machine installation:

- Multi-meter



2.3 Safety Information

The electrical machines are operating appliances for use in current heavy industrial facilities. When being operated, the machines have dangerous electrical live parts, and in some cases moving or rotating mechanical parts as well. They can therefore cause serious injuries or damages.

Make sure personnel does not perform unsafe work practices that include unqualified personnel removing covers, using tools in manners that are inconsistent with this manual, operation of the machine by an untrained individual and removing safety features.

Those responsible for safety in the plant must therefore guarantee:

- Only qualified people work with the machine.
- Machine manual delivered with the machine is always available to such people and that service personnel are obliged to observe the procedures contained therein.
- Unqualified persons are forbidden to work at or near the machine.
- Person working on a tool can locate and identify all safety warning labels before use.
- Operators do not wear loose clothing or jewelry while operating the system.
- Main power is off before opening A/C cabinets.
- Operators of the machine are aware of stop locations and operation in case of system malfunction.
- Earth wires are not disconnected from the machine.
- Internal parts of the controller to be never touched during operation.
- Wires and / or connectors are not unplugged when the power is on.
- Machine is not connected to a voltage or frequency other than what is indicated in the specifications.
- Power to the machine is not cut-off during production.

Qualified people are, by reason of their training, experience, and these instructions, as well as their knowledge of the relevant standards; regulations; accident prevention regulations and operating relations, have been authorized by people responsible for plant safety to carry out the necessary work and are able to recognize and avoid possible dangers.



Note: To prevent breakdowns, it is necessary that the prescribed maintenance, inspection and revision measures be carried out regularly by experienced servicing staff (see above).

Changes to normal operations (increased power consumption, higher temperatures, increased noises etc. or responses from the monitoring equipment) can indicate that the tool functions are impaired. To prevent breakdowns, which can directly or indirectly lead to serious injuries or damages, the maintenance staff must be informed without delay.

Unauthorized modification of the tool can lead to personnel injury and machine damage.

If abnormal conditions are observed, activate the stop on the machine immediately.

If in doubt, switch the machine off immediately and contact equipment owner at site for further support.

For any Environment, Health, and Safety related issue, contact local EHS representative at the site for immediate support.

For any tool and equipment related issues which may need attention of AEM, contact local AEM representative for the site or AEM Singapore at +65 6551 0315.

2.4 Safety Devices

In this machine, all possible sources of accidents have been considered and can be avoided using adapted protection devices.

2.4.1 Hazard Labels & Locations

Safety labels are found on the doors and cover parts to draw attention to hazards present behind or under those protections.

All intervention in these zones must be done only by qualified service personnel, who are trained for this task and are responsible for the maintenance of this machine.

All hazardous labels conform to ANSI Z535 requirement.



Figure 2-1 Hazard labels & location





2.5 Safety Hazard

Important safety information for handling and operating the equipment such as information about installation, operation, maintenance, servicing, and inspection is provided in each chapter.

2.5.1 Potentially Hazardous Energy Sources

Table 2-3 Potentially hazardous energy sources

Electrical Power Supply					
	Electrical Power Supply	Voltage	Max Current	Phase	Frequency
01	AEM 1kW Thermal Retrofit System	415VAC	63A (26145W)	3 Phase	50/60Hz
	PDU	415VAC	40A (16600W)	3 Phase	50/60Hz
	Chiller	415VAC	63A(26145W)	3 Phase	50/60Hz
02	Main EGW Supply Flow Rate	0 ~ 50 LPM (minimum), -30 °C temp			
03	Main EGW Supply Pressure	5.5 Bar			
04	Main Air Supply	5.5 – 9 Bar, oil free, -30C or lower dew point			

2.5.2 Precaution Before Automatic Operation Mode

- Keep all personnel away from the machine as the movement could cause serious injury when struck by moving parts.
- Ensure that the machine is free from any foreign objects.
- Ensure that all panel doors, which are equipped with safety interlock, are closed.



2.5.3 Hazardous Energies Listing

Table 2-4 Type of hazardous energy

YES	NO	Energy Type	YES	NO	Energy Type
		1) Electrical		√	8) Ionizing Radiation
√		a) Low voltage (50 – 600V)		√	9) Non-ionizing
	√	b) High voltage (>600V)		√	a) Ultra-violet
	√	2) Chemical - Explosion, pressure, extreme heat, fire, corrosive, reactive, oxidizer, toxic		√	b) Infra-Red
√		3) Pressure (hydraulic/pneumatic)		√	c) RF / Microwave
	√	4) Vacuum		√	d) Laser
√		5) Mechanical – Capable of crushing, pinching, cutting, snagging, striking		√	e) Magnetic
√		6) Thermal (high temperature) – Surface temperature, hot liquids, steam		√	10) Stored – Flywheels, springs, differences in elevation, elevated parts that could drop, capacitors, batteries
	√	7) Thermal (cryogenic) – Contact with super cold surface or with a cryogenic liquid			

2.5.4 Draining of EGW Line

Personnel must take note that there may be EGW hazard when performing service or maintenance. Personnel must shut off the EGW inlet valve at customer's facilities, cutting off incoming EGW sources before draining EGW hoses. Unplug the EGW inlet/outlet quick connect and reconnect to the QD for air supply and reservoir tank to drain and purge all EGW inside the thermal head.

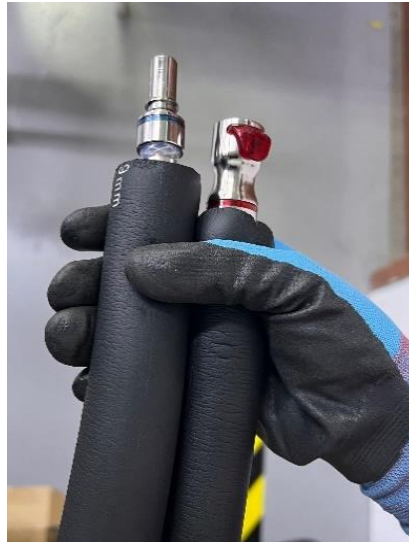


Figure 2-2 EGW draining

2.5.5 Main Power High Voltage Supply Hazard

This equipment contains high voltage supply in certain areas of electrical components. This is a hazardous situation where it can cause serious material damage or carry a risk of personal injury.

In consequence, it is strictly forbidden to modify, add or take off any parts and/or other elements that constitutes the machine.

Those responsible for safety in the plant must therefore guarantee:

- That only qualified persons work with the machine.
- That the machine manuals delivered with the machine are always available to such persons and that they are obligated to observe the procedures contained therein.



2.5.6 Standard 3 Point Safety Check

When checking the voltages within the tool, always validate the multimeter being used by performing the following checks:

- Set the multimeter to the function to be used in the tool.
- Set the range for the measurement to be used.

Note: The multimeter should indicate 0 volts (or amperage or infinity resistance).

- Measure a known working voltage source such as a working power outlet.

Note: The meter should indicate normal supply voltage:

- 230 VAC at United States.
- 230 to 240 VAC at Singapore & Malaysia.
- 220 VAC at China, Israel & Vietnam.
- Do not continue until the correct voltage is indicated.
- Switch to another multimeter if needed or try a different outlet.

Note: This verifies that the multimeter is functional before performing the test measurement.

- Perform the test measurement in the tool.
- Record the reading if needed.
- Again, measure a known working voltage source such as a working power outlet.

Note: The meter should indicate normal supply voltage (See above for normal supply voltage).

Note: This verifies that the multimeter is functional after performing the test measurement.



2.6 Environmental Consideration

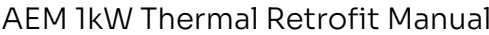
Important environmental consideration for handling materials used in installation, operation, maintenance, servicing, and inspection is provided in each chapter.

2.6.1 Warning of Using Isopropyl Alcohol

Isopropyl Alcohol (IPA) is a highly flammable liquid & can form an explosive mixture with air at elevated temperatures. It can cause eye & skin irritation if under contact in either vapour or liquid form. Follow the following precautions before & during use of Isopropyl Alcohol.

Note:

- The maximum concentration of IPA is 10%.
- Wear hand gloves (chemical resistant) while using IPA.
- Wear chemical goggles for eye protection while using IPA.
- Pour about 5 to 10 ml of IPA into a disposable container & use cotton bud to clean the parts specified for preventive maintenance requirements.
- After cleaning the parts, dispose of the container & cotton buds by sealing in a plastic cover. Need to ensure the disposal methods are in accordance to the local environmental regulation norms and/or Customer Site Disposal Procedure.
- Ensure IPA is handled in a well-ventilated area.
- For further information on IPA, refer to [12.1.1 Isopropyl Alcohol](#).

[illegible]



3. System Overview



3.1 Facility Requirements

3.1.1 Electrical Requirements

Table 3-1 Electrical requirements

Item	Phase Type	Voltage	Wiring Configuration	Full Load (Amps)
PDU	3 phase, 5 wire	415VAC	L1,L2,L3,N,E	63A
Chiller	3 phase, 5 wire	415VAC	L1,L2,L3,N,E	63A
Socket	Single phase	240VAC	P+N+E, SINGLE PHASE	13A

3.1.2 Gas Requirements

Table 3-2 Gas requirements

Item	Connection Type	Internal Regulator	Operating Pressure (PSI)
	Connect Size	Internal Shutoff	
OFA – 2-9 BAR	One Touch Fitting	YES	100.0
	10 mm Horse	YES	

3.1.3 Liquid Requirements

Table 3-3 Liquid requirements

Item	Connection Type	Operating Pressure (PSI)	Average Flow (LPM)
	Connect Size		Peak Flow (LPM)
EGW Supply	Quick Connect	70 BAR	4 LPM
	SS-QC8-D-810EP		
EGW Return	Quick Connect	70 BAR	4 LPM
	SS-QC8-B-810EP		

3.1.4 Drain Requirements

Table 3-4 Drain requirements

Item	Connection Type	Connect Size	Internal Shutoff	Average Flow (GPM)
				Peak Flow (GPM)
EGW	Quick Connect	SS-QC8-B-810EP	-	-
				-

3.1.5 Equipment Footprint

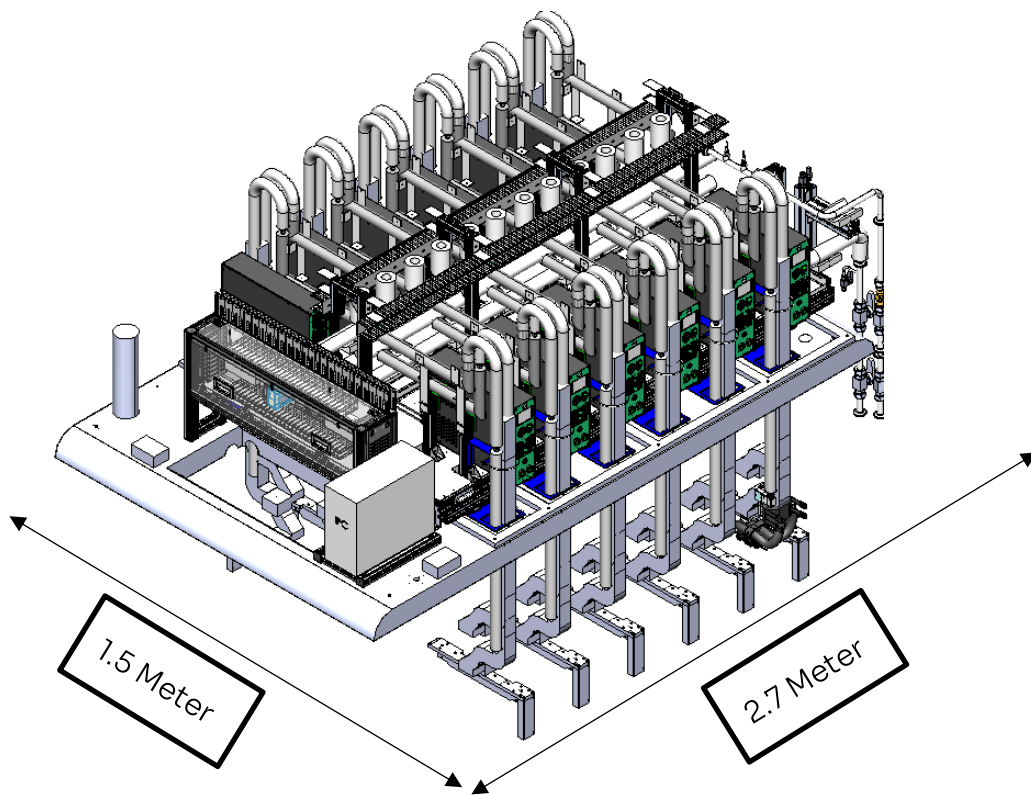


Figure 3-1 Equipment footprint

Table 3-5 Equipment footprint

Dimension	Value
Length (mm)	2.7 M
Width / Depth (mm)	1.5 M
Piping Height (mm)	Approximately 2.8 M

3.2 Mechanical Specification

This section provides the mechanical specification required to operate the system.

The table below lists the mechanical specifications. These specifications set a standard for operations. Equipment performance is gauged by the operation specifications set here.

Table 3-6 Mechanical specification

Piping

S/NO	ITEM	SPECIFICATION
1	Operating Temperature	110°C ~ -30°C
2	Relative Humidity	40% -60%
3	Weight	250KG
4	Flowrate	3.7 l/min
5	StorageTemperature	-25°C to 55°C
6	Storage Humidity	5% -95%. Atmospheremustalwaysbenon-condensing

S/NO	ITEM	SPECIFICATION
1	Operating Temperature	12°C to18°C
2	Relative Humidity	40% -60%
3	Maximum NoiseLevel	65 dB
4	Machine Colour	White +Black
5	Machine Weight	62KG
6	Altitude	Up to 1000 m
7	MachineStorage Temperature	-25°C to55°C
8	Machine Storage Humidity	5% -95%. Atmospheremustalwaysbenon-condensing

3.3 Machine Component

3.3.1 Chiller

Chiller- TCm4400W-AEM is 3 phase refrigeration system that uses water as a coolant to control process temperatures of fabrication equipment. An ozone-friendly refrigerant is used in the refrigerant coils to absorb the heat. This heat is then transferred to the water later taken to the cooling tower. Chiller has a display screen touch. This feature makes it stand out and instead of a fixed or rigid compressor has a variable compressor. Chiller uses computer programming to control its various operations like speed, voltage, water level, and temperature.

It controls the cooling by controlling the temperature to adjust it according to the requirement. Chiller is designed and programmed in such a way that it adjusts its voltage according to the compressor load.

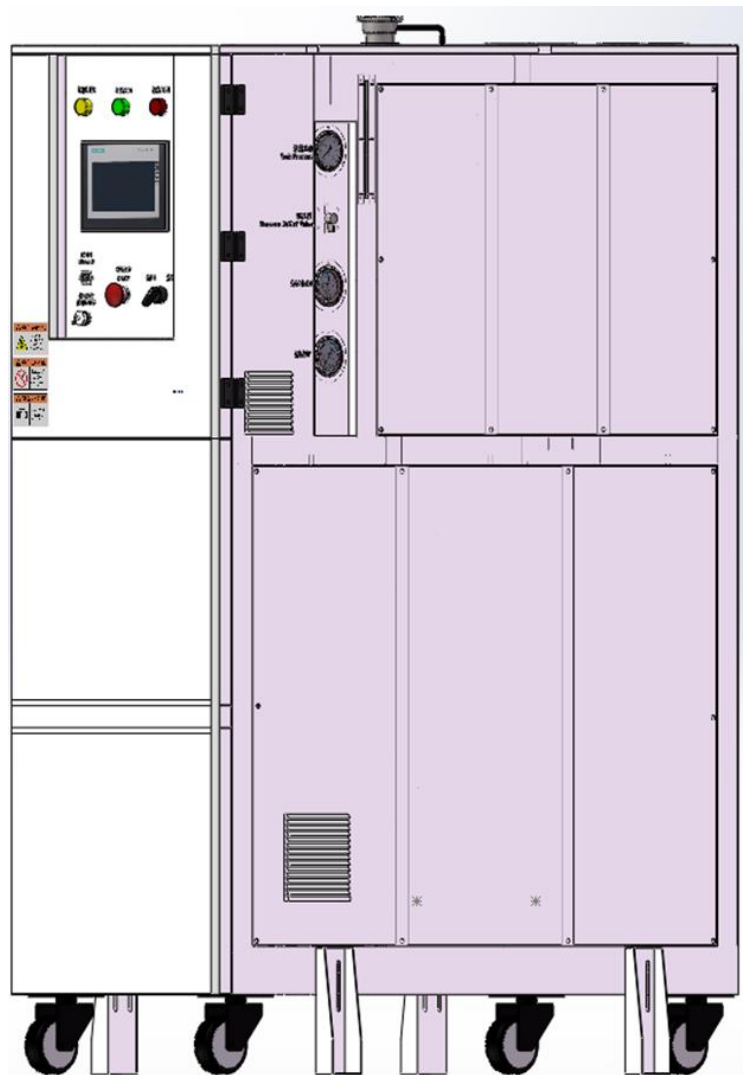


Figure 3-2 Chiller- TCm4400W-AEM

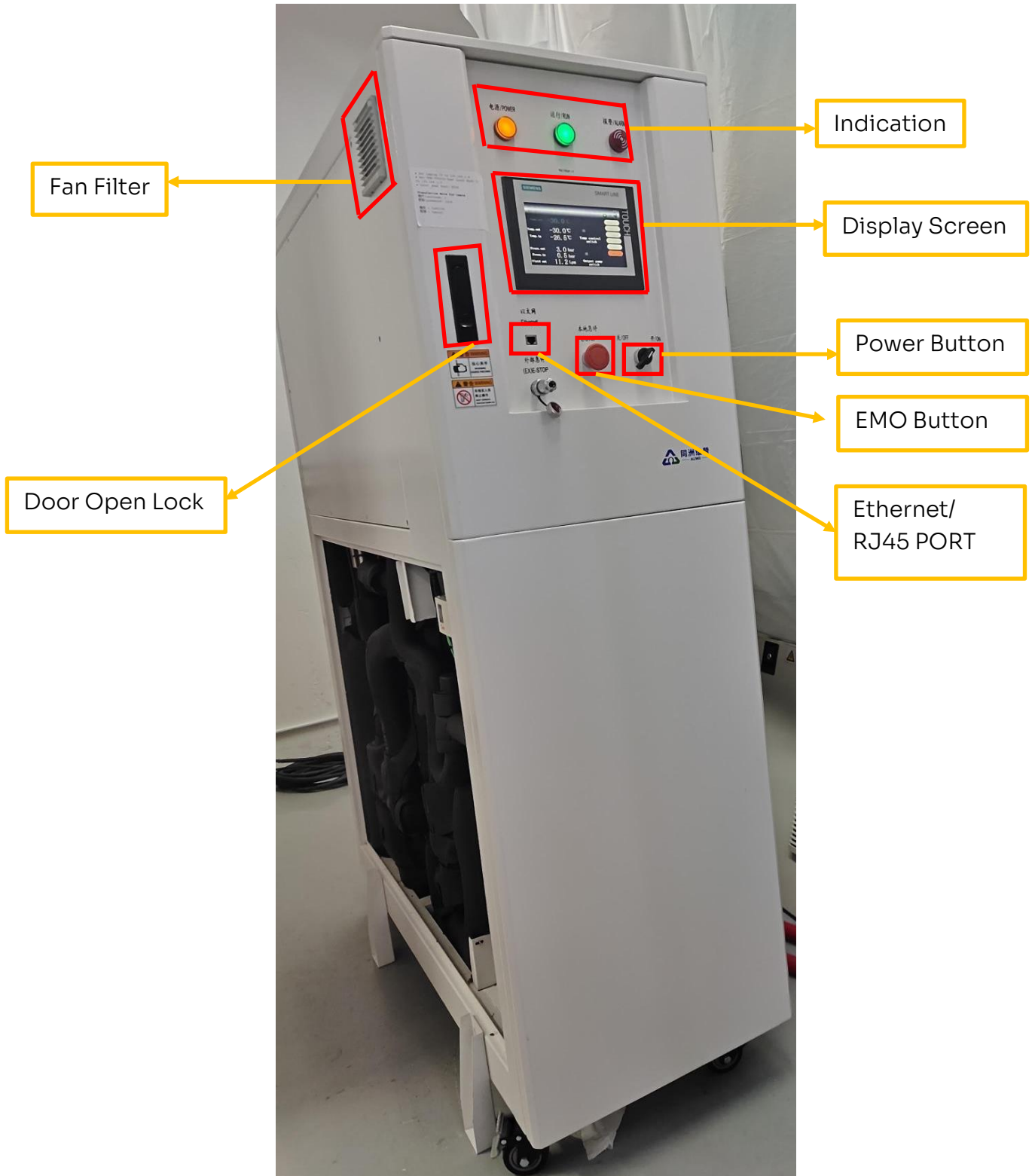


Figure 3-3 Label of parts in chiller (Front)

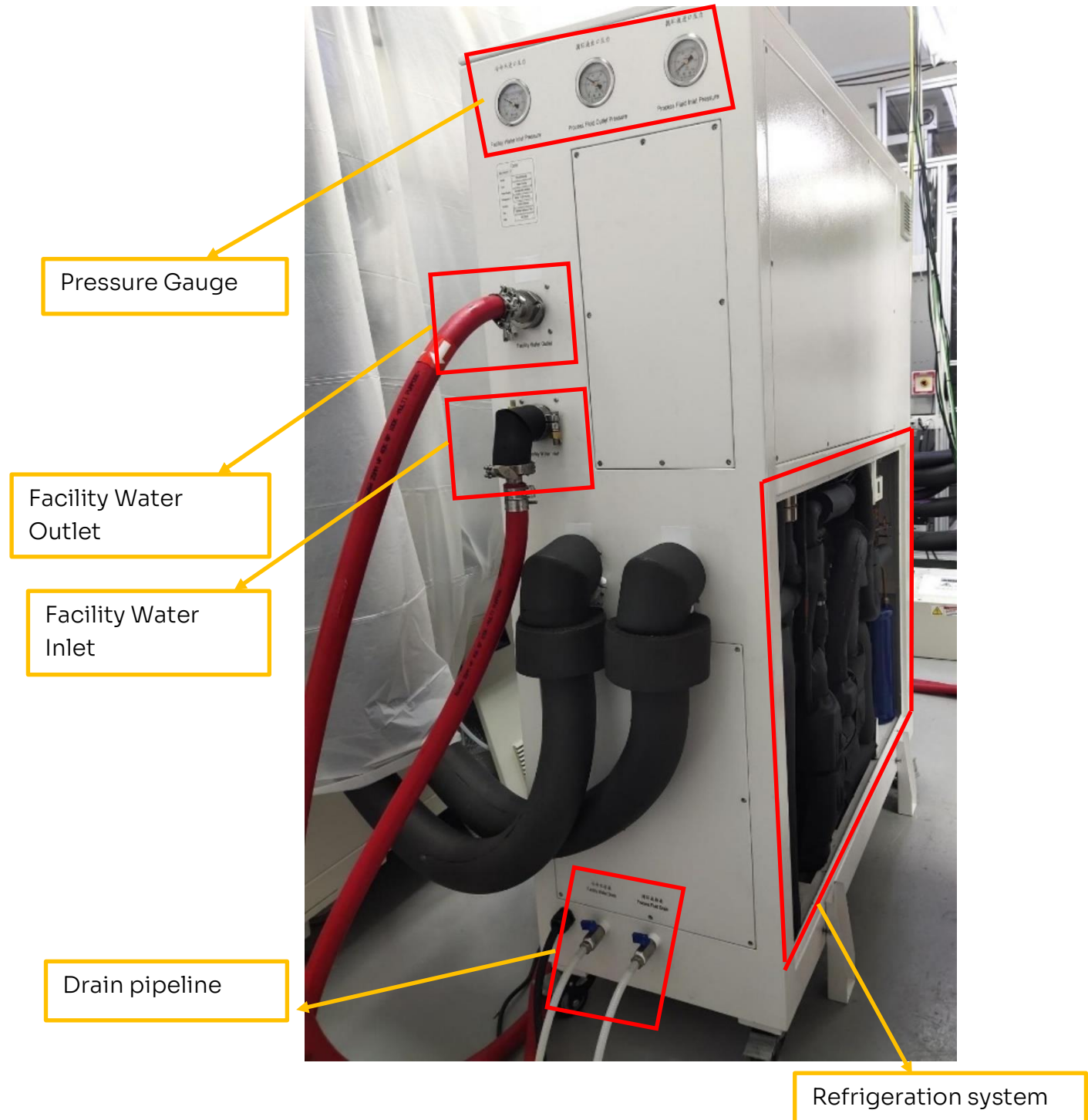


Figure 3-4 Label of parts in chiller (Back)



3.3.1.1 Chiller Specification

Table 3-7 Chiller Specification

	Model	CHL-LC-EGW-9KW-00
Process fluid	Process Re-Circulating Fluid	60% Ethylene Glycol + 40% DI water + 2% inhibitor
	Process Fluid Cooling Capacity @ 20°C	40.0kW@20°C@50Hz
	Process Fluid Cooling Capacity @ -20°C	15.0kW@-20°C@50Hz
	Process Fluid Cooling Capacity @ -30°C	9.0kW@-30°C@50Hz
	Process Fluid Heating Capacity	4.0kW (HGBP)
	Temperature Control Mode:	PID
	Temperature Stability	±1.0°C
	Temperature Range	-40°C to +35°C
	Pump Pressure Range	30~ 120psi(VFD)
	Pump Flow Range	5 to 80 l/min(@25°C)
	Pump Rated Flow Range	50 L/min @-30.0C @ 100psi@ < 40cm ² /s
	Reservoir Tank Capacity	Reservoir Tank 50L
	Reservoir Tank Material	SS304
	Reservoir Tank Feature	2" Sealed Cap
	Reservoir Tank Pressure Relief	Push button to trigger valve open to relieve air pressure trapped in reservoir
	Reservoir Tank Air Pressure Monitor	Analog pressure gauge for user to know the pressure of trapped air in reservoir tank
	Process Fluid Filtration	No Filter
	Process Fluid Drain	G 1/2" Ball Valve
	Process Fluid Internal Bypass	Adjustable Pressure Relief Valve
	Process Fluid Pressure Sensor	Outgoing and return pressure sensor + needle gauge
	Process Fluid Temperature Sensor	Outgoing and return temperature sensor
	Process Fluid pH Sensor	None
	Process Fluid Pump Type	Magnetic drive pump + VFD
	Process Fluid Flow Meter	Outgoing magnetic flowmeter
	Process Fluid Port Size and Type	ϕ 50.5 Sanitary Flange type 1/ ID25/OD32
Refrigerant	Condenser Cooling Method	Water cooled
	Cascade Stages	Single Stage
	Refrigerant	R404A



Facility Water	Facility Water Supply Flow & Pressure	"5 m ³ /h @2.5bar 83.33 L/min @ 36.2psi @EGW -30C @PCW Temp 15C"
	Facility Water Supply Temperature Range	15~25°C
	Facility Water Return Pressure	<1.0bar
	Facility Water	"hardness<550mg/l, pH between6.5~9.5"
	Facility Water Port Size	ϕ 50.5 Sanitary Flange type 1/ ID32/OD38
	Facility Water Drain	1/2" Intenal threaded ball valve
Electrical and Facility	Power Requirement (Single Power Source Input)	380 ~ 420V/50Hz, 3-phase 460V/60Hz, 3-phase
	Electrical Power Connection Type	Bare leads
	Max Current	50A
Interface to External	EMO Connection	ok
	Communication	Ethernet (modbusTCP)
Equipment Software Specification	Communication Interface	Ethernet (modbusTCP)
	Control Parameter	Same as local operation
Equipment Health hand Safety Specification	Noise Level	≈70db
	Emergency Stop Switch	Local E-stop and external E-stop.
	Leak Detection	Drip pan + level switch
	Process Fluid Reservoir Level Detection	High + Low Fluid level monitoring
	Temperature Sensor	Discharge temperature and return temperature monitoring
	Internal Flow Sensor	Flow switch for facility water
	Refrigeration Pressure High / Low Detection	High + Low Pressure Switch
	Compressor / Pump Overload Detection	Compressor / Pump Overload Protection
	Phase Reversal Relay	phase sequence protection
	Fluid No Flow Audible & Visual Alarm	Interface Audible and visible alarm
	Low/High Fluid Temp Alarm	User able to set temperature alarm for upper and lower limit



	Reservoir Relief Valve	Expansion tank + overflow pipe + air exhaust pipe
	Reservoir Manual Pressure Relief	Automatic exhaustion of air
Equipment other Feature	HMI Display	7" Screen
	Seismic Bracket	Provided
	Caster Wheels	caster wheels with brake
Equipment Maintenance Specification	Semi-Annual Maintenance	1 year warranty
	Annual Maintenance	1 year warranty
Equipment Certification or Compliance Requirement	CE	Designed to Comply
	EMC Directive (2004/108/EC)	Designed to Comply
	Pressure Directive (97/23/EC)	Designed to Comply
	SEMI S2	Designed to Comply
	SEMI F47	Designed to Comply
	SEMI S8	Designed to Comply
Equipment Operation Condition	Ambient/Facility Water Temperature Range	10°C to 20°C
	Ambient Relative Humidity Range	30% to 70%
Dimensions and Weight	Equipment Length	1600mm
	Equipment Width	850mm
	Equipment Height	1800mm
	Equipment Weight	≈700kg
Accessory	Power Cable (W Type, 2000V, 90C) 33' Long	10m power cable
	Process Fluid Drain Hose	3m fluid drain hose

3.3.1.2 Display Screen

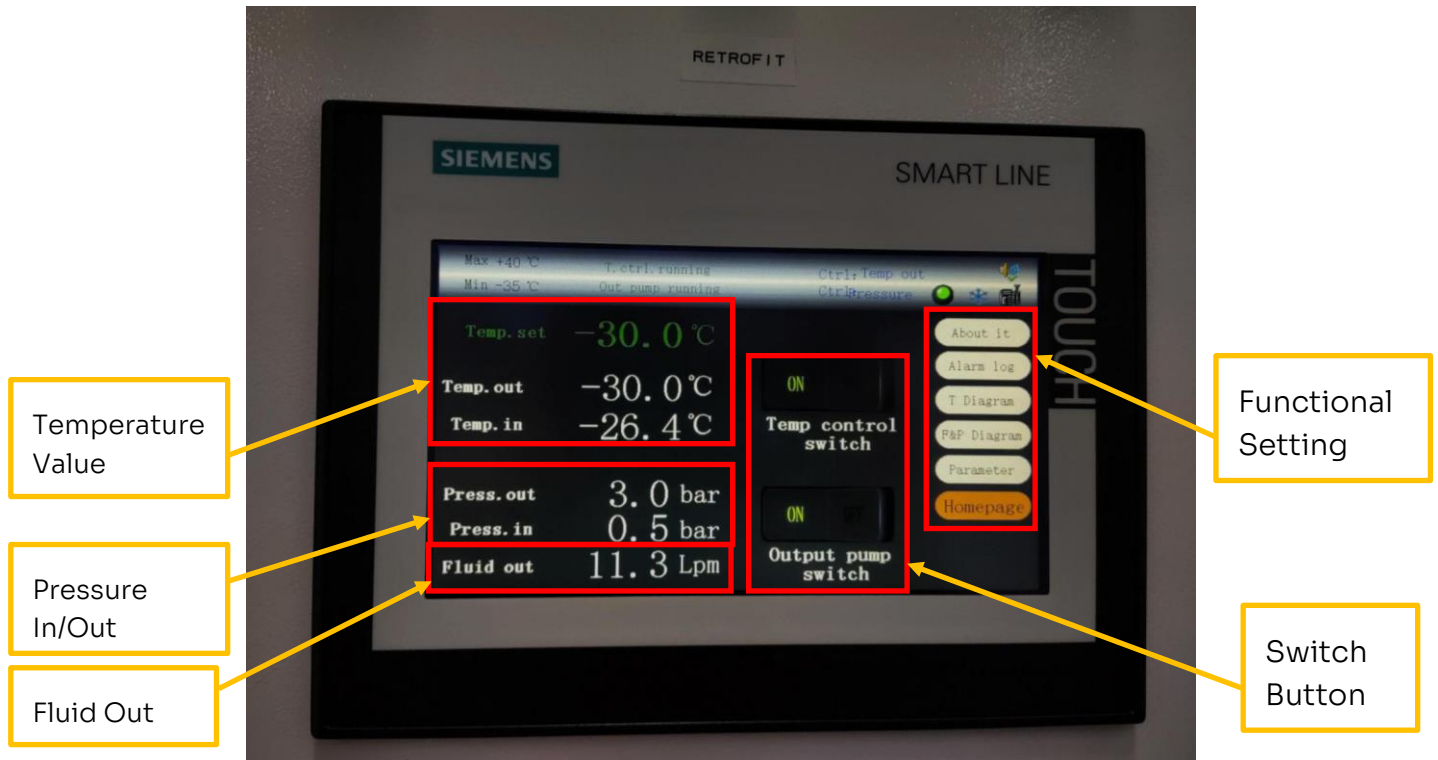


Figure 3-5 Chiller Display touch screen

3.3.1.2.1 Functional Setting

i. Homepage

Display all the actual and set values, consider as a front page.

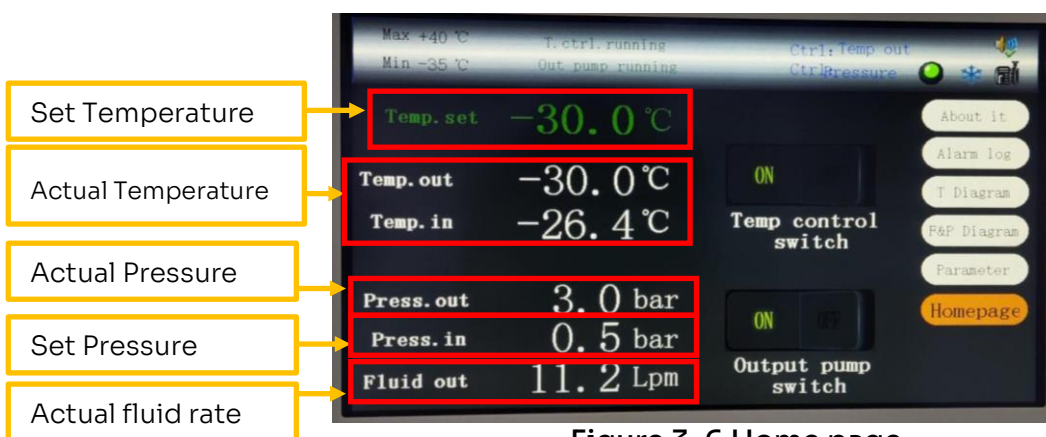


Figure 3-6 Home page

ii. Parameter

All the parameters should be set to Parameter Display Screen.

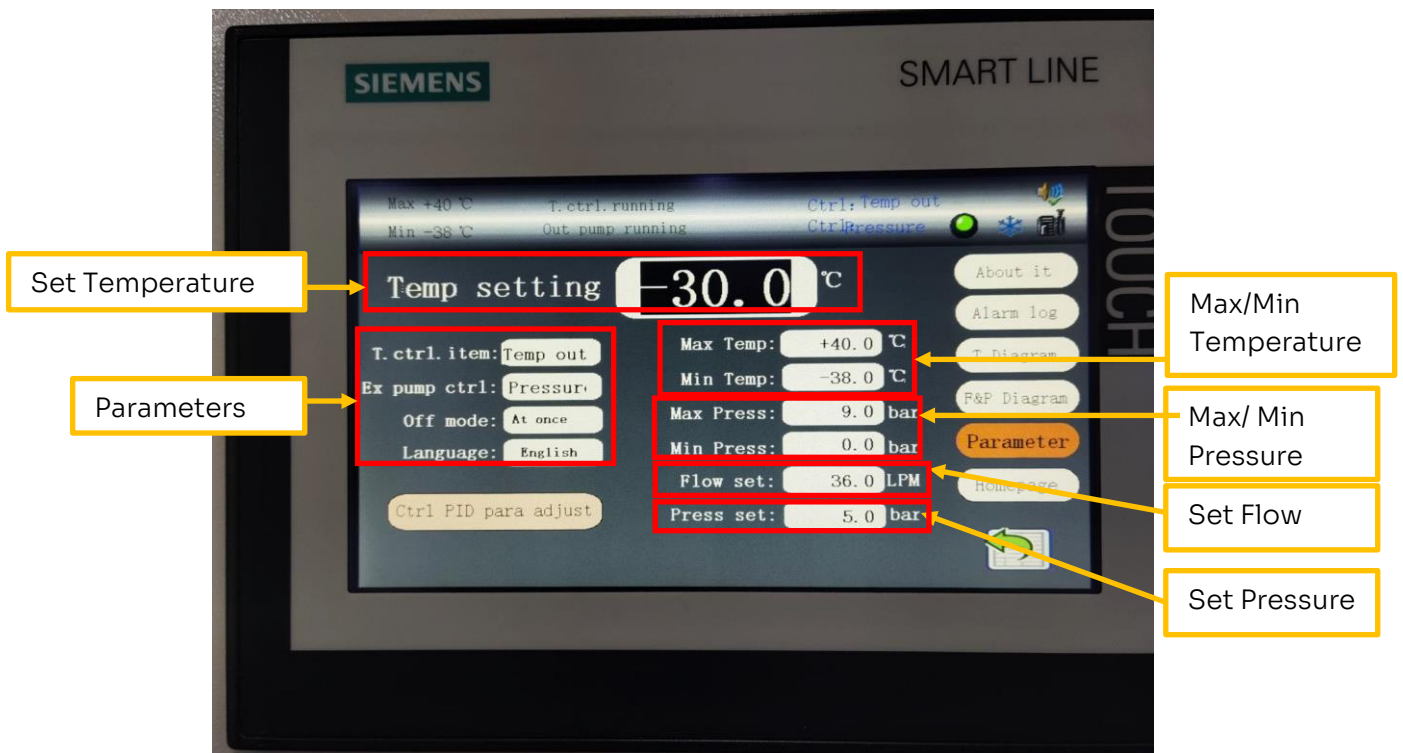


Figure 3-7 Parameter

iii. Alarm log

Display all the current and previous alarm popping up.

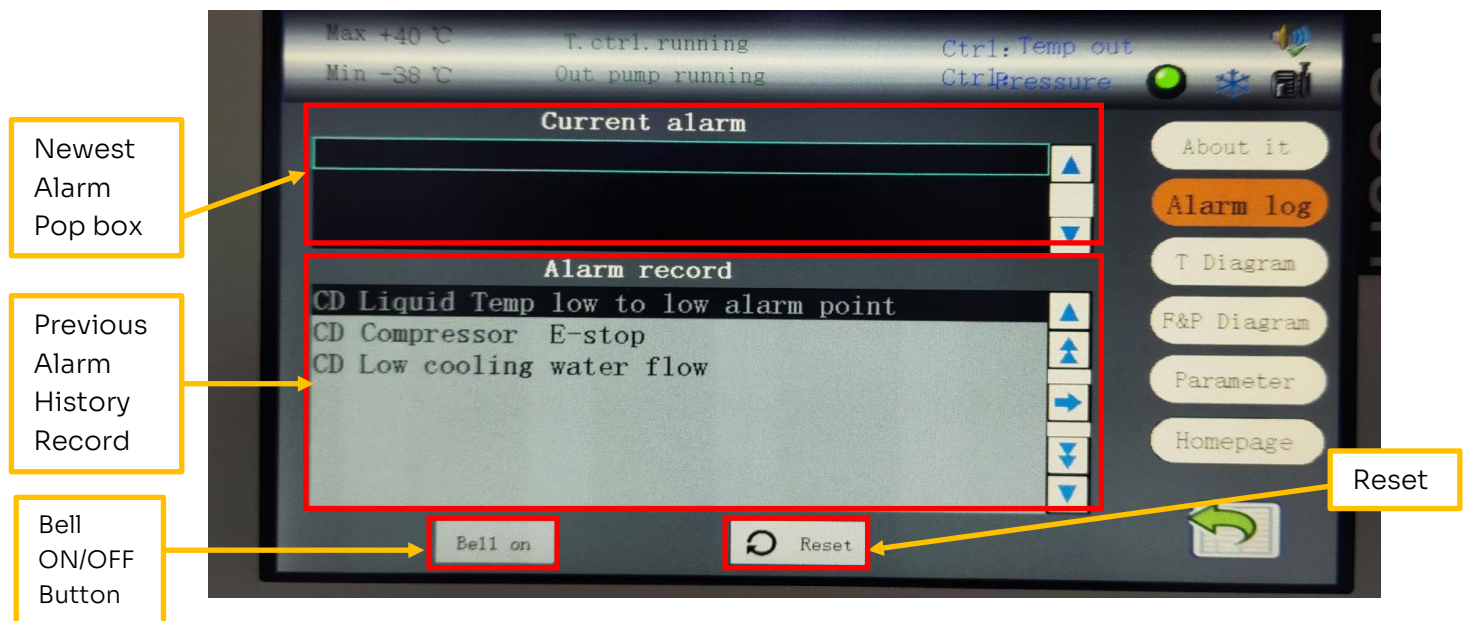


Figure 3-8 Alarm log screen

iv. About it

Display overall function and parameters of chiller.

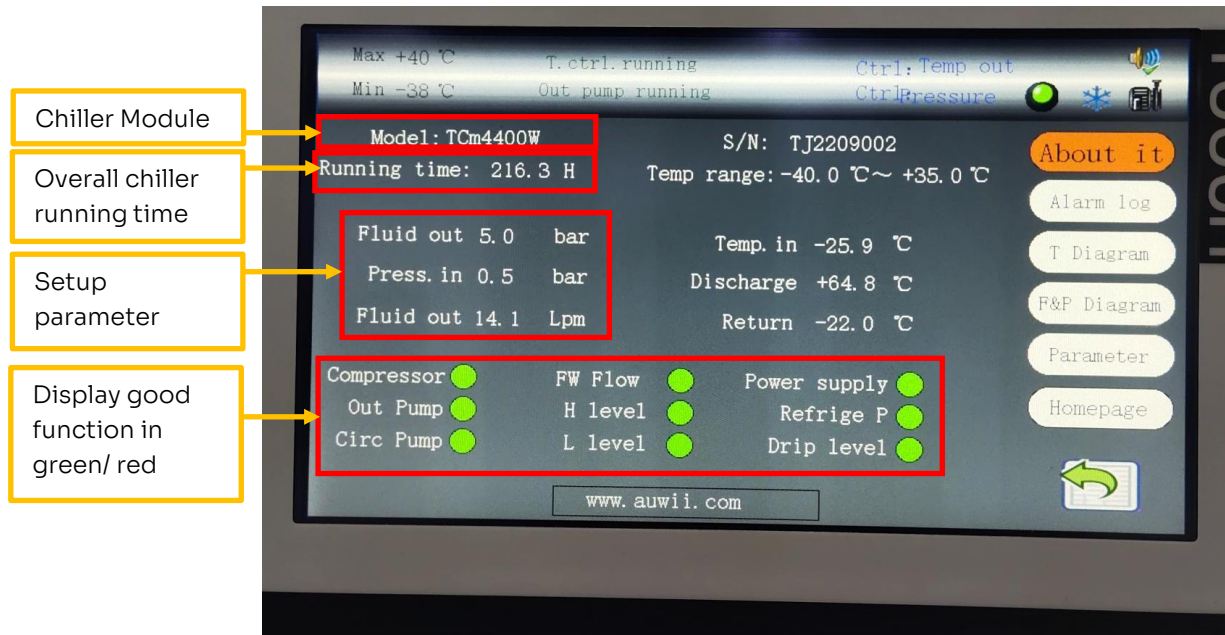


Figure 3-9 About it Page

3.3.2 Thermal Controller

The Controller Box provides thermal and electronic pressure control. Each Module consist of 12 controllers.

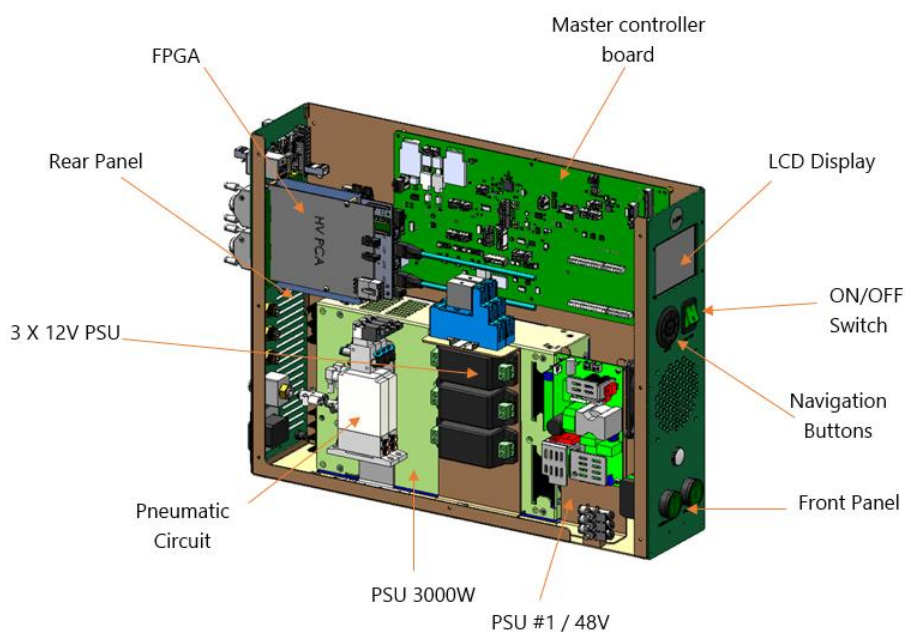


Figure 3-10 Controller box (without cover) components

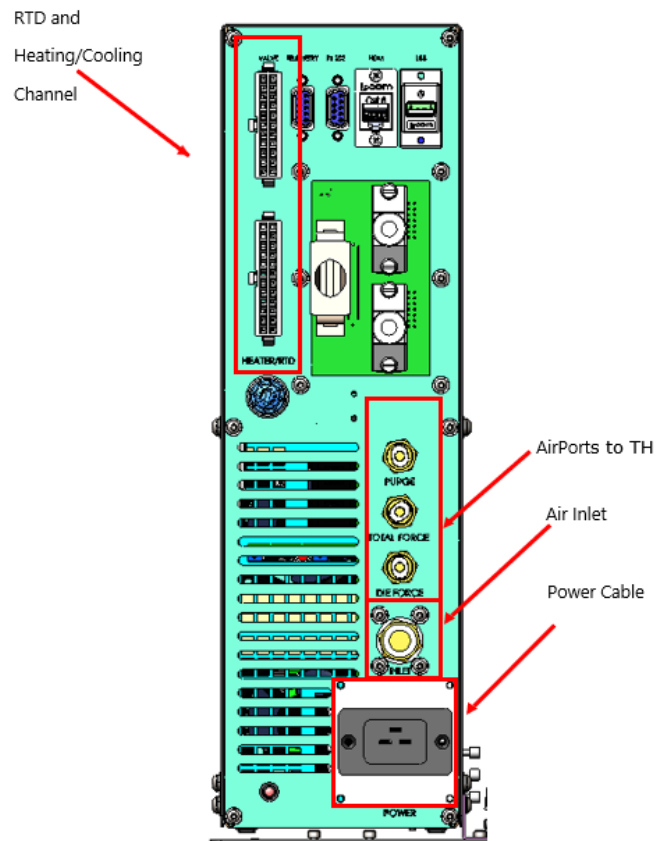


Figure 3-11 Controller box rear panel



Figure 3-12 Retrofit PDU Connection Layout

3.3.3 Thermal Head & Kit

The kit includes the nest base with top and bottom layers, the heatsink, heat let and the PCB connectors on the kit.

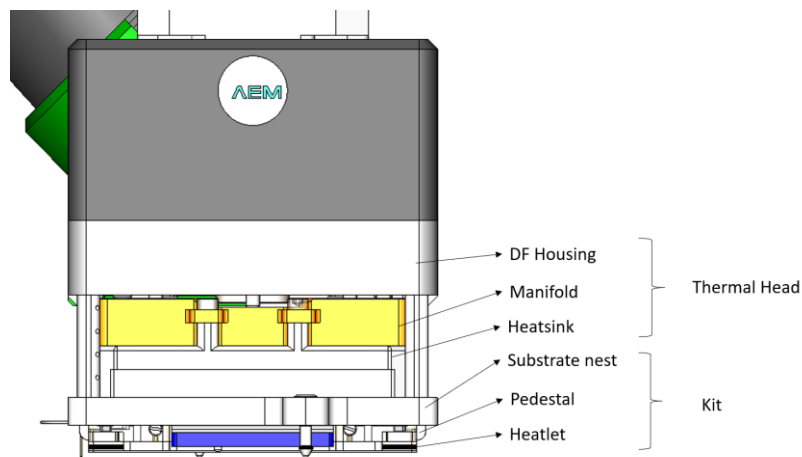


Figure 3-13 Thermal Head Components

Thermal head includes the DF housing, manifold and heatsink. The kit is mounted on the thermal head and contains pedestal, heatlet and substrate nest.

When thermal head goes down, it is engaged with the device. It has die-force that push the chips and substrate force that push the substrate. It provides heating via heating cable on the heatlet and cooling via cooling liquid that flow through the manifold to maintain the desired temperature.

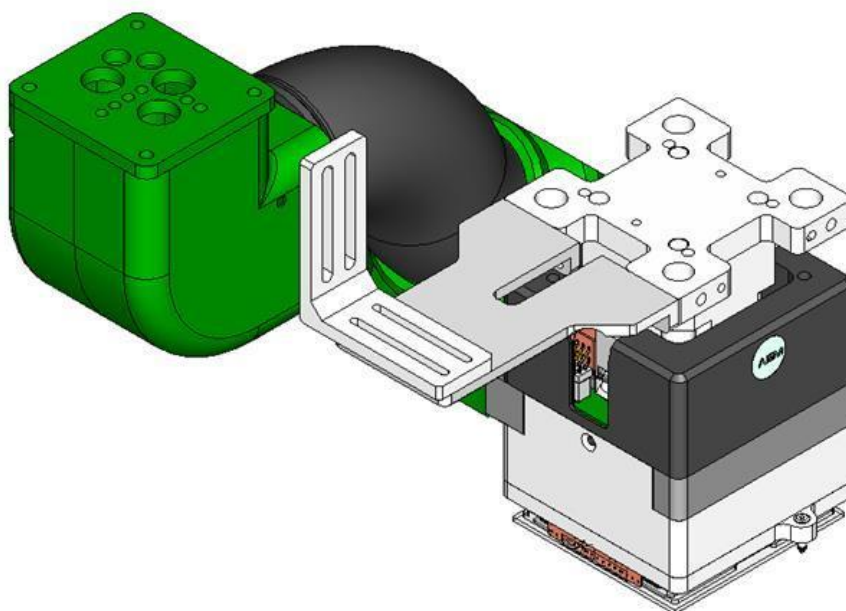


Figure 3-14 Thermal Head ISO View

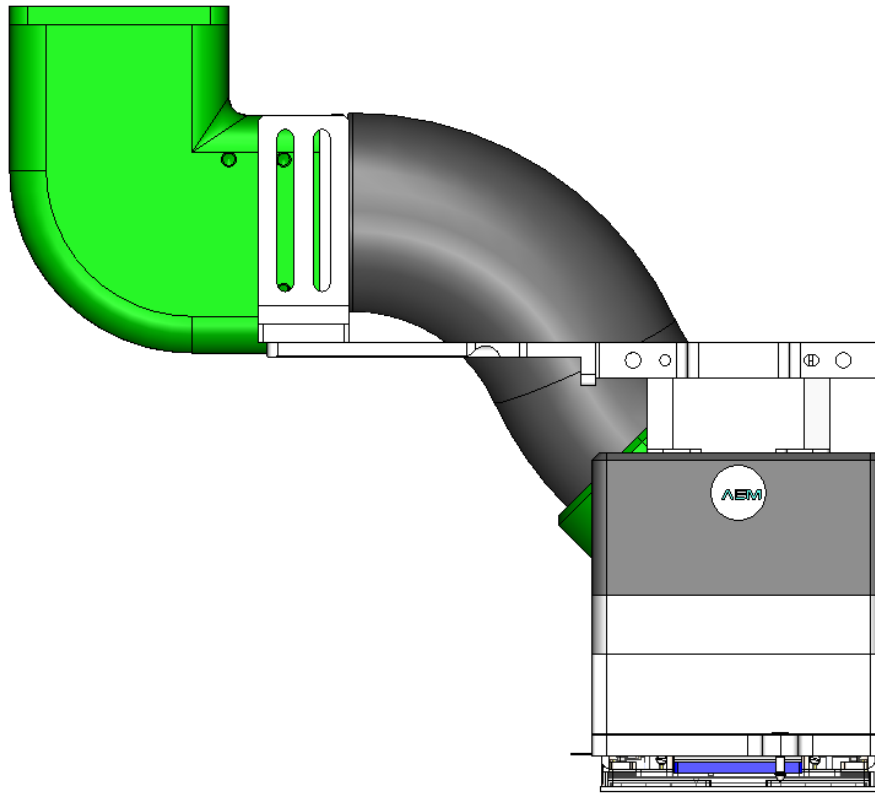


Figure 3-15 Thermal Head Back View

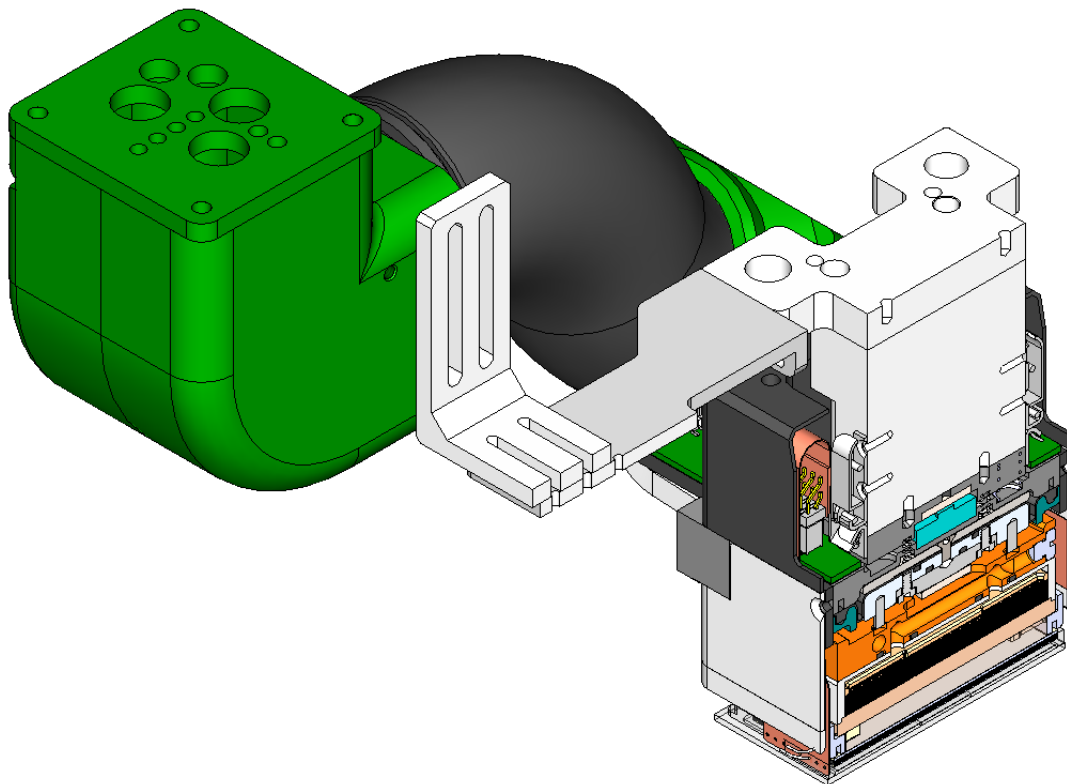


Figure 3-16 Thermal Head section view

3.3.4 Heatlet with RTD

PWM function for heater and monitors temperature (-25°C to 125°C) with the RTD within $\pm 0.5^{\circ}\text{C}$ tolerance. The heatlet has a safety feature that automatically cuts off heater power if temperature exceeds 144°C .

Note: There is a 1wire –Eeprom to save the information of the kit ID at the connector of each kit.

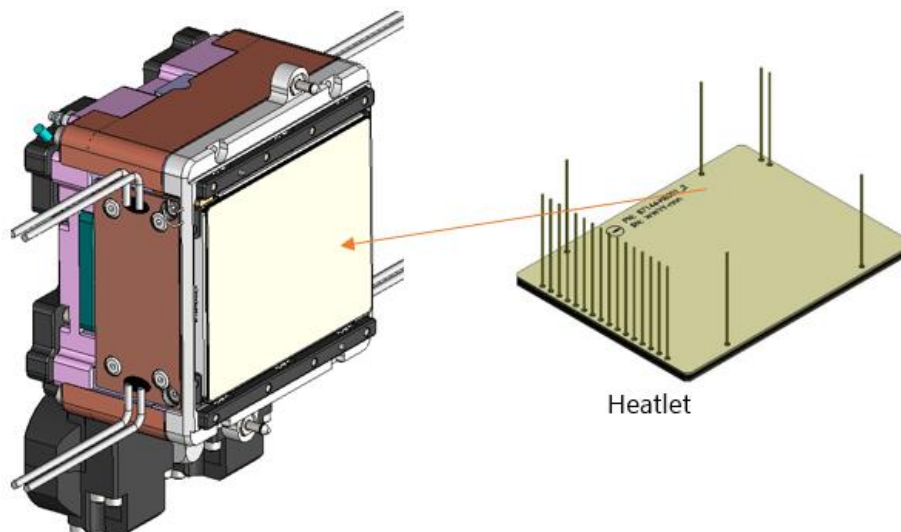


Figure 3-17 Thermal Head with Heatlet

The heatlet single zone with a total of $200 \times 8 = 1600\text{W}$. 1 RTD is used to measure the temperature.

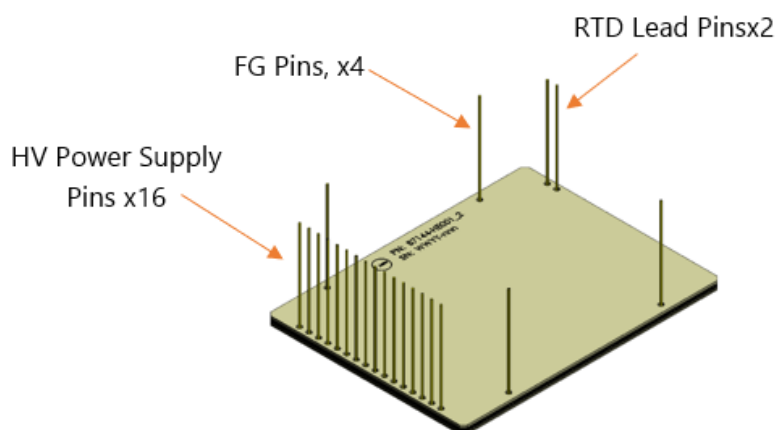


Figure 3-18 Heatlet

3.3.5 Piping

The piping system consists of two pipelines used to convey fluid by supply and return to the chiller. The pipeline can withstand $-30^{\circ}\text{C} \sim 110^{\circ}\text{C}$ with flowrate of 3.7 liter per minutes.



Figure 3-19 Piping system of retrofit



Figure 3-20 Piping system installed at machine

3.3.5.1 Piping Flow Direction

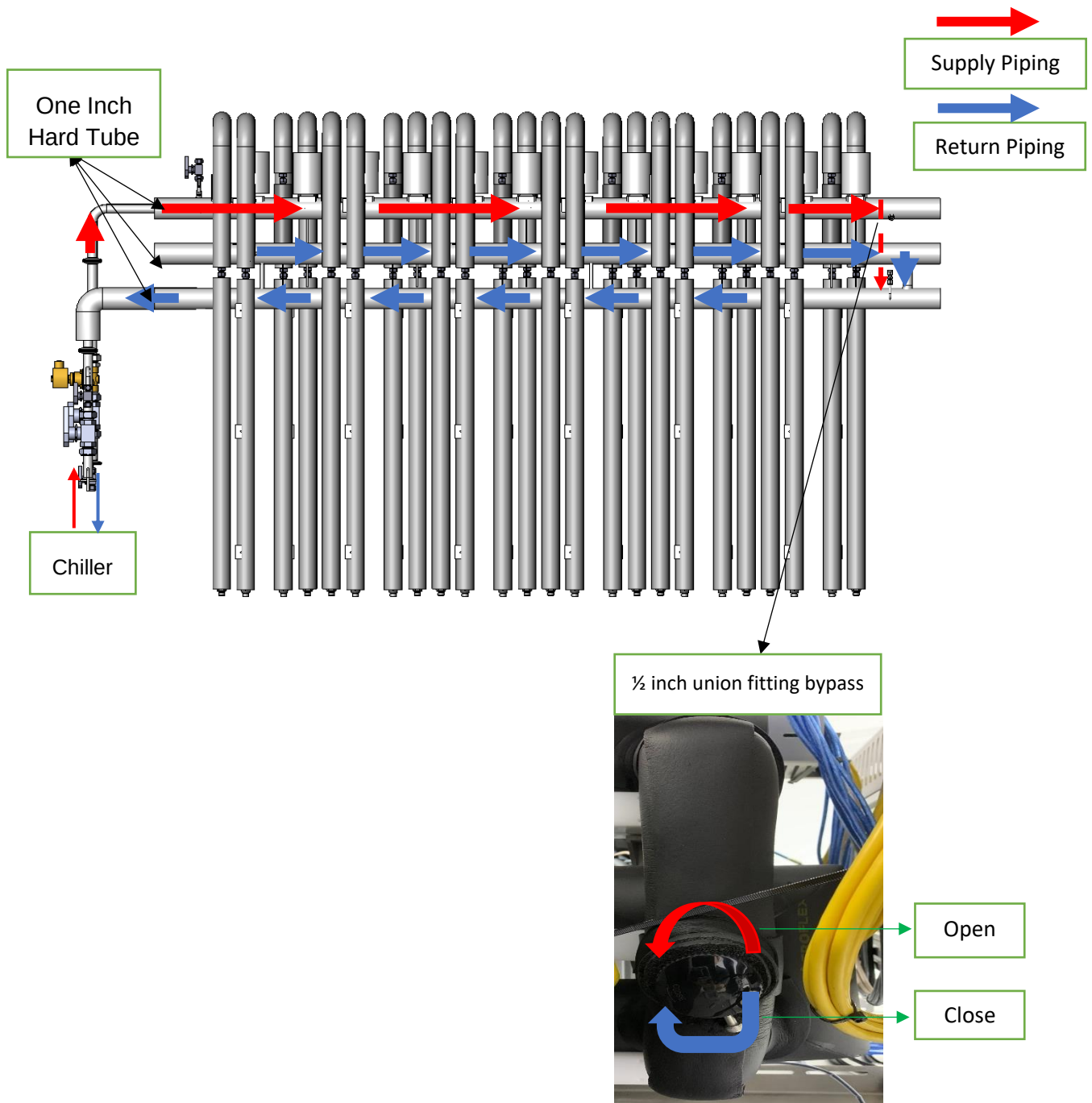


Figure 3-21

3.3.5.2 Branch Piping

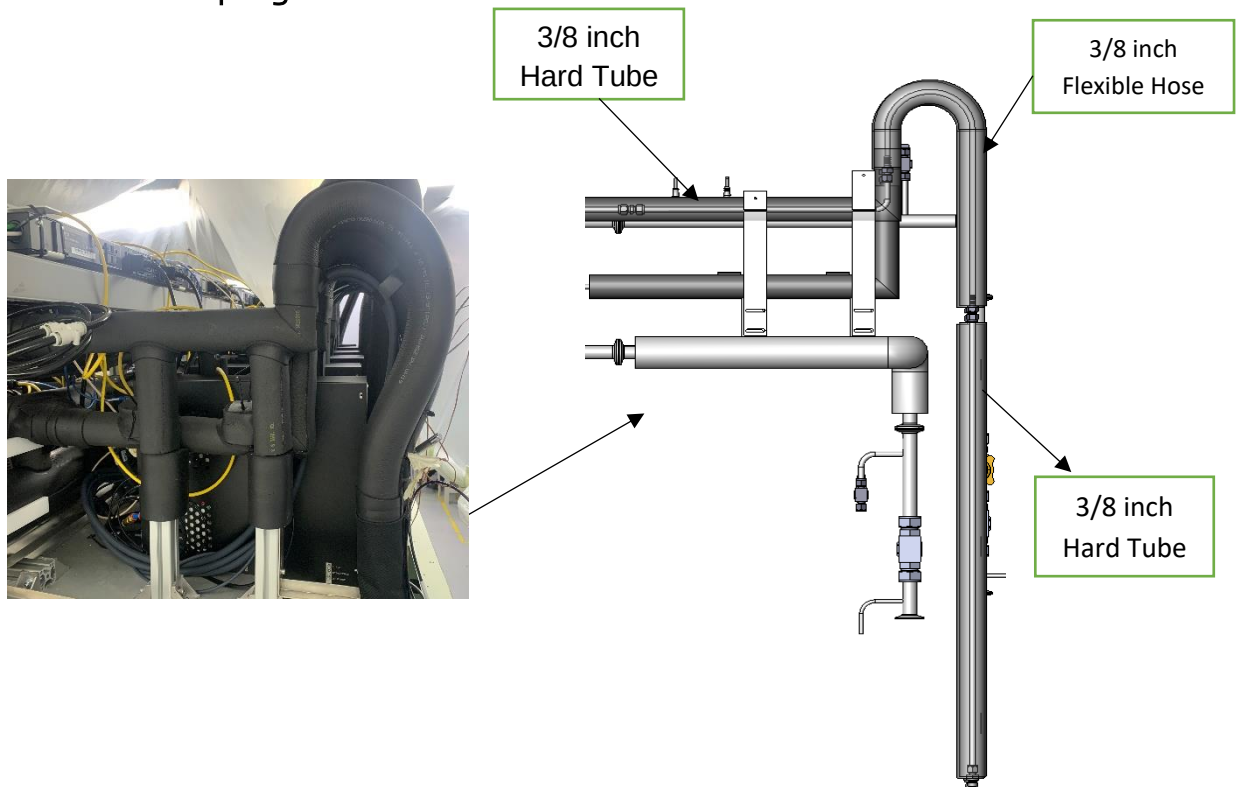


Figure 3-22

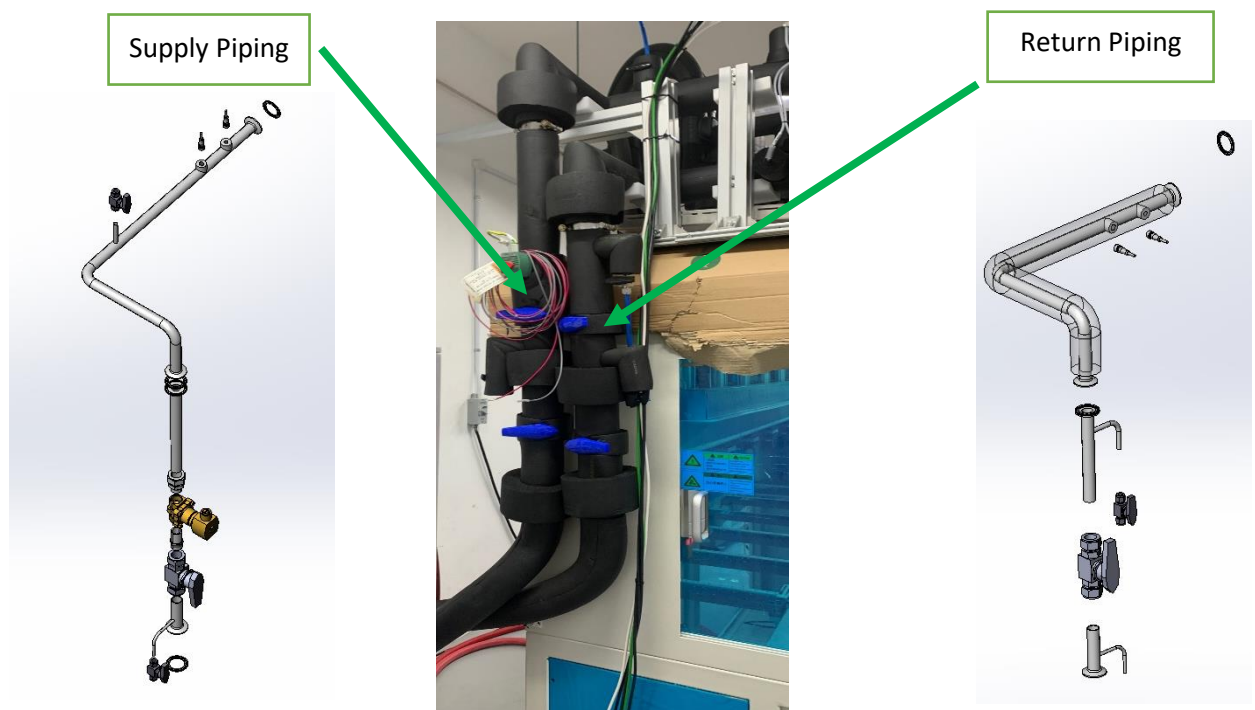


Figure 3-23

3.3.5.3 Piping Sensor

3.3.5.3.1 Pressure Sensor

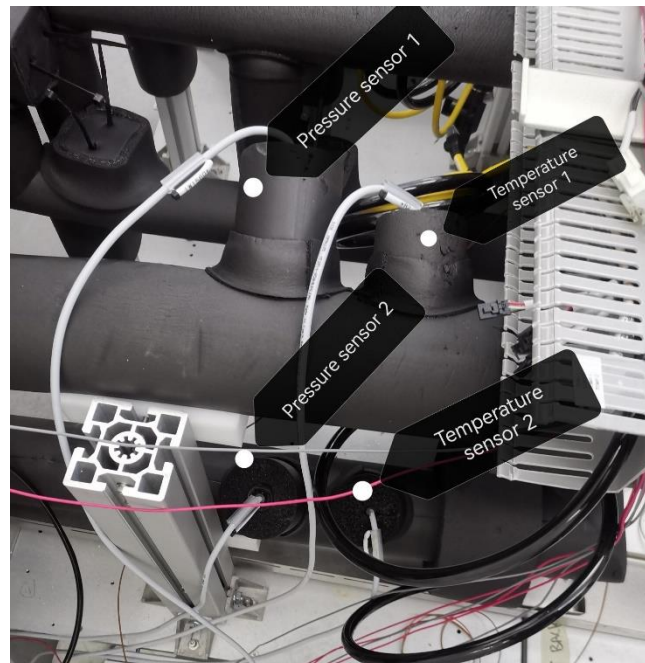


Figure 3-24 Location of Pressure Sensor

3.3.5.3.2 Drip Tray Leak Sensor

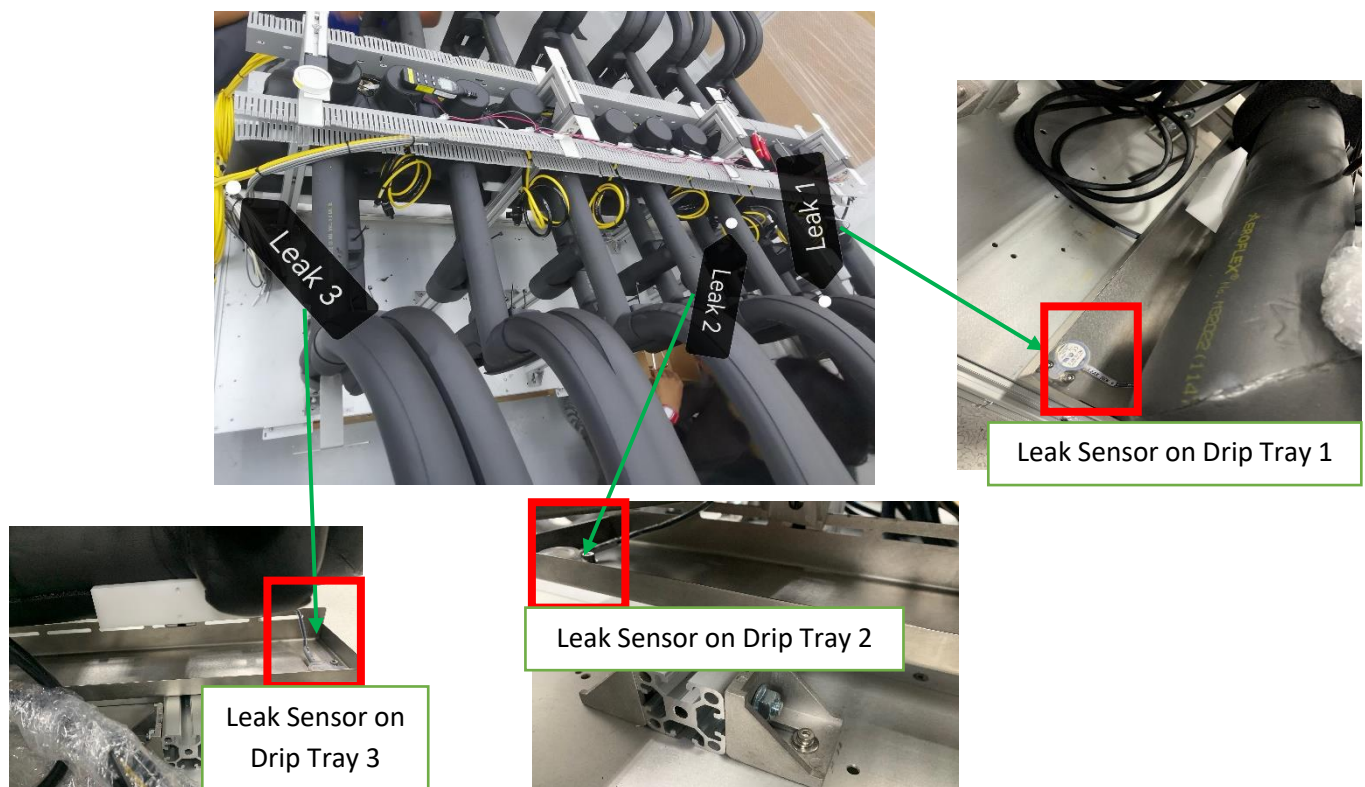


Figure 3-25 Location of Drip Sensor

3.3.5.4 Drip Tray

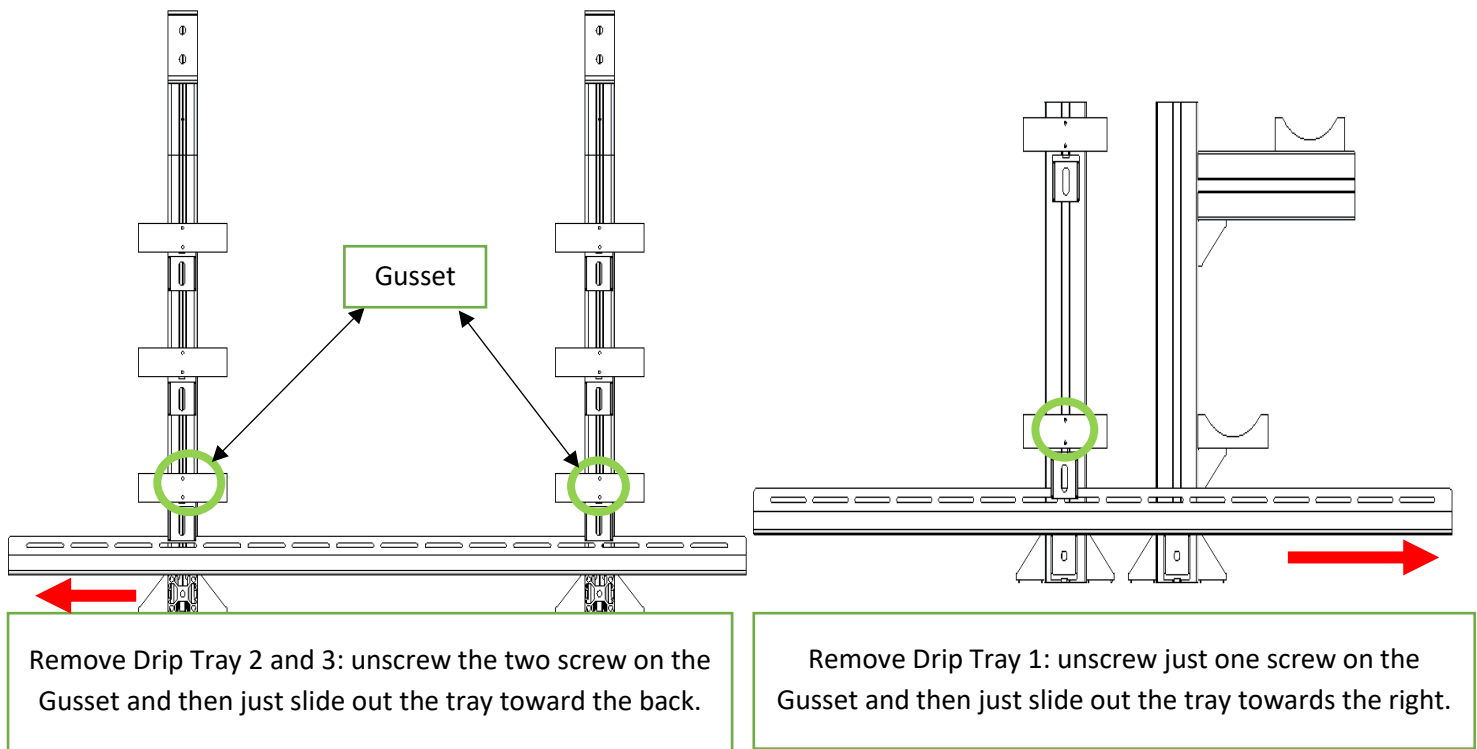


Figure 3-26

3.3.6 EE Panel

EE Panel contain the primary controls for electrical power distribution.

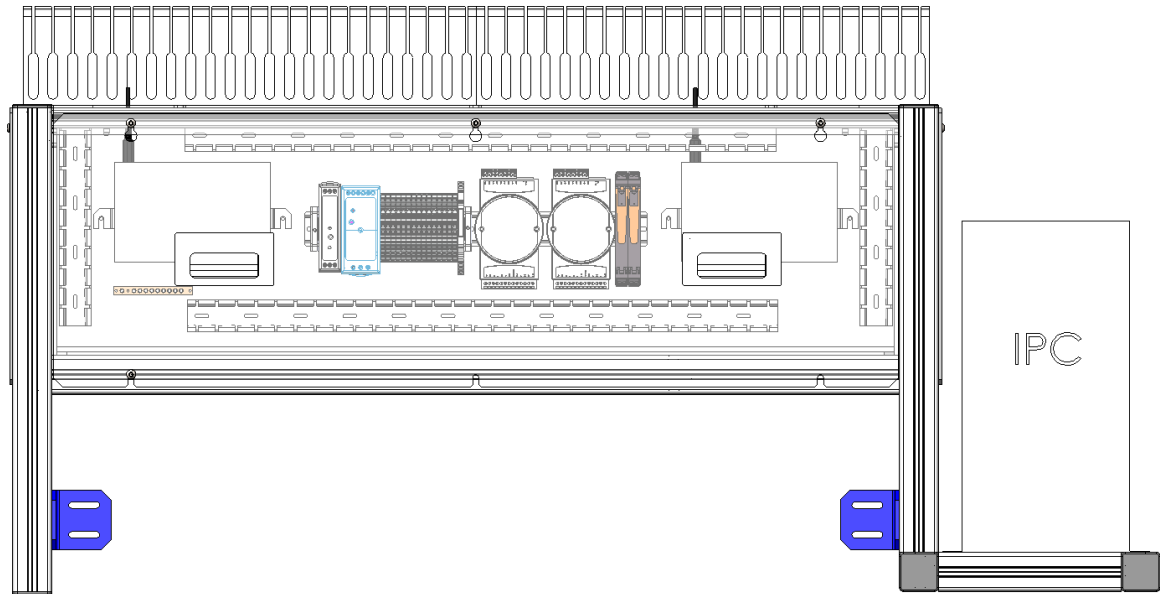


Figure 3-27 Layout of EE Panel

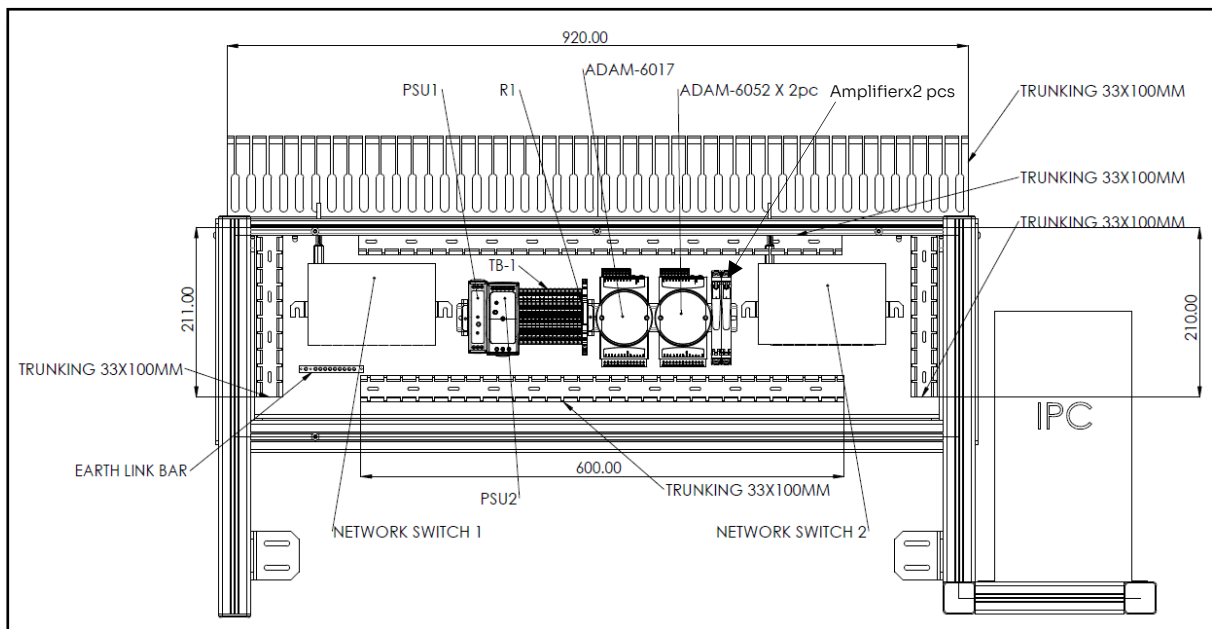
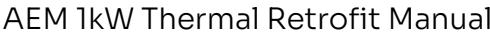


Figure 3-28 EE Panel Schematic



Figure 3-29 EE Panel

[illegible]



4. Installation



4.1 Installation

4.1.1 Shipping Containers

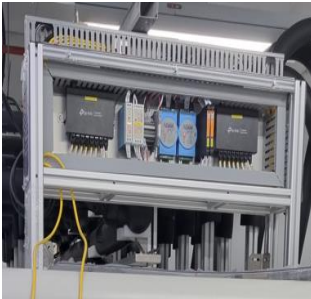
Containers Dimensions:

Crate no.	DESCRIPTION	DIMENSION (LxWxH)(mm)	QTY	WEIGHT (KG)	CRATE WEIGHT (KG)
Crate 1	RETROFIT PIPING	2360x2050x1400	1	250	150
Crate 2	CONTROLLER BOX ASSEMBLY	1350x1220x780	12	12	100
Crate 3	54X71 RESIDENT MAIN ASSEMBLY THERMAL HEAD	1350x1220x780	12	5	100
Crate 4	MAIN FACILITY COMPARTMENT ASSEMBLY	1350x900x900	1	60	100
Crate 5	Chiller	1600x840x1980	1	550	50
Crate 6	Glycol and tube	600x600x400	4	20	20

Table 4-1 Packing List

Crate No.	Part/Assembly Number	Description	QTY	Crate Pictures	Content Pictures
Crate 1	B7152-AHF00	RETROFIT PIPING	1		
Crate 2	B5461-AHA02	CONTROLLER BOX ASSEMBLY	12		



Crate No.	Part/Assembly Number	Description	QTY	Crate Pictures	Content Pictures
Crate 3	B7144-AHB00	54X71 RESIDENT MAIN ASSEMBLY THERMAL HEAD	12		
Crate 4	1. N3838-AHA00 2. ECP32774 3. ECP32088	1.MAIN FACILITY COMPARTM ENT ASSEMBLY 2. TOUCH MONITOR 24 INCH 3. IPC for Dual CTU	1		
Crate 5	-	Chiller	1		

Crate No.	Part/Assembly Number	Description	QTY	Crate Pictures	Content Pictures
Crate 6	1. WAC61620 2. PTB001 3. PSV34067	1. Ethylene glyco water (EGW) 2. 4mm Pneumatic tubing 3. 8mm Pneumatic tubing	3	NA	

1. Check the condition of AEM 1kW Thermal Retrofit System for any major damages and ensure all the components are in good condition.
2. Connection layout: Check facility supply and connect specified quick connect to facility.
3. Make sure the connections of rear panel are correct.

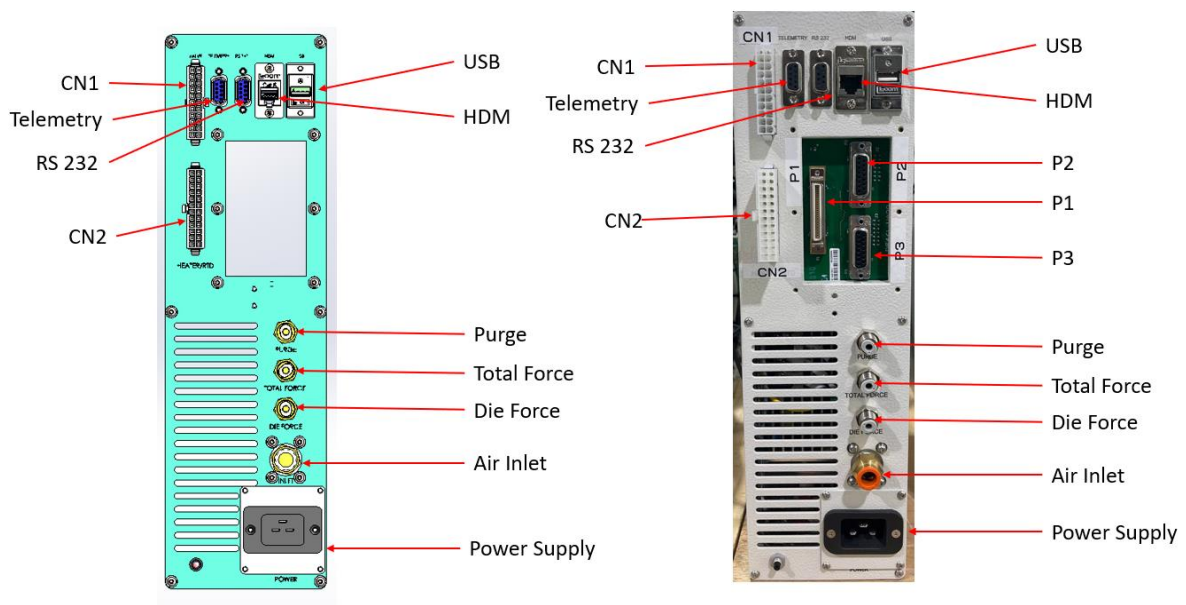
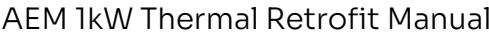


Figure 4-1 Connection and example after connection

[illegible]



5. User Interface

5.1 Network Topology

Note: If GUI is required, LAN Cable must stay connected with a powered-ON PC/Mac/Router before power on multiple stations.

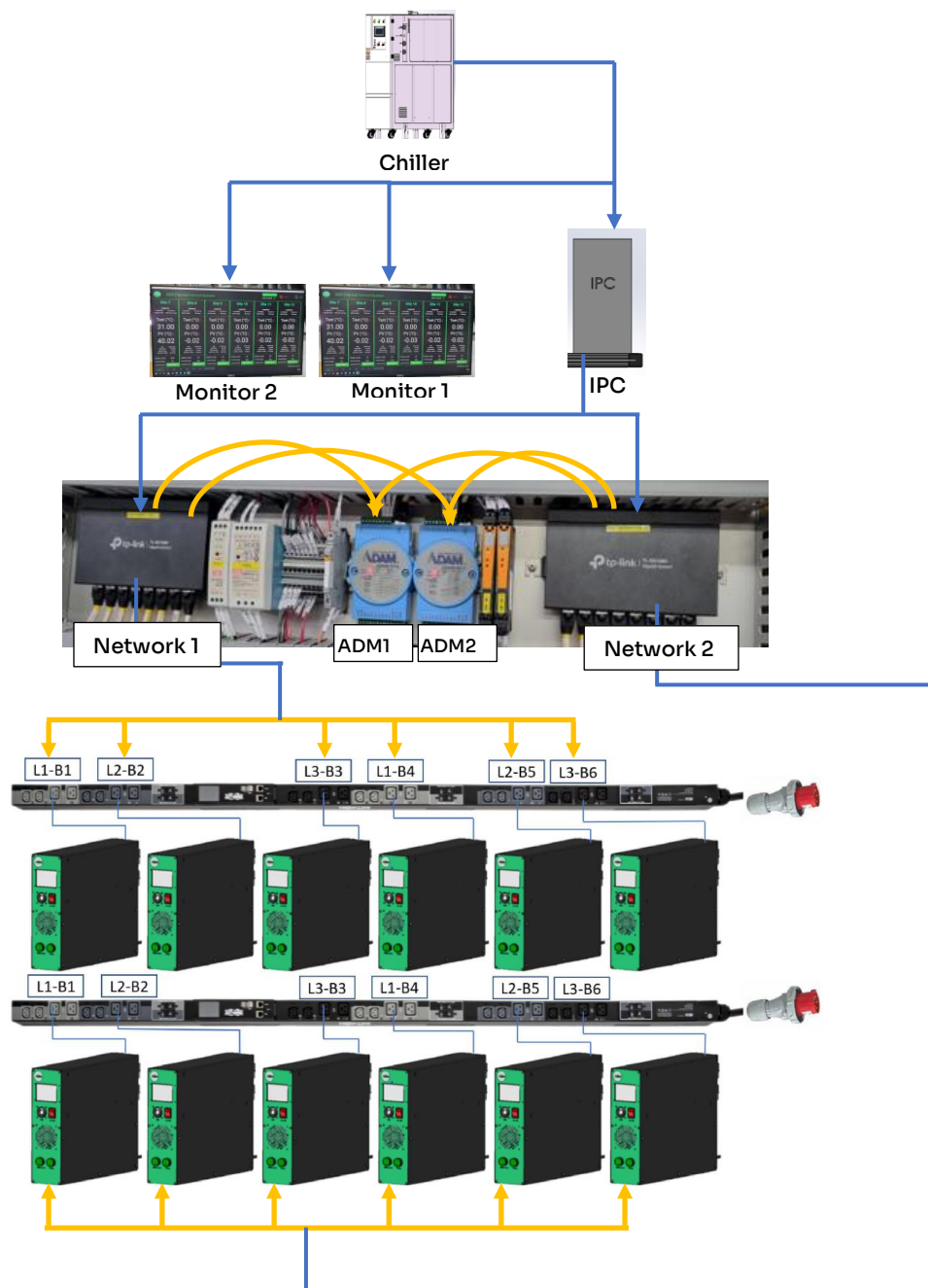


Figure 5-1 Flowchart of cable using ATC

5.1.1 ATC LCD

ATC LCD will display all the actual and current force value for each respective controllers.



Figure 5-2 ATV LCD screen

Table 5-1 ATV LCD icon function and description

Index	Function	Definitions
1	Ready	Ready state is ON when there is no Error.
2	Die Force (Actual) Die Force (Desired)	Current force on Die Actual Die force set by User
3	Cylinder Force (Actual) Cylinder Force (Desired)	Current force on Cylinder Actual Cylinder force set by User
4	Heatlet Zone Info	Set Temperature: Desire Temperature set by User (°C) Process Value: Thermal Head Temperature (°C) Mode: Following are the system mode: • Heater Following mode (HTF) • DUT Following Mode (DTF) Heater Power: Power applied on Thermal Head Liquid Valve: Value of liquid flowrate from chiller
5	Alarm Description	Display errors occur in ATC system



5.2 User Interface

Start the application by opening:

Go to Download and click on the Retrofit folder.

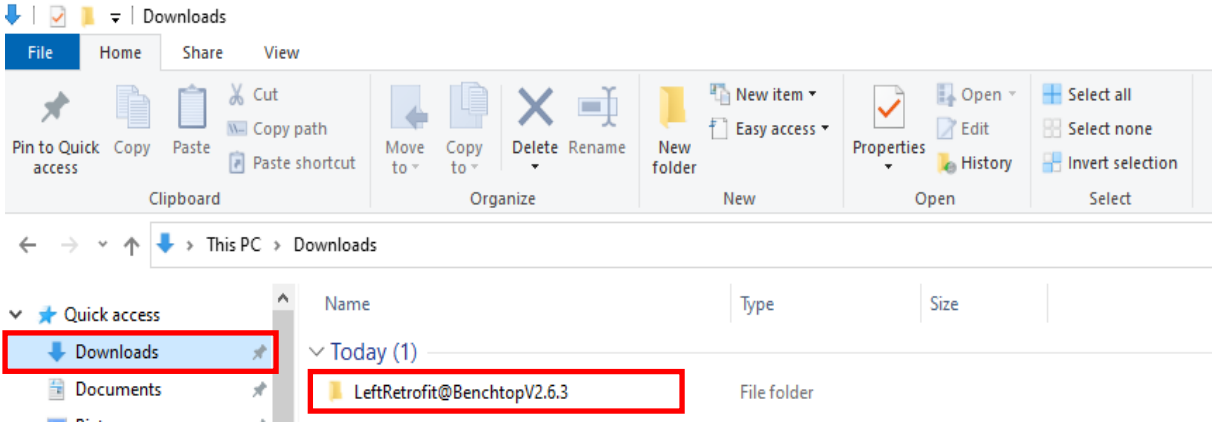


Figure 5-3 Launched Thermal Control System

Start the application by opening:

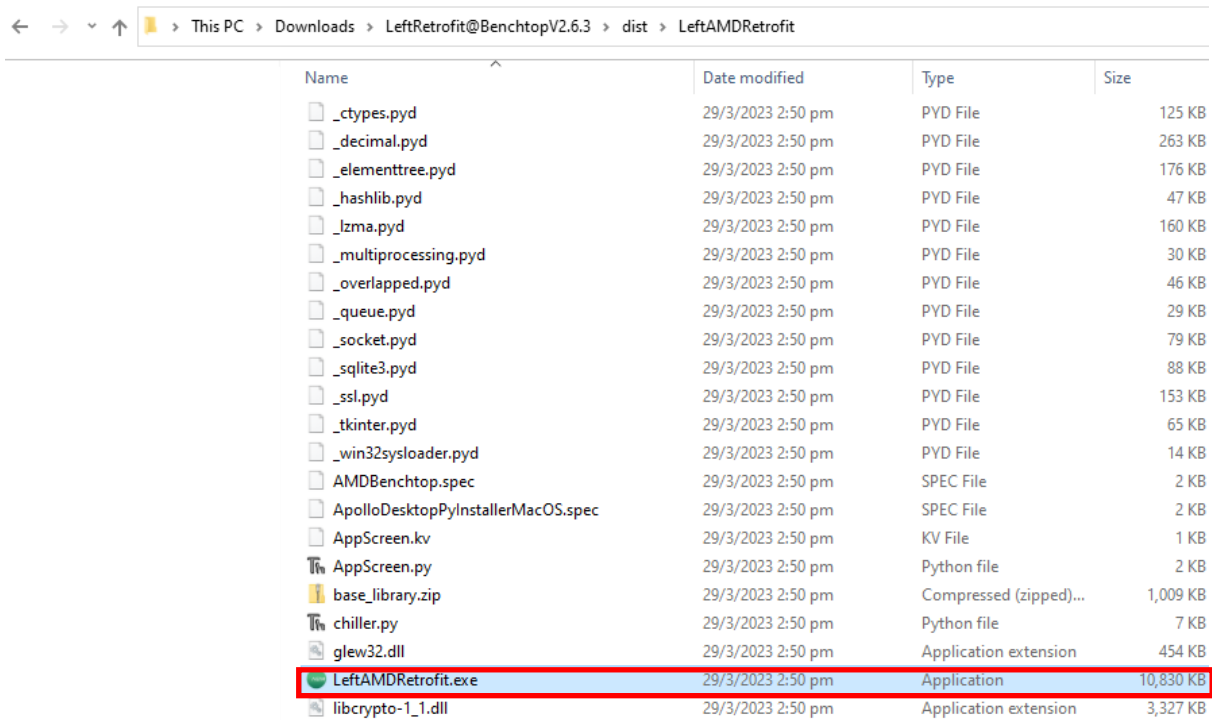


Figure 5-4 “LeftAMDRetrofit.exe” in the folder [LeftRetrofit@BenchtopV2.6.3](#)

5.3 GUI for AEM 1kW Thermal Retrofit System

5.3.1 Common Toolbars

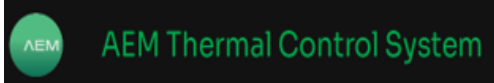
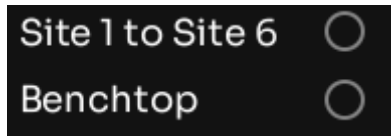
Top Toolbar

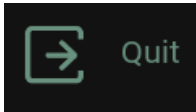
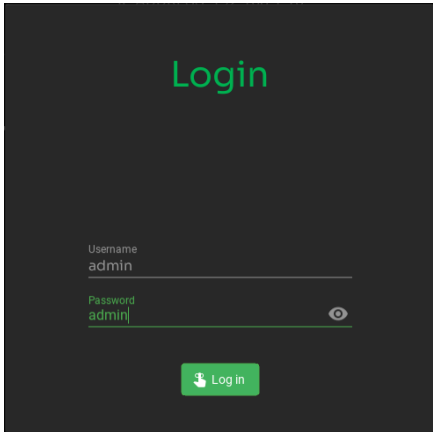


Figure 5-5

The toolbar above is the topmost label that displays peripheral connection statuses. SW automatically connects to peripherals during the launch. If a peripheral is not connected after the launch, user may retry by clicking the 'Benchtop' checkbox button.

Table 5-2

GUI Name / Illustration	Description
TITLE DISPLAY 	AEM Thermal Control System is a static title assigned in the topmost label which will be in every screen of the UI. It displays the title of the project.
MODE TAB  	Mode in here allows user to switch from Operator Mode to Engineering mode. All the controls of the parameter will be enabled only in Engineering mode. Operator mode only for view the response of the systems from the dashboard screen but not to change the variables. Default Mode – Operator Mode.
PROJECT SWITCH TAB 	Enable user to use single GUI for multiple projects. User can select the number of stations need to be connected by selecting the project name. Site 1 to Site 6 – 12 Station System Benchtop – 3 Station System
E-STOP TAB 	Emergency Stop Button tend to cut off heater powers across all zones, stop running scripts including test script and FPGA scripts. All the process will be cut-off immediately when this emergency button is clicked.

GUI Name / Illustration	Description
QUIT TAB 	Quit Button allows users to exit the GUI but all running processes will be continued. User can reactivate the GUI and continue the testing. Useful if the GUI hangs.
POP-UP WINDOW – LOGIN 	This window will only be visible when user click the Operator Mode tab change to Engineer Mode.

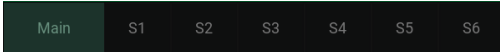
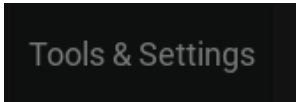
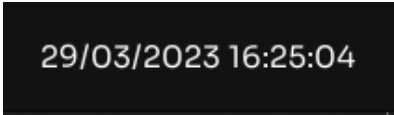
Bottom Toolbar



Figure 5-6

The toolbar above is a common toolbar which is located at the bottom of the GUI window. In this window, user can switch from main screen to individual station screen to control the variables. Current date and time will be always displayed at the right bottom corner of the GUI.

Table 5-3

GUI Name / Illustration	Description
SCREEN TAB 	This tab allows operators to switch screen from Main Screen to Individual Station Screen. 3 individual screens are presented in Benchtop GUI. S1 = Station 1 S2 = Station 2 S3 = Station 3 S4 = Station 4 S5 = Station 5 S6 = Station 6
TOOLS & SETTINGS TAB 	This tab allows engineers to do settings for the thermal head. Not for operator usage.
DATE & TIME DISPLAY 	Live date and time will be displayed here.



5.3.2 Main Screen

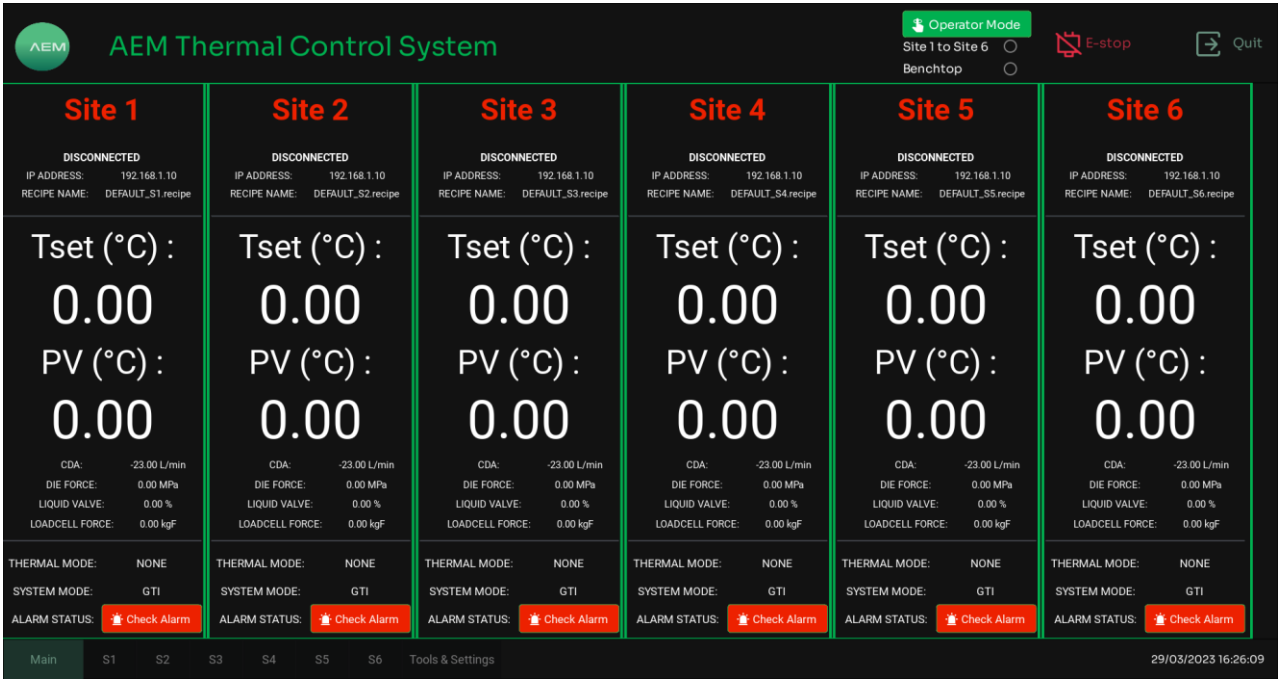


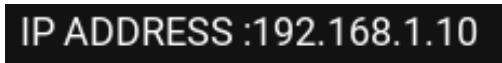
Figure 5-7

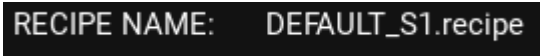
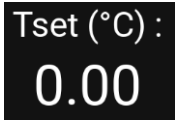
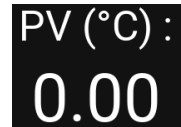
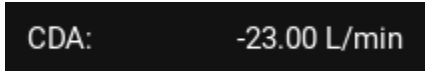
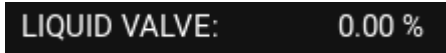
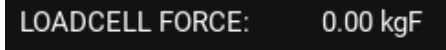
The Main Screen of Benchtop GUI displays the 3 station systems and its responses. When a user clicks the Benchtop check button at top tool bar, all 3 stations will be automatically connected and visible in connection display.

Table 5-4

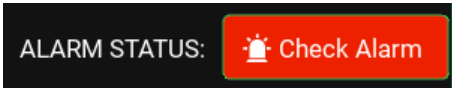
GUI Name / Illustration	Description
COLUMN	Each column consists of individual station details.



GUI Name / Illustration	Description												
SITE ID 	Topmost label in the column that indicates the Site ID. Orders of Site IDs are static. Site ID will turn into GREEN highlight when there is no alarm or GUI connects with the relevant site. RED highlight indicates presence of alarm or disconnection from the ATC system.												
CONNECTED / DISCONNECTED 	Connected / Disconnected displays the communication between SW and ATC in the system. Connection will be established automatically when physical connections are assigned. Users can't make any changes here, only for display. Users can manually change the state of this UI button in individual screen.												
IP ADDRESS 	IP address displayed here is based on user's setting in Tools & Settings (IP Config screen). Default IP addresses for Benchtop are : Site 1: 192.168.1.10 Site 2: 192.168.1.20 Site 3: 192.168.1.30 Default IP addresses for Retrofit are : <table><tr><td>Site 1: 192.168.1.21</td><td>Site 7: 192.168.1.27</td></tr><tr><td>Site 2: 192.168.1.22</td><td>Site 8: 192.168.1.28</td></tr><tr><td>Site 3: 192.168.1.23</td><td>Site 9: 192.168.1.29</td></tr><tr><td>Site 4: 192.168.1.24</td><td>Site 10: 192.168.1.30</td></tr><tr><td>Site 5: 192.168.1.25</td><td>Site 11: 192.168.1.31</td></tr><tr><td>Site 6: 192.168.1.26</td><td>Site 12: 192.168.1.32</td></tr></table>	Site 1: 192.168.1.21	Site 7: 192.168.1.27	Site 2: 192.168.1.22	Site 8: 192.168.1.28	Site 3: 192.168.1.23	Site 9: 192.168.1.29	Site 4: 192.168.1.24	Site 10: 192.168.1.30	Site 5: 192.168.1.25	Site 11: 192.168.1.31	Site 6: 192.168.1.26	Site 12: 192.168.1.32
Site 1: 192.168.1.21	Site 7: 192.168.1.27												
Site 2: 192.168.1.22	Site 8: 192.168.1.28												
Site 3: 192.168.1.23	Site 9: 192.168.1.29												
Site 4: 192.168.1.24	Site 10: 192.168.1.30												
Site 5: 192.168.1.25	Site 11: 192.168.1.31												
Site 6: 192.168.1.26	Site 12: 192.168.1.32												

GUI Name / Illustration	Description
RECIPE NAME 	Recipe set in the individual recipe screen will be displayed in this label. Users can't make any changes in this main screen but can ensure that the set recipe is the preferable one.
SET TEMPERATURE 	Tset is the temperature set by user in recipe screen or TJset value receiving in GPUTJ mode. This will be in large scale to ensure higher visibility to the users. Degree Celsius (°C) is the unit measured in for the set temperature.
PROCESSED VALUE 	PV is the process value reads from the heater which responds to the set temperature setted by user. GUI will receive this process value through ATC system which is measured in Degree Celsius (°C). PV in GUI should be correlated to the PV in ATC controller system.
COMPRESSED DRY AIR 	Indicates the actual incoming dry air for each individual station which measured in Litre per minute.
DIE FORCE 	Indicate the pressure value applied on the Thermal Head
LIQUID VALVE 	Indicate the percentage of liquid flowrate from chiller.
LOADCELL FORCE 	Loadcell Force is measuring weight as a change in pressure of the internal filling fluid
SYSTEM MODE 	There are 2 system modes for the benchtop / retrofit thermal system, GTI mode and AEM mode. Mode will be displayed in this label based on the system operation. Default mode: GTI.



GUI Name / Illustration	Description
	Users can't change the mode here (dashboard screen), only changeable in individual site screen.
THERMAL MODE 	Under GTI mode, there are 3 thermal modes; Tcase, TDiode and SBTSl. Under AEM mode, there are 2 thermal modes; HTF and DTF. Primary modes will be received from the GTI mode. RTD temperature control mode (HTF and DTF) works in AEM mode only.
ALARM STATUS 	Alarm section lists out the alarm alerts user about 'Force NG' by showing RED indicator. It occurs when the set force for engage is low. Over temperature indicates the read temperature exceed the cut-off temperature. HV OFF will be RED when high voltage is inactive. GREEN and system healthy indicate there's no alarm activated, and system is ready to initiate the process.

5.3.3 Common Toolbars On Individual Station Screen

a. TOP TOOLBAR



Figure 5-8 Top label of individual station screen

The toolbar above is located below the top toolbar of the Main Screen. It is static for all the individual station screens.

Table 5-5

GUI Name / Illustration	Description
STATION ID 	First label that indicates the station ID. User can't change the station ID, only for display.
CONNECTED / DISCONNECTED CONTROL 	Connected / disconnected button here is controllable and user can manually connect to the ATC system by clicking this button. GREEN BUTTON indicates CONNECTED RED BUTTON indicates DISCONNECTED
IP ADDRESS SETUP 	User can setup their preferred IP address here and press the connection button to establish connection to the station.

b. LEFT TOOLBAR

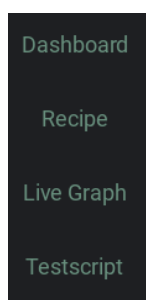


Figure 5-9 Menu of Left Toolbar

The toolbar above is located at the top left of the individual station screens. It is static for all individual station screens. This is the directory that will lead to dashboard, recipe, live graph, and test script screens.

5.3.4 Individual Station – Dashboard Screen

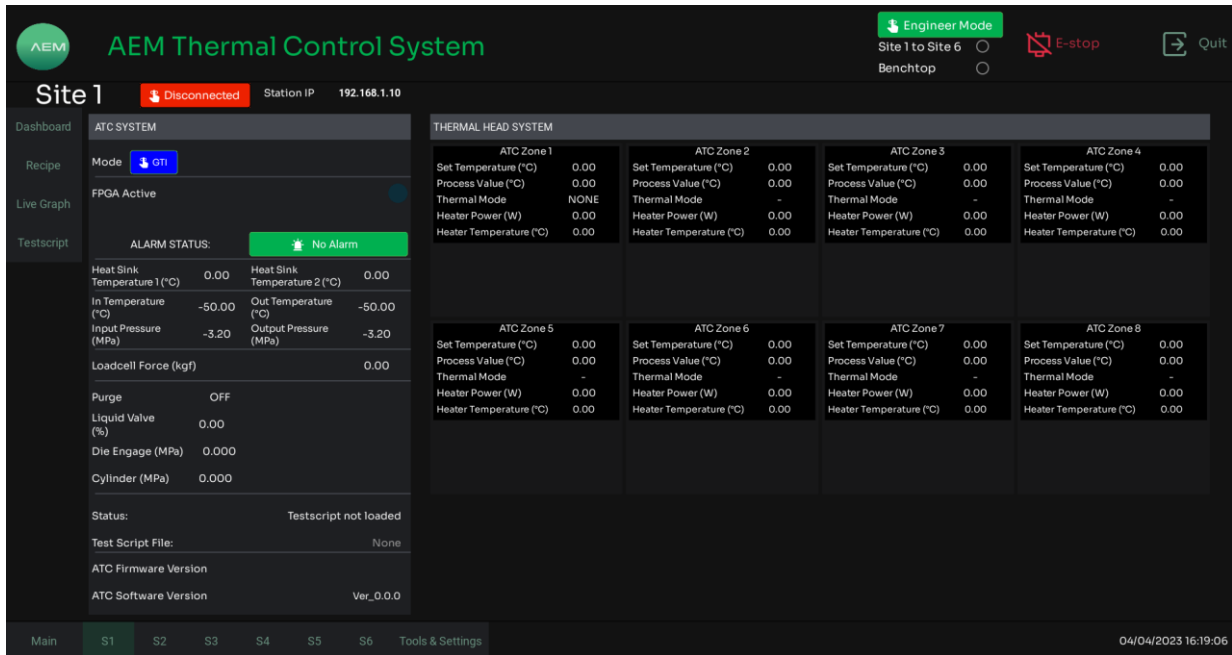
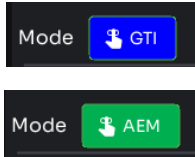
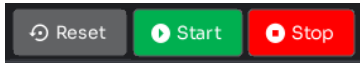


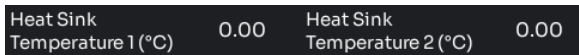
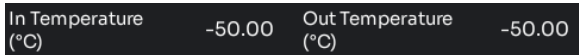
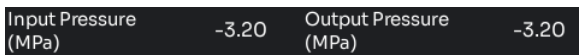
Figure 5-10

Individual Station Dashboard Screen is designed to be the main display and control screen for the users. User performs essential controls and read on this screen. It allows users to read ATC parameters and statuses.

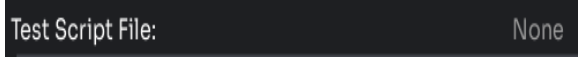
Table 5-6

GUI Name / Illustration	Description
SYSTEM MODE CONTROL 	There are two modes which are GTI and AEM. GTI mode where command will be received from Master Controller and user is not allowed to take control of the parameters. In AEM mode, user is allowed to set the parameters and read the response by their preference. Default: GTI mode.
SYSTEM OPERATION CONTROL 	Buttons in system section allow users to reset, start and stop the running script. Green indication shows its active. Activation and deactivation of these button will be reflected in the Main Screen System Status display.

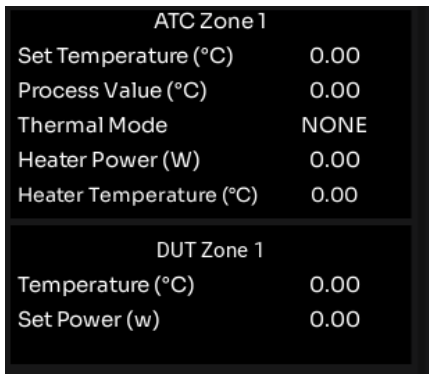


GUI Name / Illustration	Description
FPGA ACTIVE 	FPGA Active section enables users to identify FPGA status via GREEN light indication when its active.
ALARMS 	Alarms section alerts users in Red when any abnormalities occur. GREEN and alarm status in 'NO ALARM' indicates system is healthy and ready to initiate the process. 8 type of alarm code will be presented in the check alarm pop up window. There are: • FPGA High Voltage is Turn Off condition • Temperature is too high • Die Force is not engaged • Low CDA flow Rate • Ethylene Glycol Leak Detected • Timeouts occur and switch to IDLE mode • Loadcell Force <30kgf and heatsink temperature <25°C • Chiller Alarm triggered.
HEATSINK TEMPERATURE 	Cold plate temperature displays the internal temperature of the heater.
TEMPERATURE 	Inlet and Outlet temperature of chiller fluid in Degree Celsius.
INPUT & OUTPUT PRESSURE 	Input and Output of chiller fluid in MPa
LOADCELL FORCE 	Actual pressure applied on DUT measured in kgf



GUI Name / Illustration	Description
PURGE 	User can enable/disable dry air purge via UI by clicking ON/OFF buttons in this section. GREEN light indicates the button is activated.
LIQUID VALVE 	User can control the liquid valve flow rate at this section. Auto mode allows the flow rate to be function in dynamic way. Manual allows users to set specific flow rate and maintain it throughout the process. Users can set the flow rate in manual mode in recipe screen.
DIE ENGAGE 	DIE can be attached and detached via UI by clicking Engage/Disengage button in this section. It also displays the force of engaging in MPa for Diaphragm Engage. GREEN light indicates the button is activated.
CYLINDER 	Display measured force of cylinder engaged controlled by handler.
TEST SCRIPT STATUS 	ATC station connection status can be seen in this section. Loaded: Test script opened & loaded in 'Test Script' screen, yet script not executed Running: Test script executed, system operating Stop: Test script execution stopped Complete: Execution complete Disconnect: Execution interrupt by a stop
TEST SCRIPT FILE 	Reflects the file name opened & loaded from the 'Test Script' screen.



GUI Name / Illustration	Description
VERSION DISPLAY 	Firmware Version: Shows Micro Controller Firmware Version of the controller box. Software Version: Micro Controller Software version. User should ensure all controllers have same firmware and software version.
THERMAL HEAD 	ATC ZONE 1-8 Displays the measurement of Measured Temperature (°C), Power (W), Mode and Set Temperature (°C) from ATC for 1 zone in Thermal Head. The preferred values can be set in 'Test script' screen. These data interpret as line graph in real time in 'Live Graph' screen. Only 1 zone is enabled in benchtop system. ATC Zone 1 = Heater Channel 1 = Enabled ATC Zone 2 = Heater Channel 2 = Disabled ATC Zone 3 = Heater Channel 3 = Disabled ATC Zone 4 = Heater Channel 4 = Disabled ATC Zone 5 = Heater Channel 5 = Disabled ATC Zone 6 = Heater Channel 6 = Disabled ATC Zone 7 = Heater Channel 7 = Disabled ATC Zone 8 = Heater Channel 8 = Disabled DUT ZONE 1-8



GUI Name / Illustration	Description
	Displays the measurement of Temperature (°C) & Set Power (W) from read from DUT for 1 zone. The preferred values set in 'TTV Recipe' screen. These data interpret as line graph in real time in 'TTV Graph' screen. DUT Zone 1 = Heater Channel 1 = Enabled DUT Zone 2 = Heater Channel 2 = Disabled DUT Zone 3 = Heater Channel 3 = Disabled DUT Zone 4 = Heater Channel 4 = Disabled DUT Zone 5 = Heater Channel 5 = Disabled DUT Zone 6 = Heater Channel 6 = Disabled DUT Zone 7 = Heater Channel 7 = Disabled DUT Zone 8 = Heater Channel 8 = Disabled

5.3.5 Individual Station – Recipe Screen

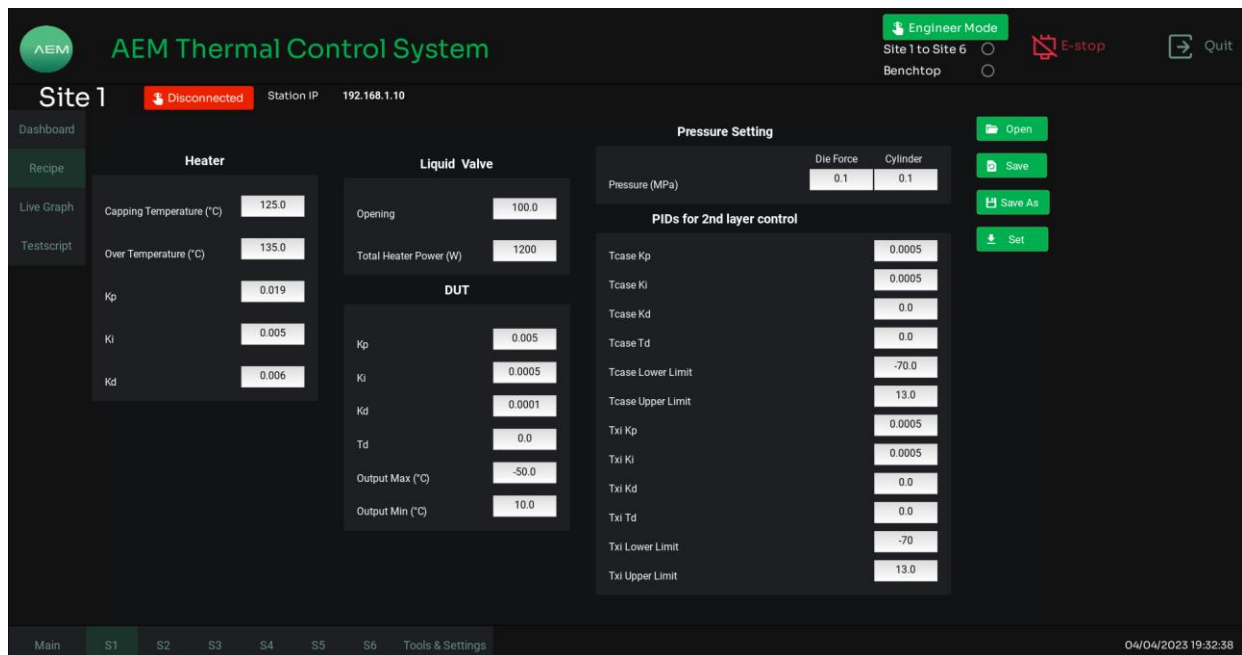
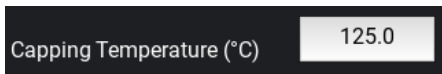
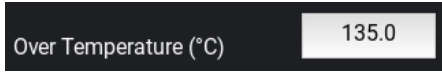


Figure 5-11

Individual Station Recipe Screen is the controller screen for the system. It contains 6 sections which in specific are: Heater, Liquid Valve, DUT, Pressure Setting, PIDs for 2nd Layer Control and Control Buttons.

Heater

Table 5-7

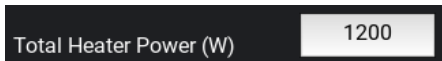
GUI Name / Illustration	Description
CAPPING TEMPERATURE 	Set the maximum limit heater temperature (°C) can reach. Default value: 125.0
OVER TEMPERATURE 	Set the limit temperature (°C) to trigger thermal cut-off (TCO) Default value: 135.0
Kp 	PID Parameter (Proportional) for the Heater Heater PID value 0-1 Proportional term / Error force Default value: 0.019



GUI Name / Illustration	Description
Ki 	PID Parameter (Integral) for the Heater Sum of error over time Default value: 0.005
Kd 	PID Parameter (Derivative) for the Heater Based on rate of change of error Cause the heater to respond faster as error increases. Default value: 0.006

Liquid Valve

Table 5-8

GUI Name / Illustration	Description
OPENING 	Set Range: 0.0-1.0 Adjust the liquid valve opening based on this percentage. User manually assign the opening diameter of liquid valve. Inefficient in liquid valve 'Auto' mode
TOTAL HEATER POWER 	Automatic Mode Reading from FPGA exceed the set value of Total Heater Power = Heater deactivates Reading from FPGA lower than set value in this section, Total Heater Power (W) = Heater activates back



DUT

Table 5-9

GUI Name / Illustration	Description
Kp 	PID Parameter (Proportional) Heater PID value 0-1 Proportional term / Error force
Ki 	PID Parameter (Integral) Sum of error over time Least value
Kd 	PID Parameter (Derivative) Based on rate of change of error Cause the heater to response faster as error increases
Td 	Delta Time
OUTPUT MAX 	Set maximum DUT output temperature (°C)
OUTPUT MIN 	Set minimum DUT output temperature (°C)



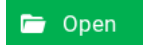
Pressure Setting

Table 5-10

GUI Name / Illustration	Description
PRESSURE 	Set the initial pressure (MPa) of diaphragm and cylinder. Die force is the centre force of diaphragm.

Control Buttons

Table 5-11

GUI Name / Illustration	Description
OPEN 	Open recipe from the folder. User can load default recipe from accessing into this button, select the preferred file and load it.
SAVE 	Save on-screen values on current file selected.
SAVE AS 	Save on-screen values to a different file as a new file.
SET 	Set the values referring to the opened file. Set the mode HTF/DTF.

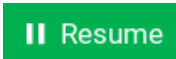
5.3.6 Individual Station – Live Graph Screen

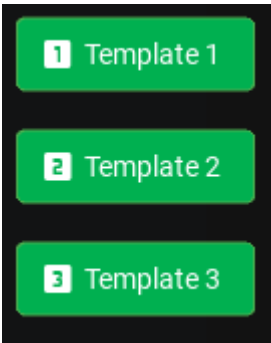
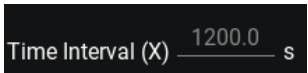
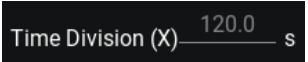
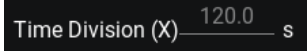
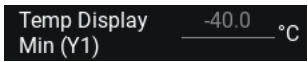
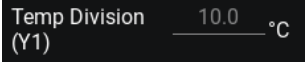
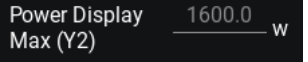
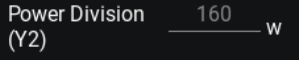


Figure 5-12

Individual Station Live Graph Screen displays real time readings in lines. Left y-axis indicates Temperature (°C) and right y-axis displays Power (W) versus x-axis that represents real-time intervals. Intervals and divisions can be modified. Legends on the graph indicates the parameters under measure with the system.

Table 5-12

GUI Name / Illustration	Description
PAUSE/RESUME  	Click 'Pause' to freeze and 'Resume' to unfreeze the current view of the graph.
SAVE 	Click 'Save' button, goto your preferred directory, input file name & save.
CLEAR 	Click 'Clear' button to remove the graph plotting on the screen.

GUI Name / Illustration	Description
TEMPLATE 	User can obtain pre set value for graph layout.
TIME INTERVAL (X) 	Users can edit & set preferred time interval at this section.
TIME DIVISION (X) 	Users can edit & set preferred time division at this section.
TEMP DISPLAY MAX (Y1) 	Users can edit & set maximum temperature to be displayed at y1-axis at this section. Y1: Left axis
TEMP DISPLAY MIN (Y1) 	Users can edit & set minimum temperature to be displayed at y1-axis at this section. Y1: Left axis
TEMP DIVISION (Y1) 	Users can edit & set temperature division at y1-axis at this section. Y1: Left axis
POWER DISPLAY MAX (Y2) 	Users can edit & set maximum power to be displayed at y2-axis at this section. Y2: Right axis
POWER DISPLAY MIN (Y2) 	Users can edit & set minimum power to be displayed at y2-axis at this section. Y2: Right axis
POWER DIVISION (Y2) 	Users can edit & set power division at y2-axis at this section. Y2: Right axis

GUI Name / Illustration	Description																																																																																																				
CHECK BUTTON <table><tr><th>ID</th><th>PV</th><th>TH</th><th>TD</th><th>TSET</th><th>PH</th><th>PD</th><th>PSET</th><th>TX</th><th>LQV</th></tr><tr><td>All</td><td>☒</td><td>☒</td><td>☐</td><td>☒</td><td>☒</td><td>☐</td><td>☐</td><td>☐</td><td>☒</td></tr><tr><td>1</td><td>☒</td><td>☒</td><td>☐</td><td>☒</td><td>☒</td><td>☐</td><td>☐</td><td>☐</td><td>☒</td></tr><tr><td>2</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☐</td><td>☒</td></tr><tr><td>3</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>4</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>5</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>6</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>7</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>8</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr></table>	ID	PV	TH	TD	TSET	PH	PD	PSET	TX	LQV	All	☒	☒	☐	☒	☒	☐	☐	☐	☒	1	☒	☒	☐	☒	☒	☐	☐	☐	☒	2	☒	☒	☒	☒	☒	☒	☒	☐	☒	3	☒	☒	☒	☒	☒	☒	☒	☒	☒	4	☒	☒	☒	☒	☒	☒	☒	☒	☒	5	☒	☒	☒	☒	☒	☒	☒	☒	☒	6	☒	☒	☒	☒	☒	☒	☒	☒	☒	7	☒	☒	☒	☒	☒	☒	☒	☒	☒	8	☒	☒	☒	☒	☒	☒	☒	☒	☒	Check and uncheck the button depending on the preference of users on graph line measurement. PV = Process Value TD = DUT Temperature TSET = Set Temperature PH = Heater Power PD = DUT Power PSET = Set Power TX = Heatsink LQV = Flow rate of liquid valve
ID	PV	TH	TD	TSET	PH	PD	PSET	TX	LQV																																																																																												
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5.3.7 Individual Station – Test Script Screen

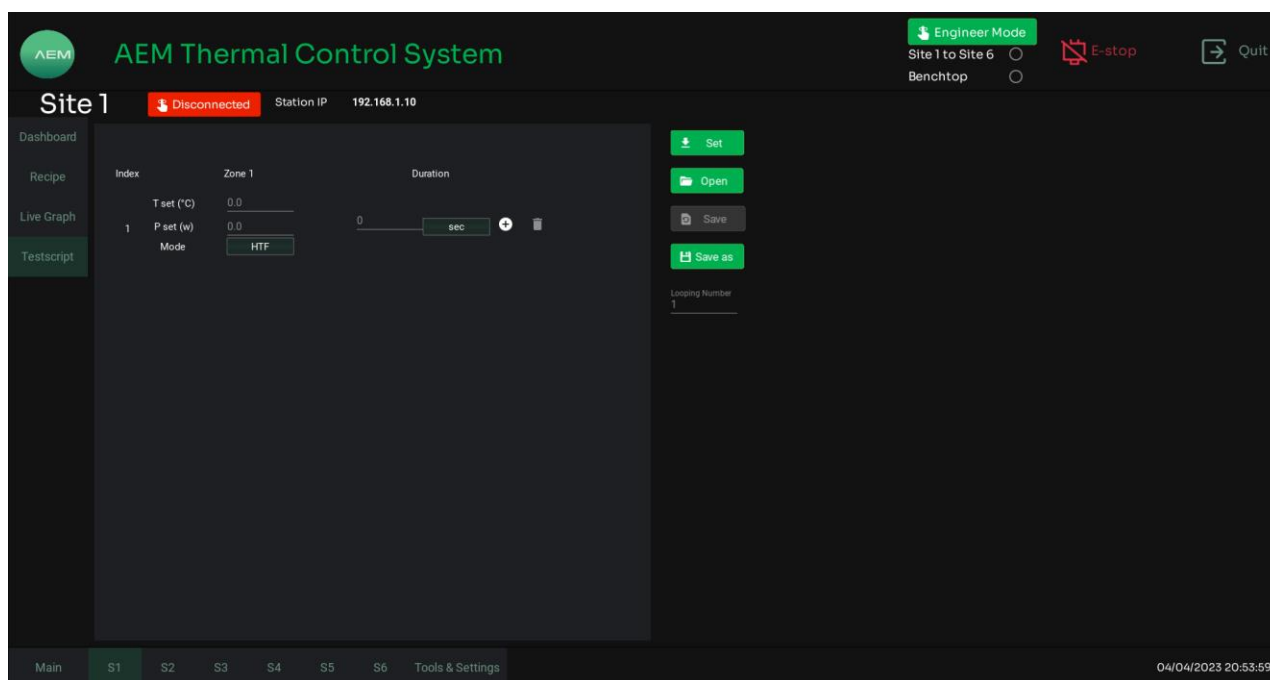
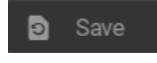
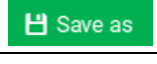
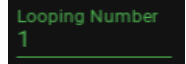


Figure 5-13

Individual Station Test Script Screen allows user to read, edit, save existing test script, or create new test script. Users can open the test script they desired and set at this screen. In 'Test Script Screen' user can set Tset and Pset values.

Table 5-13

GUI Name / Illustration	Description
TSET 	User's preferred temperature for thermal head zones can be set here. For benchtop station only 1 zone can be set the temperature and others are disabled.
PSET 	User's preferred power for the DUT zones can be set here.
DURATION 	Duration for each cycle for the set temperature and pressure.
MODE 	HTF & DTF mode can be set at here as well which aligned with the recipe screen mode.
ADD 	Add another row of data.
DELETE 	Remove entire row of data.
SET 	Load the values of Tset & Pset to the thermal head via ATC. Main screen 'Test Script Status' shows 'Loaded' if set successfully.
OPEN 	Open new test script from folder. Load back the previous / default test script.
SAVE 	Set the value to the thermal head via ATC.
SAVE AS 	Save on-screen values to a different file.
LOOPING 	Executing the same set of instructions a given number of times or until a specified result is obtained.

5.3.8 Tools & Settings - Chiller

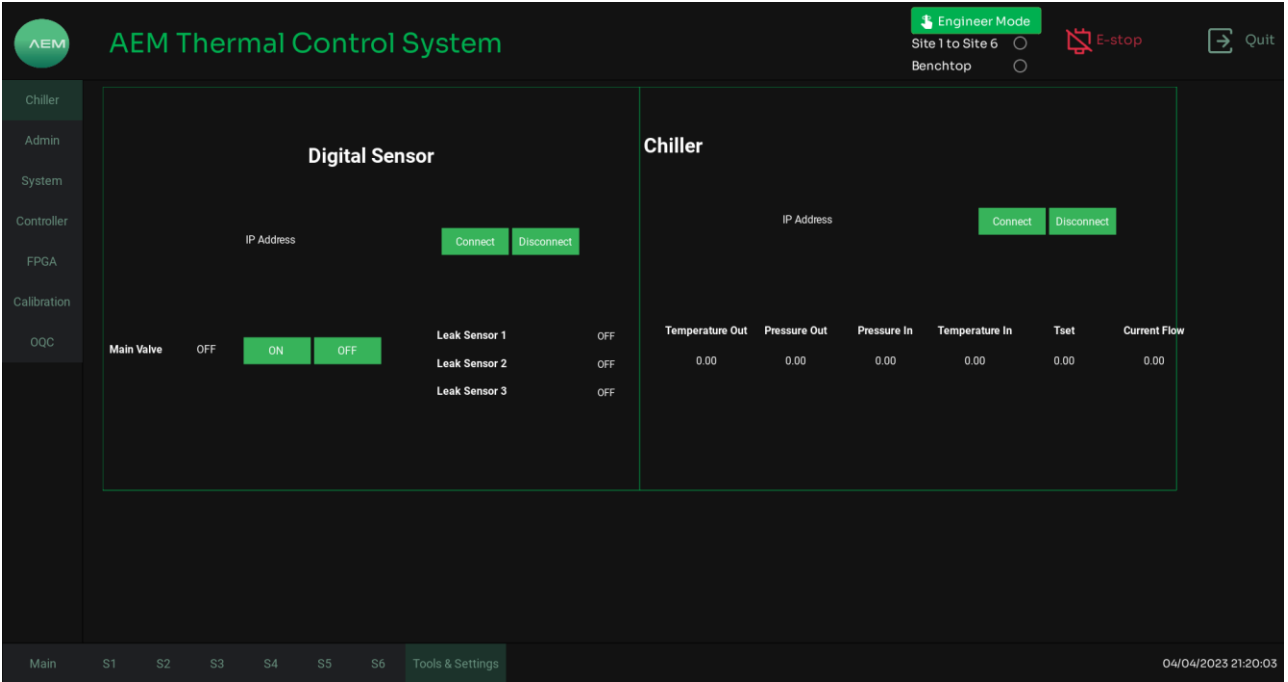


Figure 5-14

Table 5-14 DIGITAL SENSOR

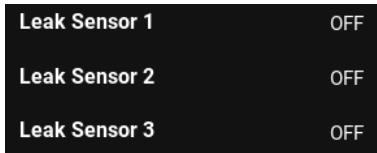
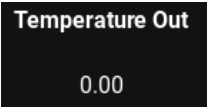
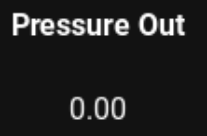
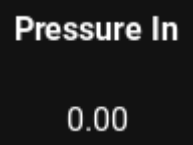
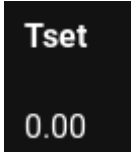
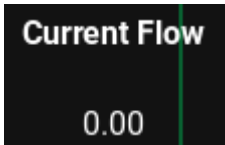
GUI Name / Illustration	Description
IP ADDRESS 	User can connect to the IP Address of leak sensor by clicking “Connect button while manually disconnect the leak sensor by clicking the button.
MAIN VALVE 	User can Turn ON or OFF chiller main valve by clicking to this button.
LEAK SENSOR 	This bar display all the sensor status of leak sensor. If the status shows ON ,leak is detected while Off means no leak is detected.



Table 5-15 CHILLER

GUI Name / Illustration	Description
IP ADDRESS 	User can connect to the IP Address of chiller by clicking “Connect button while manually disconnect the chiller by clicking the button.
TEMPERATURE OUT 	User can monitor actual temperature out from chiller.
PRESSURE OUT 	User can monitor actual pressure out from chiller.
PRESSURE IN 	User can monitor actual pressure in from chiller.
TEMPERATURE IN 	User can monitor actual temperature in from chiller.
TEMPERATURE SET 	User can monitor setting temperature of chiller.
CURRENT FLOW 	User can monitor current flowing from chiller

5.3.9 Tools & Settings – Admin Screen

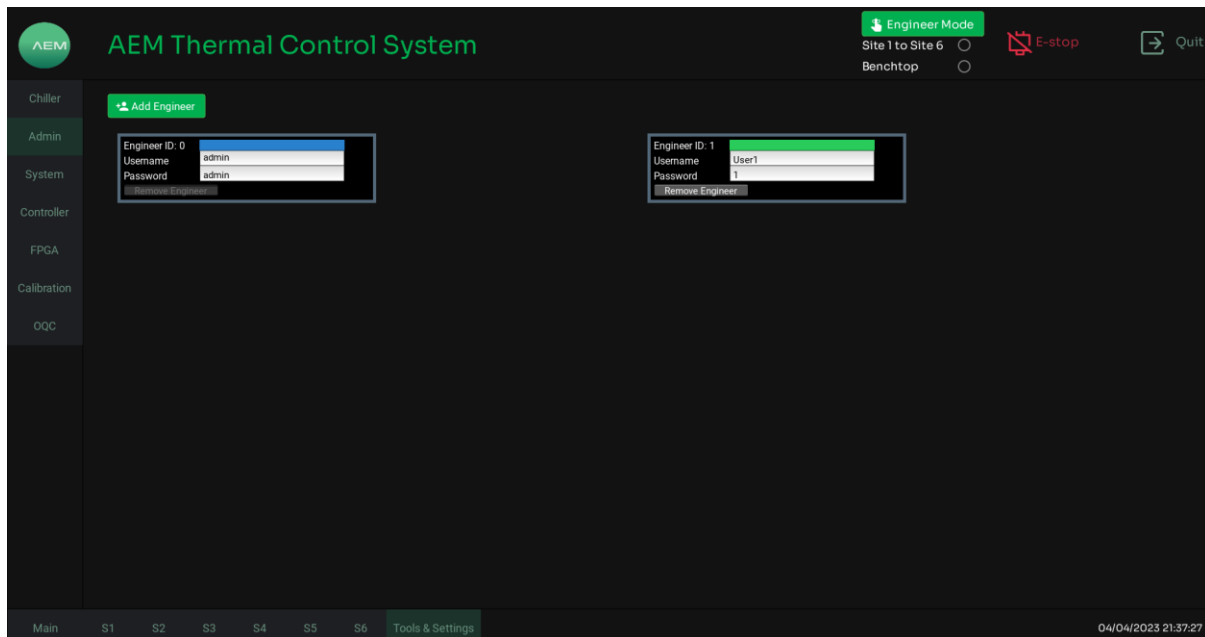
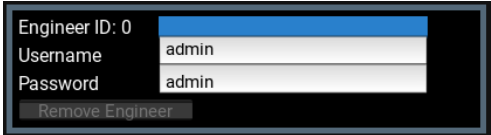
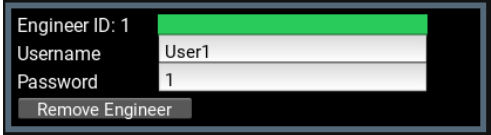


Figure 5-15

Table 5-16 Description of Admin Screen

GUI Name / Illustration	Description
ADD ENGINEER 	Adding authorised engineer who working on the UI system.
ENGINEER DETAILS ID 	Displays details of engineers added to the UI such as ID, username and password.
ACCESS ENGINEER DETAILS 	Display details of engineers who is using the UI system.
REMOVE ENGINEER 	Authorised engineer can delete out engineer not working on the system.



5.3.10 Tools & Settings – System Screen

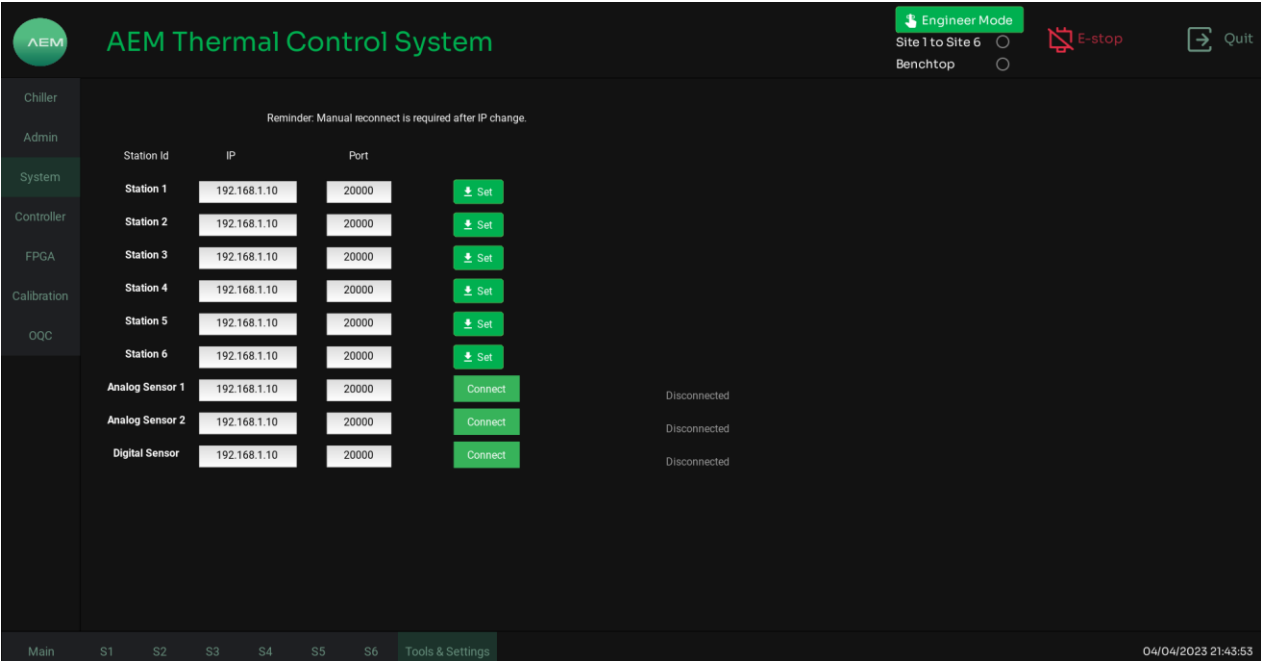


Figure 5-16

5.3.11 Tools & Settings – Controller

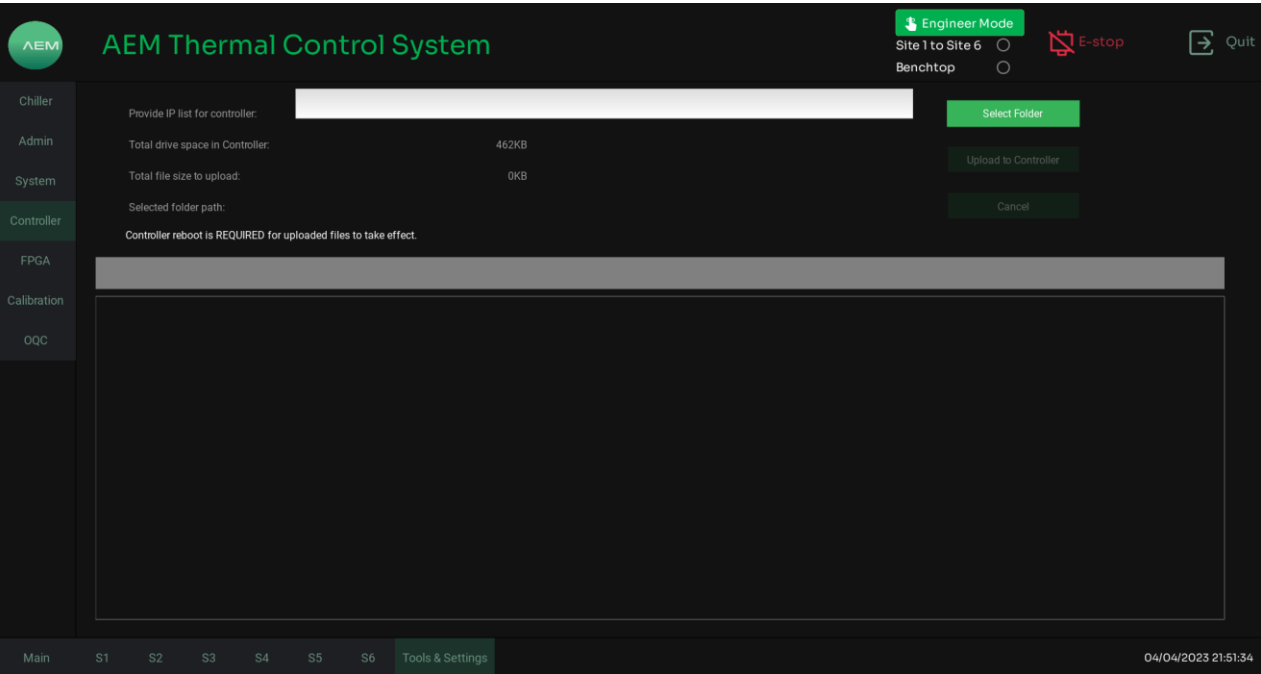


Figure 5-17



5.3.12 Tools & Settings – FPGA

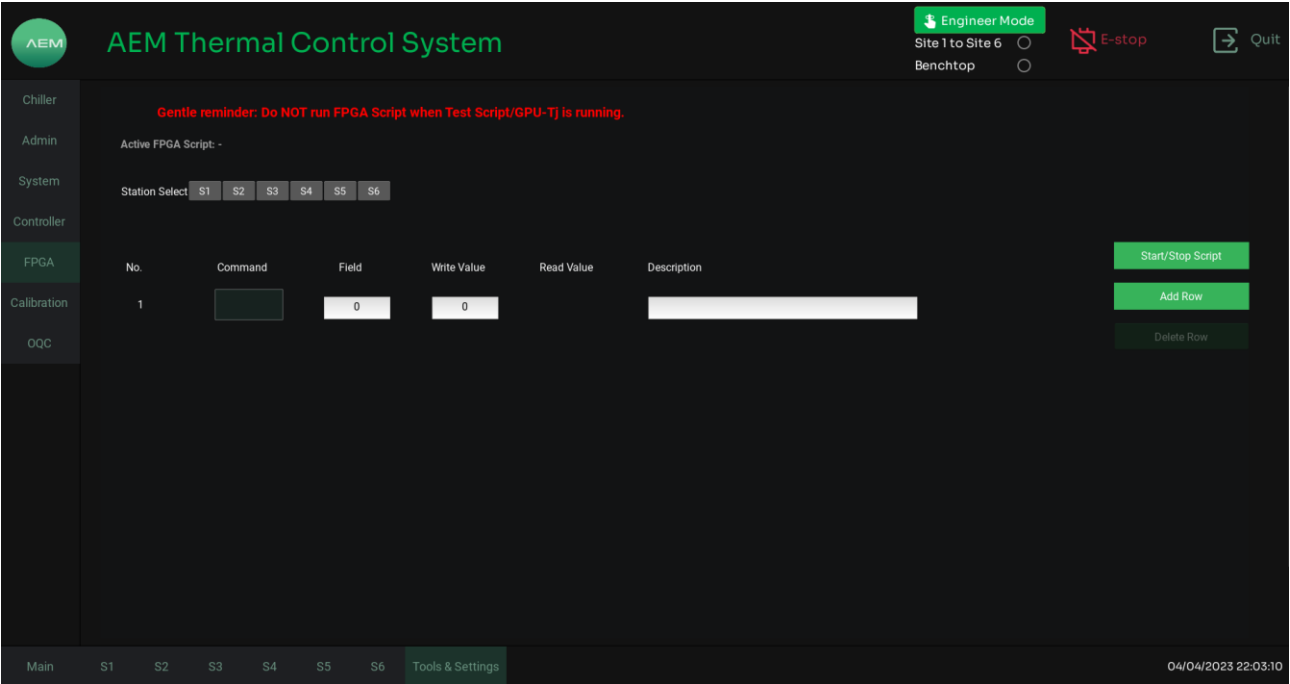


Figure 5-18 FPGA Screen

5.3.13 Tools & Settings – Calibration Screen

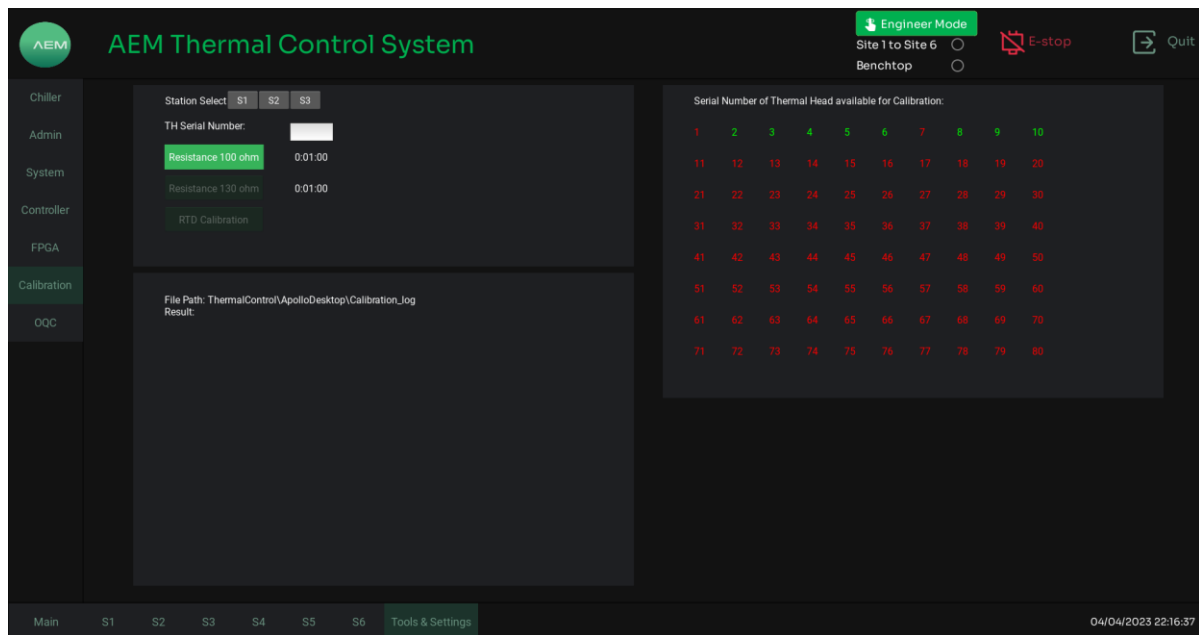
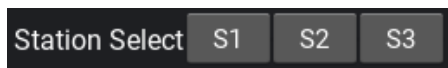
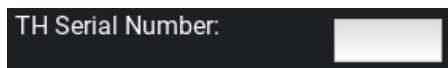
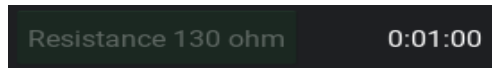


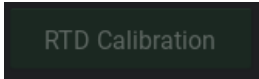
Figure 5-19

Calibration screens allow engineers to perform calibration for the thermal head. Not advisable for the users to use this screen without supervision.

Table 5-17

GUI Name / Illustration	Description
STATION SELECTION 	User able to select the station calibrating on.
SERIAL NUMBER INPUT 	Each thermal head defines for specific serial number. Engineers should enter serial number which is attached to the station at the given input section.
BUTTON TO CALCULATE RESISTIVE ERROR OF 100 OHM 	Engineers should attach the CALSET of 100 Ohm and click this button to start the calibration process. Offset value will be measured during this time. Timer will indicate the 1 minute of this process completion.
BUTTON TO CALCULATE RESISTIVE ERROR OF 130 OHM 	Once the first step and second step completed, this third button will be enabled, and engineers should attach 130 Ohm of CALSET to perform the second



GUI Name / Illustration	Description
	cycle of offset value calculation. Timer will indicate the 1 minute of this process completion.
RTD CALIBRATION BUTTON 	Last button will be enabled when the 3 steps are completed. Clicking this button will generate hex codes which will be load into the ATC system to achieve stable and reliable thermal head temperature values.

5.3.14 Tools & Setting - OQC Screen

5.3.14.1 Single Station OQC Test

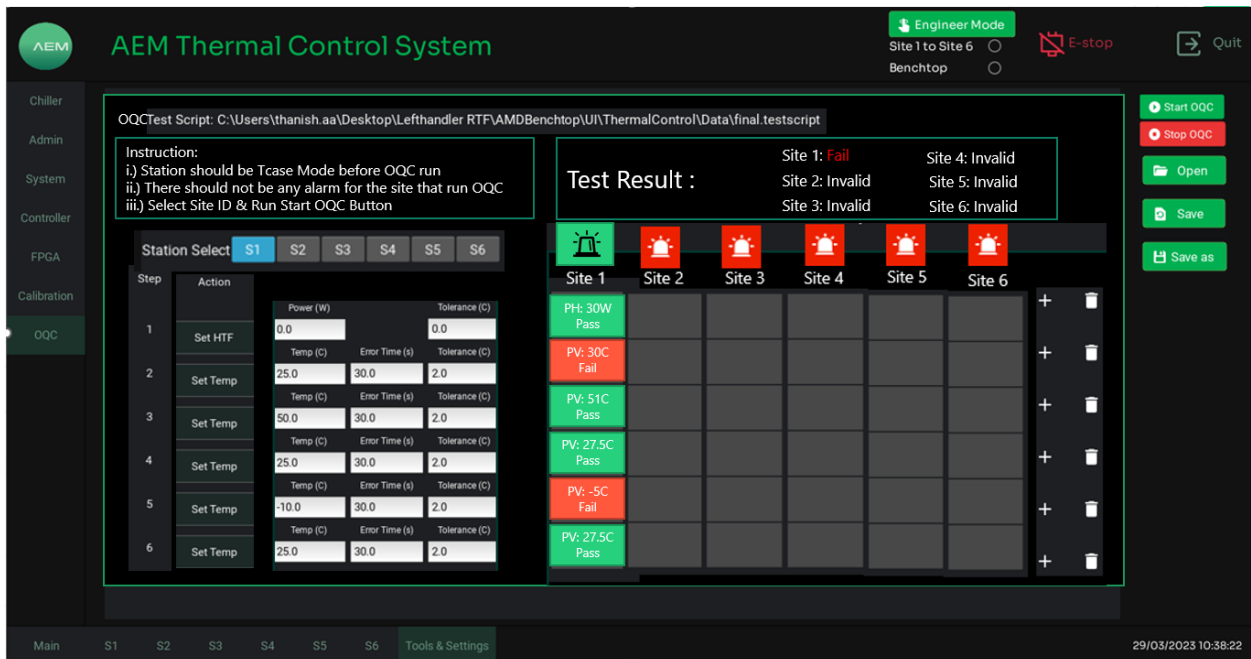


Figure 5-20

5.3.14.2 Multiple Station OQC Test

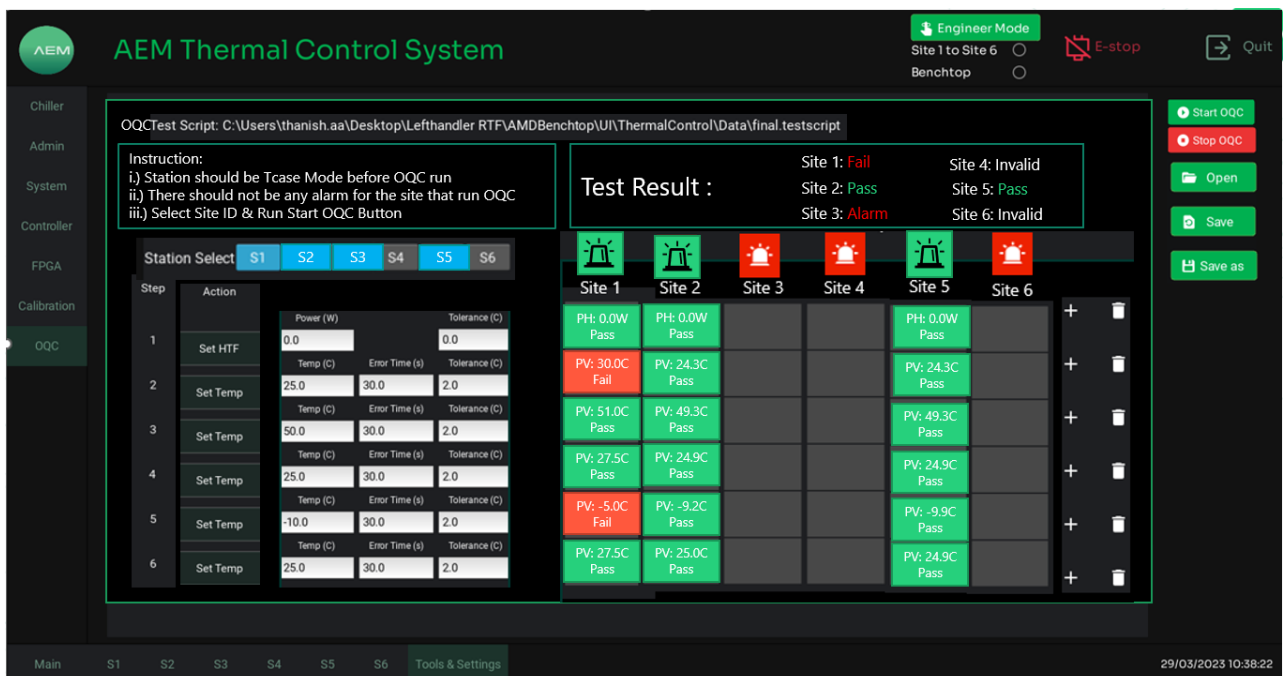
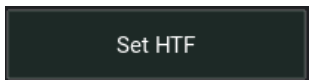
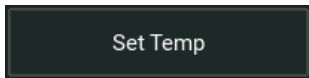
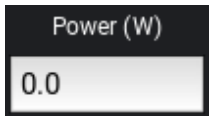
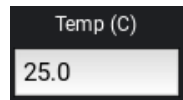
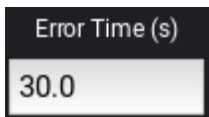
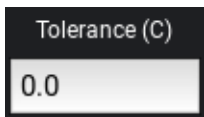
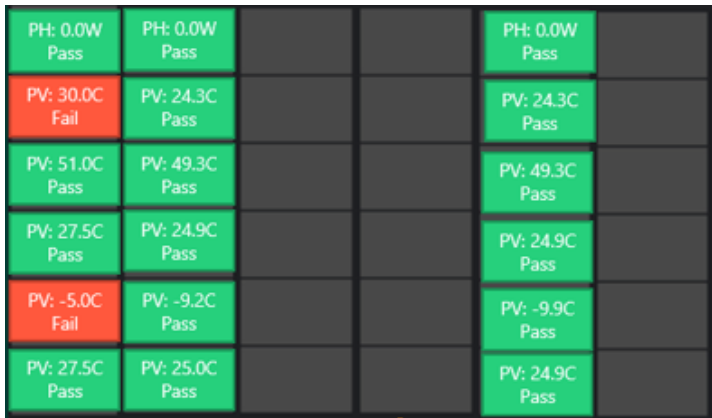


Figure 5-21

Description

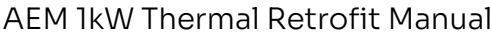
Table 5-18

GUI Name / Illustration	Description
STATION SELECTION 	User can select individual station or multiple station at once to run the OQC test simultaneously.
SET HTF 	User can select in the drop-down box. Set HTF will allow to set the Heater Power.
SET TEMP 	User can select in the drop-down box. Set Temp to enter desired set temperature.
POWER 	User can set the power required at the box given.
TEMPERATURE 	User can set the desire temperature manually.
ERROR TIME 	User can set the parameters at the input box given here. Error time is the grace time for the actual reading match to the set temperature.
TOLERANCE 	User can set the parameters at the input box given here. Tolerance can be adjusted based on user preference.
ALARM INDICATOR 	Alarm Indicator which turns red when there's any alarm generated in the relevant site. It won't be displaying Alarm Code, but just indication whether

GUI Name / Illustration	Description
	there's alarm or not. User have to enter individual/main screen (Check Alarm) feature to acknowledge the alarm description. Green alarm indicates good to run OQC test.
STATUS 	Result of each set parameters will be displayed in box-tabular form. In each box, actual reading will be displayed and set the background colour green if the result passed and red for fail.
SUMMARY OF TEST RESULT 	Test Result section will display the overall result according to the sites that has been run. Even if one step fails, the overall result will be shown as fail. If there's an alarm for the site selected by the user, it will display 'Alarm'. 'Invalid' for the sites which has not been selected for the OQC test.
ADD & DELETE 	Add and Delete options are enabled for user to modify the OQC script. Scroll view will be applied if the OQC script is longer the screen size.
START OQC 	User shall click the Start OQC Button, to run the test. Liquid valve will be always in Auto Mode. It runs each step for every station selected



GUI Name / Illustration	Description
	simultaneously. User can't start the run with any alarm triggered in the station.
STOP OQC 	Stop OQC will allow user to stop the test at any stage of the run. User must click Start OQC to restart the test after their preferable changes.
OPEN 	Open buttons enable user to choose the OQC script they desire.
SAVE 	Save button enable user to save the changes which has been modified in the pre-saved script.
SAVE AS 	Save As button enable user to create a new script and save in the folder.

[illegible]



6. Operation

6.1 Operation

This section provides information on the layout of the operation panel as well as the function and location.

6.1.1 Thermal Controller

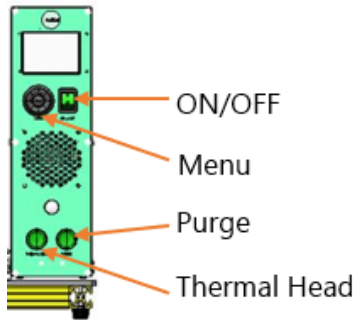


Figure 6-1

6.1.1.1 Push Button

Table 6-1

Push Button	Description
ON / OFF	Energizes / Deenergizes the tool
Thermal Head	Engage / Disengage the TH Total / Die force
Purge	Turns on compressed air for EGW Purging
Menu	Navigates the GUI

6.1.2 Monitor

The monitor is used to display and monitor the operating controller system of all 12 thermal controllers. Each column displays 12 controller connection status, running temperature, and force values.



Figure 6-2

6.2 Machine Operating Procedure

The following descriptions state the list of hazardous energies and guide the user in step-by-step procedure in operating the machine.

6.2.1 List of Potentially Hazardous Energy Sources

Refer to [Section 3.1](#) for facility requirements of the AEM 1kW Thermal Retrofit System.

6.2.2 Power ON Procedure



ENSURE THAT ALL PERSONNEL ARE KEPT AWAY FROM MACHINE AS ANY MOVING PARTS CAN CAUSE SERIOUS INJURY

Check and ensure that the incoming power supply is according to the specifications provided before switching on the power. Refer to [Table 3-1](#) for correct specifications.

1. Turn ON the switch located at EE panel station by pressing the ON switch.
2. Turn On the thermal controller green ON/OFF button for each controller.

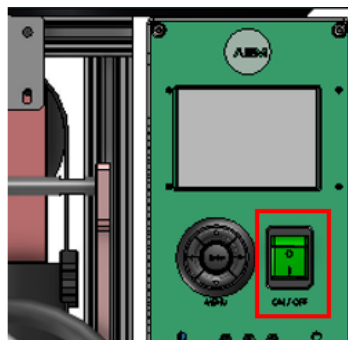


Figure 6-3 Switch ON the Thermal Controller

3. Turn On the Chiller switch Button.



Figure 6-4 Chiller Turn On button

6.2.3 Power OFF Procedure

Step 1 Ramp up chiller current temperature to 20 degrees from touchscreen UI or at chiller control panel.

Note: Do not drastically set temperature to 20°C. Slowly increase the temperature.

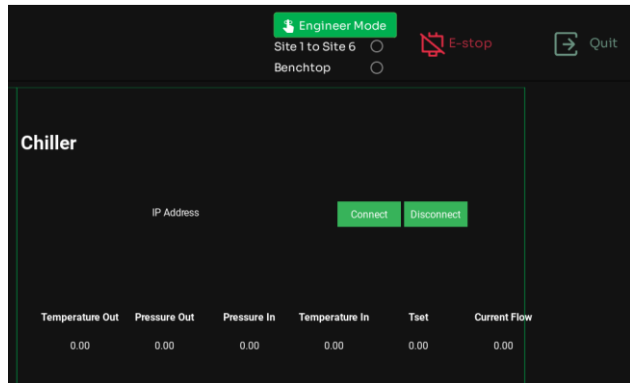


Figure 6-6 UI Tool & Setting Screen

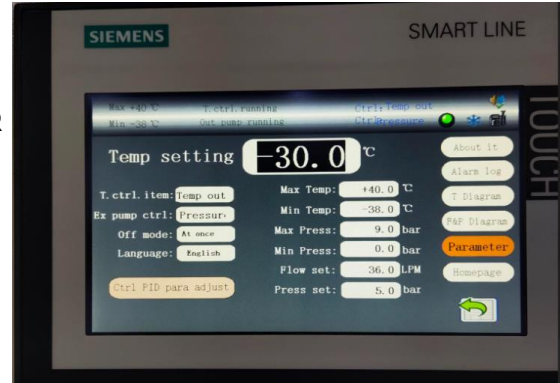


Figure 6-5 Chiller Parameter Screen

- From UI, select “Shut Down” test script
- “Set” and “Start” the test script
- Test script: 5°C/30mins until set temp is reached.

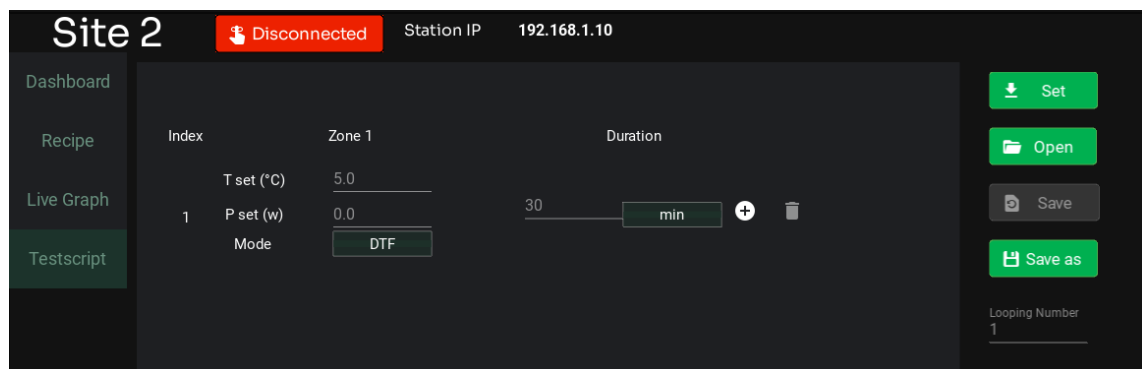


Figure 6-7

Step 2 Turn off Chiller compressor and pump from Chiller control panel.



Figure 6-8

- Step 3** Once set temp is reached, wait for 6 hours.
- Step 4** Ensure no condensation occurs on Thermal heads and Piping during this period.
- Step 5** Turn off dry air main supply.



Figure 6-9 Turn OFF the chiller power supply

- Step 6** Turn off 12x controllers power off button.



Power Button

Figure 6-10 Switch Off all the controller Power Button

- Step 7** Shut down IPC from touchscreen monitor.



Figure 6-11 Turn Off the button

- Step 8** Turn off main power switch. Observe LOTO procedure refer to the section [6.2.4 Lock Out /Tag Out \(LOTO\)](#).



Figure 6-12

Note: For Handler shut down procedure, kindly refer to AMD guidelines

6.2.4 Lock Out / Tag Out (LOTO)

Lockout is used to:

- Warn other workers that electrical equipment or other systems are being worked on presently.
- Prevent the inadvertent start-up of the system or equipment during maintenance activities.
- Protect both the worker maintaining the equipment and any other personnel in the area.

Below shows the Lockout procedure:

1. De-activate the “AIR” Push Button.
2. Press “OFF” power button.
3. Main isolator switch turned to "OFF" so that the machine is isolated from the energy source.



Figure 6-13

4. Display standard tag(s) with the required information on it and lock out the isolator switch with pad lock.



Lock out and Tag out the Isolator Switch using hasp & pad lock

Figure 6-14

5. The tag must be filled with the information of the personnel who is performing the maintenance work on the machine.



Figure 6-15

6. To isolate the CDA energy, locate the main service unit in the side enclosure of the machine.



Figure 6-16

7. Turn the hand valve (RED color) in the counterclockwise direction to cut off the main CDA to the handler. The display will change to 0 bar (value in red LED) immediately.

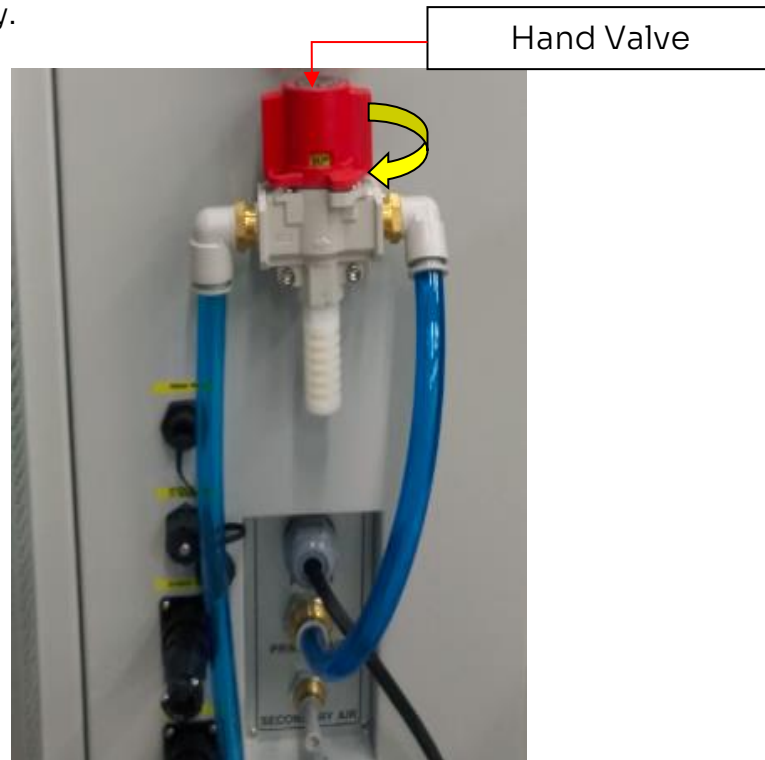


Figure 6-17

8. Display standard tag(s) with the required information on it and lock out the hand valve with hasp and pad lock.

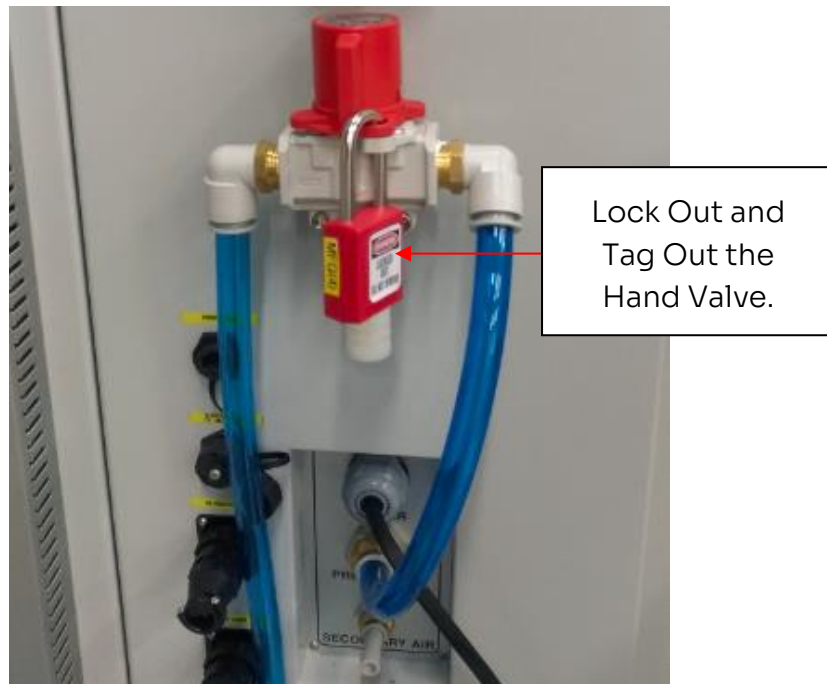


Figure 6-18

9. The tag must be filled with the information of the personnel who is performing the maintenance work on the machine.



Figure 6-19

10. Engage the push button or other operating controls to ensure that the machine is de-energized.
11. Disconnect the machine from the main power supply and the compressed air.

Electrical LOTO

1. Press “OFF” power button.
2. Disconnect Plug 1 and Plug 2 from main power supply then place inside the plug lockout.



Figure 6-20

3. Lock out with pad lock.



Figure 6-21

6.2.5 Verify EGW & OFA are Connected

1. Verify that the following lines are connected to the tool:

- EGW Supply
- EGW Return
- Air Supply

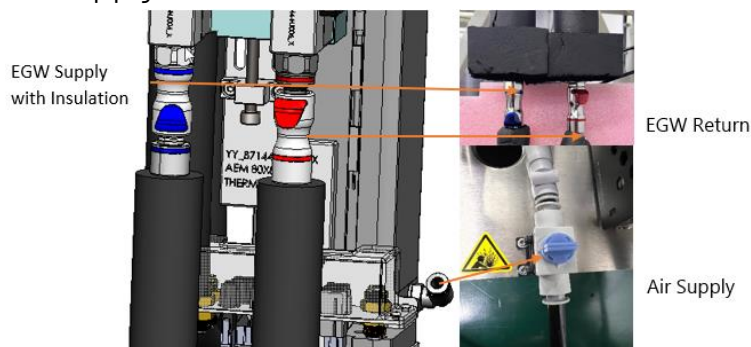


Figure 6-22 AEM 1kW Thermal Retrofit System lines

2. Make sure the EGW is circulating before conducting any testing. (**Note:** PV 1, TX1 and TX2 should be close to 10°C if the incoming EGW temperature is 10°C).
3. Check the incoming air supply by toggling the purge mode into manual. Use a hand to feel the purge air coming from the purge point at thermal head.

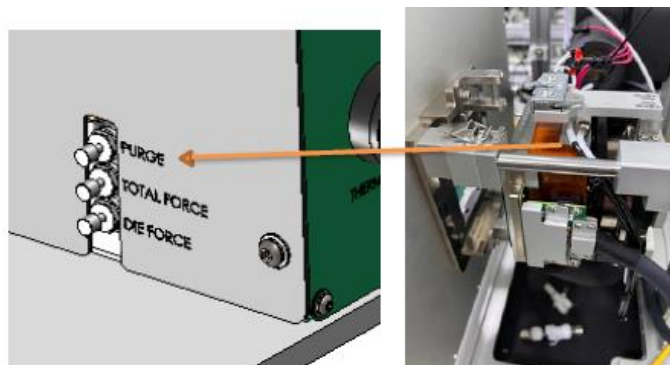
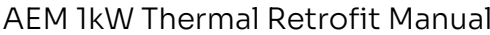


Figure 6-23

4. Check diaphragm piston functionality by pushing the Thermal Head button.

[illegible]



7. Fault Description

7.1 Error and Handling

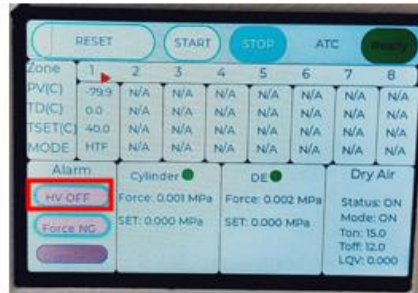


Figure 7-1

There are 3 types of error that may cause thermal control issues. The reason, effect and solution of these errors are stated in the following pages.

1. HV OFF
2. FORCE NG
3. OVERTEMP

1. HV OFF

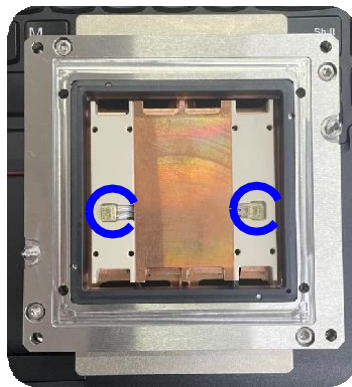


Reason:

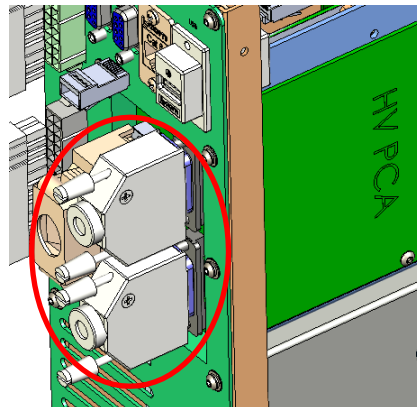
1. TCO is broken or HV board is malfunction.
2. Connector is not plugged in.
3. Interface board faulty.

Effect:

1. Zone heaters do not have power output to heat up the Thermal Heat.

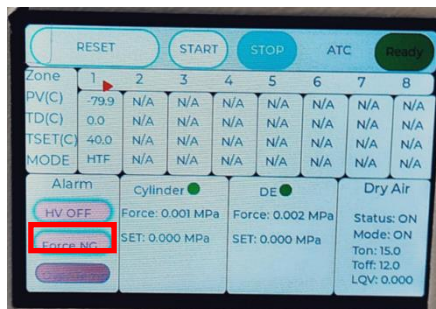


Solution:



1. Ensure TCO is not broken.
2. Connectors are plugged in.

2. Force NG



Reason

1. When Force is less than 30.0 lbf this alarm

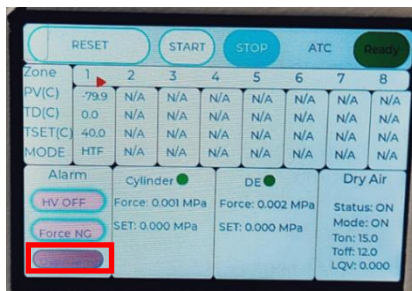
Effect:

1. Does not stop any thermal control
2. Indirect affecting thermal performance

Solution:

1. Encouraged to set Force above 30.0 lbf (Not a must for Force Setting)

3. Over Temp



Reason:

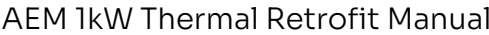
1. When temperature is above set Over Temperature Value on the recipe screen, this alarm lights up.

Effect:

1. Stop control.

Solution:

1. Ensure fluid flow is good
2. Press Reset to enable back control.



This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.



8. Conversion



Mechanical Hazards: These hazards exist in any case where a component can move in such a way as to impact or otherwise injure personnel performing maintenance functions on the equipment. To avoid these hazards always operate the equipment with the safety covers in place and door interlocks enable. When performing maintenance functions with the covers open, be sure to avoid those areas where impact with the moving parts of the equipment is possible. When removing components, it may be necessary to isolate the potential mechanical energy by restraining the mechanism using cable ties or blocking.

This chapter covers all the procedures to change over the tooling to support a different test unit type. When requirements call for a change to a new package with different sizes of units, trays and/or chassis, a conversion kit must be installed. The conversion kit does not include the common part; the kit only includes the specific design parts.



ENSURE THAT A SIGN IS PLACED INDICATING THAT THE EQUIPMENT ARE BEING SERVICED OR ADJUSTED, IN ORDER TO KEEP ANY ONE FROM TOUCHING THE OPERATION PANEL.

ONLY QUALIFIED PERSONNEL SHOULD CONDUCT CONVERSION OPERATIONS.

ALWAYS SWITCH OFF MAIN POWER SUPPLY AND COMPRESSED CDA TO THE MACHINE BEFORE DOING CONVERSION OPERATION.

8.1 Thermal Head Kit Removal Procedure

1. Turn off the power supply.

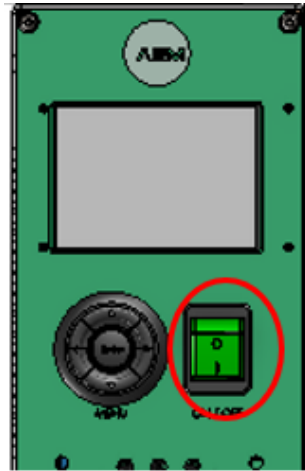


Figure 8-1

2. Disconnect the Inlet & Outlet Quick Disconnects (QDs) at the end of the thermal head hose.

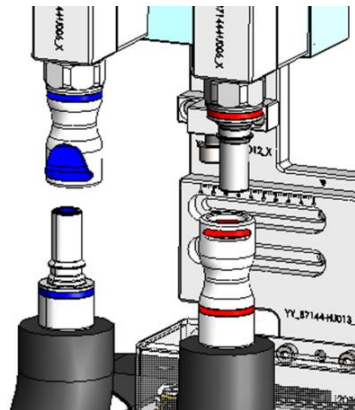


Figure 8-2

3. Disconnect the air hose and disconnect the electrical connectors.

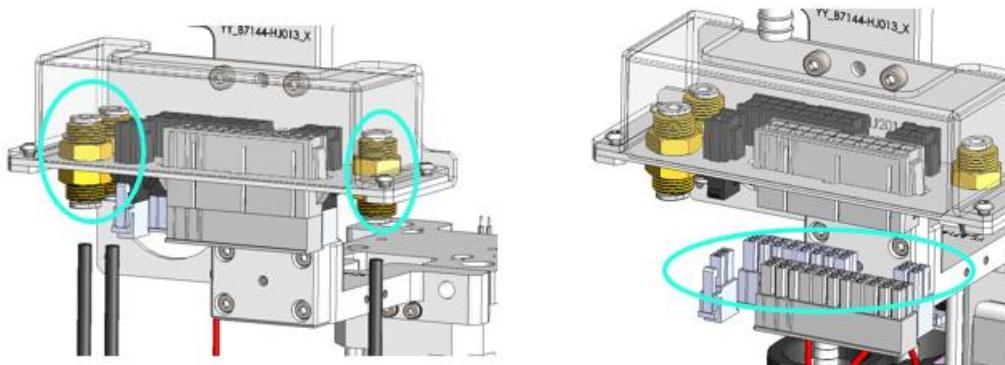


Figure 8-3

4. Catch clips on Top & Bottom of the TH are released.

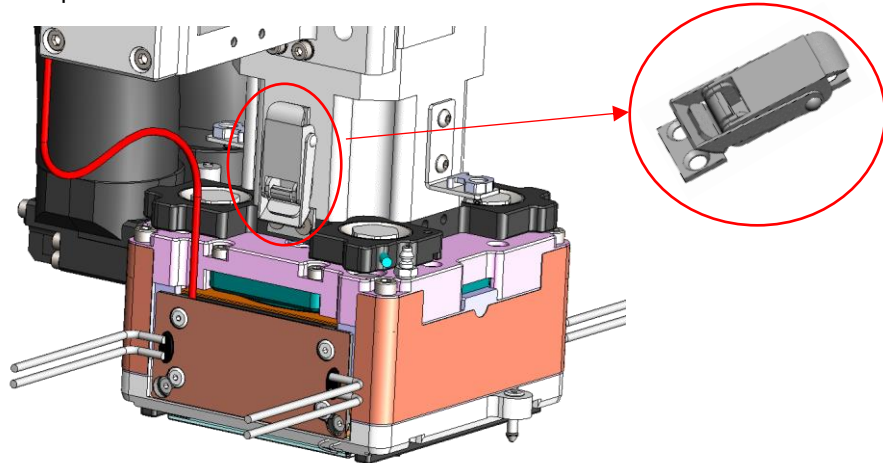


Figure 8-4

5. TH now can be removed from the bottom.

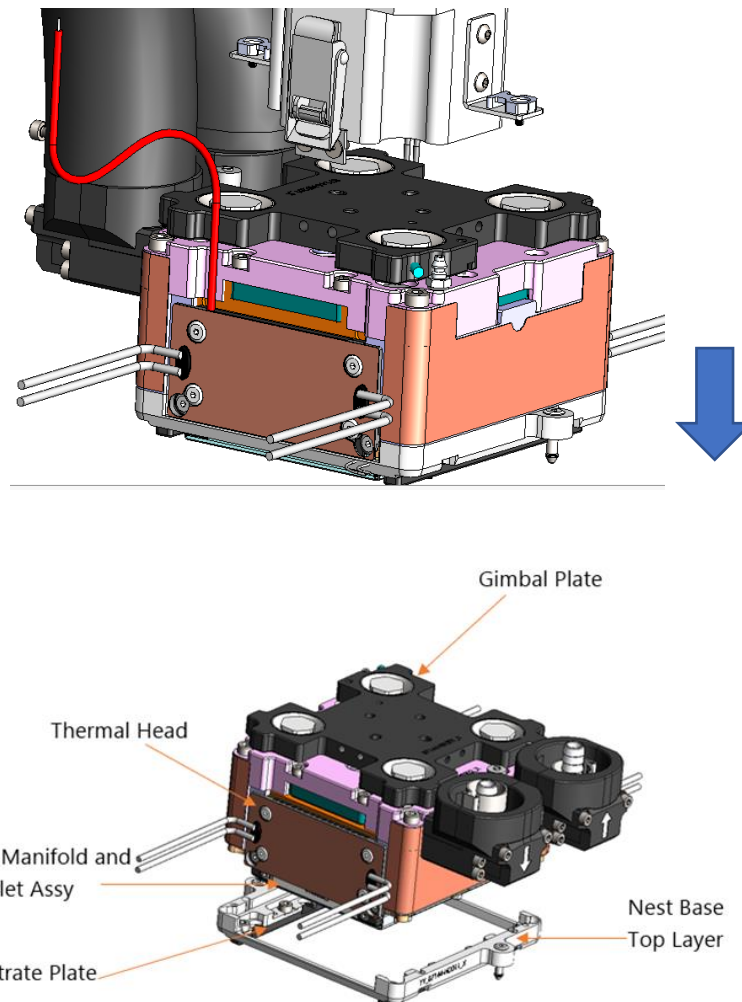


Figure 8-5

6. Remove 4x screws on kit and remove the substrate pusher plate.

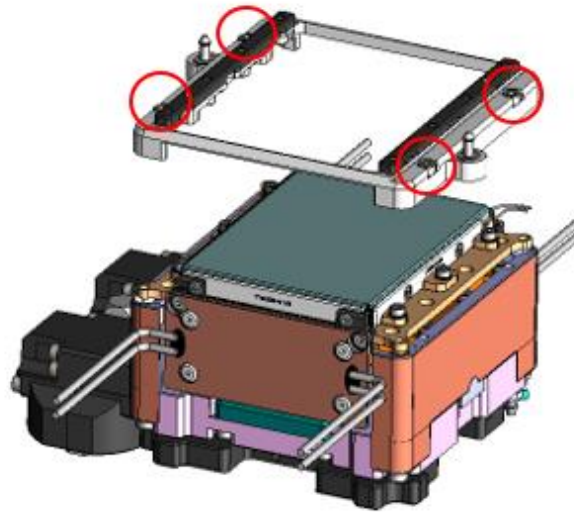


Figure 8-6

7. Kit removed from TH.

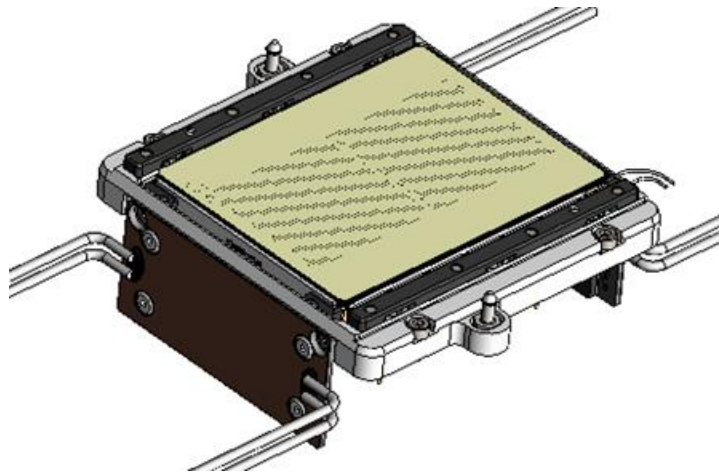


Figure 8-7

8. Change to another TH kit or TIM and reverse the step from 6 to 1.

8.2 Changing TIM

1. Remove the nest base bottom layer.

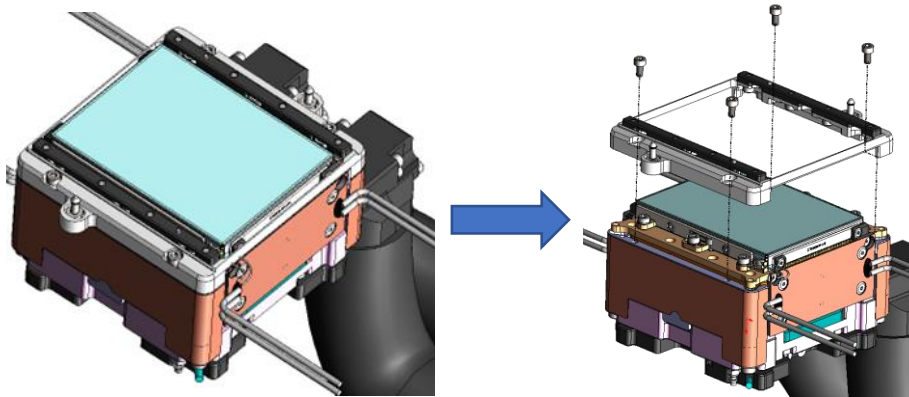


Figure 8-8

2. Remove x8 screws fastened to the TIM retainer.

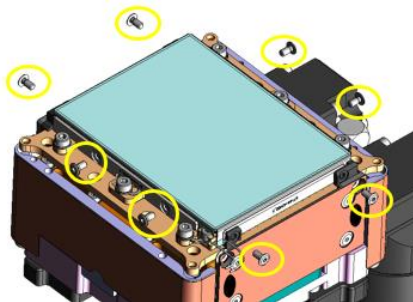


Figure 8-9

Note: Fix from the Nest Base (bottom layer) to the Substrate pusher.

3. Change TIM from the heatlet.

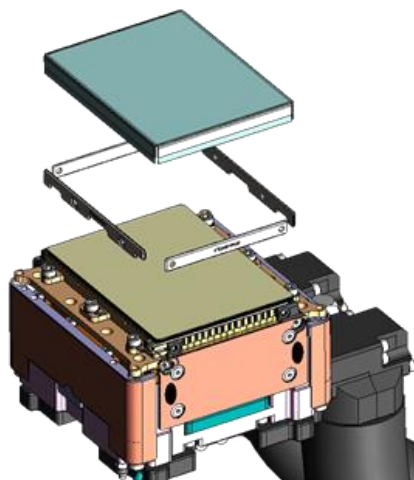


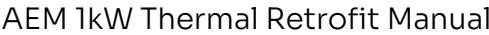
Figure 8-10



4. Reinstall the kit following the above steps in reverse.

Note: A stress test need to be conducted if new TIM is implemented.

- Make sure the station is connected to the facility (EGW and air pressure).
- Set the socket force into 140 lbf.
- Engage the socketing force to the DUT.
- Run the stress test script for 5 mins continuously (from -30°C then stay 30 sec to 110°C).
- Observe the TH and TD should be able to reach -30°C.

[illegible]



9. Troubleshooting



9.1 Equipment Malfunction

This chapter consists of a general description of possible equipment malfunctions or failures. Each description includes the remedies or possible solution of an issue that may arise. It is set with a quick guide to assist with general ideas to overcome unwanted miss failures. We strongly recommend you to first contact the supplier for further assistance if there is any doubt on how to perform activities of troubleshooting.



**ONLY QUALIFIED PERSONNEL SHOULD ATTEMPT TO TROUBLESHOOT THE
DEBUG CART MACHINE.**

TROUBLESHOOTING GUIDE TABLES Index of Contents:

9.1 General Possible Failures: a) General machine power failures b) Over temp c) Unable to extend / retract piston diaphragm	9.2 Thermal Activation Failures: a) Temperature ramp-up cannot reach set point b) Temperature ramp-up overshoots set point
---	---



Please always follow safety guidelines, SATEFY FIRST!



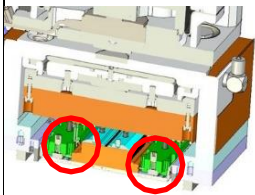
9.2 General Possible Failures

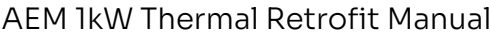
Table 9-1

Problem	Symptom(s)	Remedy / Possible Solution	Illustration / reference
a) Over temp	Over temp alarm Unable to start any testing Failed to maintain the temperature	Check Main Facility of EGW and ensure flow is good Check the assembly orientation. It must face up or towards pedestal	When temperature is above user set temp, this alarm lights up
b) Unable to extend/retract piston diaphragm	Cylinder not moving after toggle to extend or retract	Check Main Facility of air and ensure flow is good Check set force minimum must be above 20 lbs	

9.3 Thermal Activation Failures

Table 9-2

Problem Description	Possibility Rootcauses	Actions	Illustration / reference
a) Temperature ramp-up cannot reach set point	TIM pre-conditioning not performed	Perform stress test stated at Chapter 8	
	TIM damaged	Replace VCTIM	
		Perform stress test stated at Chapter 8	
	Thermal fuse blown	Replace thermal fuse	
	Heater damaged	Replace the thermal head kit	
b) Temperature ramp-up overshoots set point	EGW didn't turn on	Check Main Facility of EGW and ensure flow is good	
	Incorrect heater-RTD parameter	Retrieve heater-RTD parameter from record Confirm that correct parameters are displaying after programming	

[illegible]

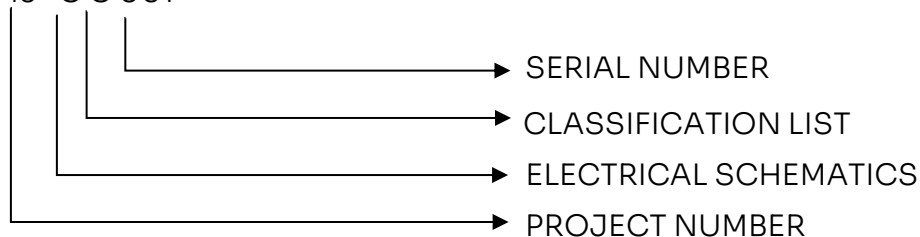


10. Electrical

10.1 Electrical Schematics Nomenclature

10.1.1 Schematics-Numbering System

B2743- C C 001



REVISION &
DATE

TITLE : TAPC 1&2 CONNECTION				 THIS IS A PROPERTY OF AEM SINGAPORE PTE LTD IT SHOULD NOT BE COPIED OR REPRODUCED WITHOUT THE WRITTEN APPROVAL OF AEM SINGAPORE PTE LTD.
DRAWING NO : B3065-CC012		REV: 0	DATE: 15/12/2019	
	INITIAL	SIGN	DATE	
DRAWN	PETHUEL ENOT		09/07/2019	
CHECKED	PETHUEL ENOT		26/10/2019	
APPROVED	ARUN KUMAR		26/10/2019	
SIZE : A3	DO NOT SCALE DRAWING	PAGE NO : 44 - 50		

10.1.2 Schematics Classification

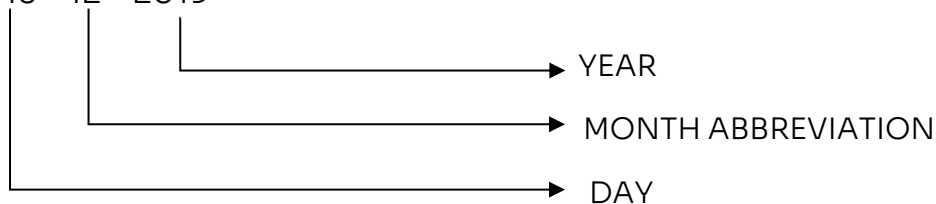
Table 10-1

Classification	Description
A	Content and drawing list
B	Control panel layout, location, sizes, etc
C	Power Circuit and DC control circuit
*D	PC interface, PLC interface, other interface, etc
*E	PLC Special I/O, Digital I/O, Analog I/O, etc
F	Motor Control Circuit for Stepper, Servo & Driver
*G	Motor Control Circuit like conveyor, speed control, AC or DC
*H	Control, connection for solenoid valve
I	Input, output terminal connection for sensor, SV, etc
*J	Module drawing consists of motors, sensors, solenoid valve, etc

*Not commonly used

10.1.3 Schematics Date Format

15 - 12 - 2019



TITLE : TAPC 1&2 CONNECTION				
DRAWING NO : B3065-CC012			REV: 0	DATE: 15/12/2019
	INITIAL	SIGN	DATE	 THIS IS A PROPERTY OF AEM SINGAPORE PTE LTD IT SHOULD NOT BE COPIED OR REPRODUCED WITHOUT THE WRITTEN APPROVAL OF AEM SINGAPORE PTE LTD
DRAWN	PETHUEL ENOT		09/07/2019	
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SIZE : A3	DO NOT SCALE DRAWING		PAGE NO : 44 - 50	

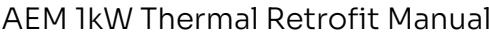


10.2 Electrical / Schematic Circuits Content

This section consists of the electrical / schematic diagrams for the AEM 1kW Thermal Retrofit System.

Table 10-2

No	Drawing No	Title
1	N3838-CC001	RETROFIT POWER SUPPLY 1, 2 & 3 CONNECTION
2	N3838-CC002	CONNECTION FOR IO MODULE ADAM-6017
3	N3838-CC003	CONNECTION FOR IO MODULE ADAM-6052
4	N3838-CC004	CONNECTION FOR IO MODULE ADAM-6017 FLOW SEN
5	N3838-CC005	RETROFIT CONNECTION FOR NETWORK SWITCH 1 & 2
6	N3838-CC006	CONNECTIONS FOR RETROFIT RJ45 CABLE
7	N3838-CC007	RETROFIT EMO SYSTEM CONNECTIONS
8	ECA61466	CABLE ASSY RETROFIT INCOMING PWR PDU 1
9	ECA61467	CABLE ASSY RETROFIT INCOMING PWR PDU 2

[illegible]



11. Maintenance



11.1 AEM 1kW Thermal Retrofit System Preventive Maintenance

Two Persons Required When Commencing Maintenance and Servicing

When doing maintenance and / or repairing work, two people are required to be present for precautionary purpose. This shall be done so that one of them can aid in the lookout for the push button panel, preventing other people from accidentally pushing the button while the other person is servicing the machine.

All employees, upon observing a machine or piece of equipment which is locked out to perform servicing, maintenance, or replacement of parts, shall not attempt to start, energize, or use that machine or equipment.

Before doing any maintenance or replacement of parts, power off the Thermal Station (refer to [6.2.3 Power OFF Procedure](#) and [2.5.6 Standard 3 Point Safety Check](#)). Follow all the safety procedures before start doing any maintenance and replacement (refer [2.2.2 General Safety Guidelines](#)) and wear all PPE (refer [2.2.3 Personal Safety Equipment](#)).

Spring Washers

Any time a Spring Washer is removed in the performance of regular maintenance activities, replace the Spring Washer with a new washer. Spring Washers should not be reused. Record any replaced Spring Washers in available procedure checklist.

General Maintenance

Specific machine components should be checked periodically to ensure they are working properly. Follow the preventive maintenance (PM) procedures given below will facilitate in minimizing production interruptions.

The procedures for regular inspections and maintenance can be classified into intervals of:

1. Semi-Annually (every 6 months)



ONLY QUALIFIED PERSONNEL SHOULD CONDUCT MAINTENANCE OPERATIONS. ALWAYS DISCONNECT THE MACHINE FROM MAINS AND COMPRESSED AIR BEFORE DOING MAINTENANCE OR CLEANING OPERATION.



ENSURE THAT A SIGN IS PLACED INDICATING THAT THE EQUIPMENT AND ROBOT IS BEING SERVICED OR ADJUSTED, IN ORDER TO KEEP ANY ONE FROM TOUCHING THE OPERATIONAL PANEL.



ALWAYS WEAR SAFETY GLASSES WHENEVER WORKING WITH COMPRESSED AIR. DIRECT AIR PRESSURE AND/OR FLYING DEBRIS FROM AIR BLOWN PARTICLES COULD CAUSE SERIOUS EYE INJURIES.

11.1.1 Daily Preventive Check

11.1.1.1 Chiller System Component Interface Inspection

1. Refrigeration system interface check

High pressure interface, low pressure interface, exhaust gas temperature detection, return gas temperature detection

2. Circulating fluid system interface check

Pressure relief valves, bypass pipes, inlet and outlet temperature, pressure detection, water leakage detection, etc. are all equipped.

3. Cooling water system interface check

The internal line of the cooling water adopts a live takeover and is equipped with a drainage connection.

4. Electrical system interface check

The electrical system is equipped with an external emergency stop. Local emergency stop, Ethernet communication and other interfaces, as well as indicators.

11.1.1.2 System general Alarm Test

1. Cooling water detection

The cooling water is disconnected, the compressor cannot be started, and the panel alarm prompts.

2. Lack of circulating fluid detection

The lack of circulating fluid will alarm.

3. Over temperature detection

By artificially changing the alarm value, there will be high temperature and low temperature alarms, and the low temperature alarm will stop the compressor.

4. Fluorine pressure detection

Adjusting the pressure switch can test the pressure protection function, and the compressor cannot be started in an alarm state.

5. Water leakage test

This feature is detected by filling the sink with water and the panel prompts.



11.1.2 Semi Annual Preventive Maintenance

Semi Annual Preventive Maintenance to be performed every 6 months.

11.1.2.1 Warning Label Check

Refer to [2.2 Safety Alert Boxes](#). Check all warning label condition and adhesion. Replace any label that is not legible or is peeling away from panel surface.

The Chiller internal wire marking is fully equipped, the user interface screen printing is complete, and the equipment safety logo is complete.

11.1.2.2 Tamper Proof Screws

Check all Tamper Proof Screws and verify Tamper Proof Screws are in place and torqued correctly. Replace any screws that should be tamper proof but has been replaced with a standard screw or if the Tamper Proof Screw has been damaged. Torque to the specification by screw size.

11.1.2.3 Spring Washers

Any time spring washer is removed in the performance regular maintenance activities replace the spring washer with a new washer. Spring washers should not be rushed. Record any replaced spring washers in available procedure checklist.

11.1.2.4 Chiller Fan Filter Clean Procedure

1. Turn OFF the chiller switch.

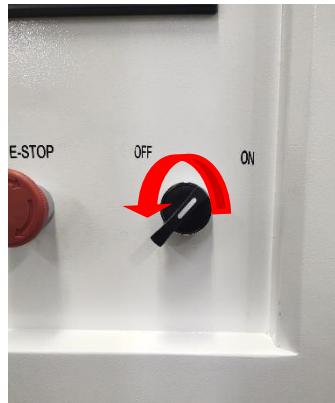


Figure 11-1 Turn OFF Chiller

2. Open the facility door.



Figure 11-2



3. Remove the filter cover.



Figure 11-3 Dusty filter

4. Loosen all the screws and nut by using spanner.



Figure 11-4 Remove all the screws attached to Fan filter

5. Use the brush to clean the dust on the filter.
6. Install the cleaned filter back the fan.



Figure 11-5

11.1.3 Annual Preventive Maintenance

11.1.3.1 Coolant Changing Procedure

1. Turn OFF the chiller switch.



Figure 11-6 Turn OFF

2. Prepare a pail and drain out all the fluid from the pipeline. Tapped switch the process fluid drain to drain out chemical substances.



Figure 11-7

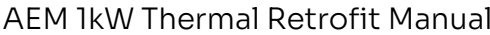


3. Open the tank cover at top of the chiller and refill the EGW fluid. Ensure follow the satefy procedure when handling with chemical fluid.



Figure 11-8

4. Ensure not exceed the fluid tank levelling. Finally close back the cover. Turn On the chiller.

[illegible]



12. Appendix



12.1 Dynalene EG (Heat Transfer Fluid)

1. Product Identifiers

Product Name: Dynalene EG (includes all concentrations/dyes)

Producer: Dynalene, Inc.

Product Number: Not available.

CAS-No.: Not available.

Identified uses of the product and uses advised against

Identified Uses: Heat transfer fluid.

Details of the chemical supplier

Company: Dynalene, Inc.

5250 West Coplay Road

Whitehall, PA 18052

USA

Telephone: +1 610-262-9686

Fax: +1 610-262-7437

Emergency telephone number

Within the U.S.: +1 800-424-9300 (CHEMTREC)

Outside the U.S.: +1 703-527-3887 (CHEMTREC)

2. Hazards Identification 2.3 Hazards not otherwise classified (HNOC) or not covered by GHS

Classification of the substance or mixture

GHS Classification in accordance with 29 CFR 1910 (OSHA HCS)

Acute toxicity, Oral (Category 4), H302

Specific target organ systemic toxicity – repeated exposure (Category 2), H373

**GHS Label elements, including precautionary statements**

Pictogram



Signal word: Warning

Hazard statement(s)

H302 Harmful if swallowed.

H373 May cause damage to organs (kidneys) through prolonged or repeated exposure (oral)

Precautionary statement(s)

Wash skin thoroughly after handling. Do not eat, drink, or smoke when using this product. Do not breathe mist/vapors/spray. Get medical advice/attention if you feel unwell.

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell. Rinse mouth.

Dispose of contents/container to an approved waste disposal plant.

Hazards not otherwise classified (HNOC) or not covered by GHS

None.

Composition/Information on Ingredients

Product mixture

Synonyms: Mixture.

Molecular Wt: Not available.

CAS-No.: Not available.

Ingredients	Classification	CAS No.	Concentration
Ethylene glycol	Acute Tx. 4, H302; STOT RE 2, H373	107-21-1	15 – 100%
Inhibitor solution (trade secret)	Not hazardous	N/A	<12%



First Aid Measures

Description of first aid measures

Skin exposure

Wash off with soap and water. Consult a physician.

Eye exposure

Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

Inhalation

If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

Ingestion

Never give anything by mouth to an unconscious person. Rinse mouth with water and consult a physician.

Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and in section 11.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

5. Fire Fighting Measures

5.1 Suitable (and unsuitable) extinguishing media

Suitable: Water spray, carbon dioxide, foam, dry chemical, any ABC class.

5.2 Specific hazards arising from the chemical

When involved in a fire, this material may decompose and produce irritating vapors and toxic gases (e.g., carbon oxides).

5.3 Advice for firefighters

Incipient fire responders should wear eye protection. Structural fire fighters must wear Self-Contained Breathing Apparatus and full protective equipment. Move fire-exposed containers if it can be done without risk to firefighters. If possible, prevent run-off water from entering storm drains, bodies of water, or other environmental areas.

6. Accidental Release Measures

6.1 Personal precautions, protective equipment, and emergency procedures

Proper protective equipment should be used. In case of an uncontrolled release, clear the affected area, protect people, and respond with trained personnel.



Avoid breathing vapors. Ensure adequate ventilation. Dynalene, Inc. Dynalene EG
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6.2 Environmental precautions

Do not let product enter drains.

6.3 Methods and materials for containment and cleaning up

Small spill: Cover with absorbent material (floor absorbent, vermiculite, etc.).
Soak up spill and place material into a drum.

Large spill: Wear protective equipment. Stop spill at source, dike the area surrounding the spill to prevent further exposure. Prevent material from entering sewer system. If necessary, absorbents such as vermiculite, clay floor absorbent may be used on spill and shoveled into drums.

6.4 References to other sections

For disposal see section 13.

7. Handling and Storage

7.1 General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of the day.

7.2 Precautions for safe handling

Use in a well-ventilated location. Open drums and other containers of this product slowly, on a stable surface. Drums and other containers of this product should be properly labeled. Keep containers tightly closed.

7.3 Conditions for safe storage, including any incompatibilities

Move drums of this product carefully, with the appropriate drum-handling equipment. Store drums and other containers in cool, dry locations, away from direct sunlight, or sources of intense heat. Storage areas should be made of fire-resistant materials. Keep containers away from incompatible chemicals.

8. Exposure Controls/Personal Protection

8.1 Control and exposure limits recommended by the chemical manufacturer

USA OSHA, Table Z-1, Limits for Air Contaminants – 1910.1000: 50 ppm, 125 mg/m³ (ethylene glycol, C value)

USA ACGIH, Threshold Limit Values (TLV): 100 mg/m³ (ethylene glycol, C value)

Eye and upper respiratory tract irritation, not classifiable as a human carcinogen.



8.2 Appropriate engineering controls

Use with adequate ventilation to minimize exposure to mists or sprays of this product. Prudent practice is to ensure eyewash/safety shower stations are available near areas where this product is used. Monitoring of oxygen level is recommended.

8.3 Individual protection measures, such as personal protective equipment

All personnel handling the product should use a personal protective equipment level D.

Respiratory protection

None needed for normal circumstances of use. If respiratory protection is needed, use only protection authorized in 29 CFR 1910.134, or applicable State regulations. Use supplied air respiration protection if oxygen levels are below 19.5% or are unknown.

Eye protection

Wear safety glasses with side shields.

Hand protection

Wear butyl rubber, natural rubber, neoprene, Nitrile rubber, or other suitable gloves for routine industrial use. Dynalene, Inc. Dynalene EG SDS Effective February 20, 2020

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Body protection

Wear impervious clothing.

9. Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

- a) Appearance Clear, liquid.
- b) Odor Odorless.
- c) Odor threshold No data available.
- d) pH 7.0 - 11.0
- e) Melting/freezing point -16.7°C (2°F), for >99% concentration.
- f) Boiling point >100°C (>212°F)
- g) Flash point 120.5°C, for 100%
None for concentrations <80%.
- h) Evaporation rate 1



- i) Flammability (solid, gas) No data available.
- j) Upper/lower flammability Upper (UEL): 15.3% (V)
or explosive limits Lower (LEL): 3.2% (V)
- k) Vapor pressure 0.08 mmHg at 25°C (77°F)
- l) Vapor density 2.14 (Air = 1.0)
- m) Relative density 1.0 – 1.2 g/cm³ at 25°C (77°F)
- n) Water solubility Soluble.
- o) Partition coefficient: $n - \log P = -1.36$ octanol/water
- p) Auto-ignition temp No data available.
- q) Decomposition temp No data available.
- r) Viscosity >1.0 cP at 25°C (77°F)

10. Stability and Reactivity

10.1 Reactivity

No data available.

10.2 Chemical stability

Stable under ordinary conditions of use and storage.

10.3 Possibility of hazardous reactions

Stable under ordinary conditions of use and storage.

10.4 Conditions to avoid

Contact with incompatible chemicals and exposure to extremely high temperatures.

10.5 Incompatible materials

Strong oxidizers, strong acids, acid chlorides, acid anhydrides, chloroformates, or strong reducing agents.

10.6 Hazardous decomposition products

Mainly carbon dioxide and carbon monoxide. Dynalene, Inc. Dynalene EG SDS
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11. Toxicological Information

11.1 Information on toxicological effects

For ethylene glycol

LD50 Oral – rat: 4,700 mg/kg

LD50 Dermal – rabbit: 10,626 mg/kg

Skin corrosion/irritation

No data available.

Serious eye damage/eye irritation

Eyes – rabbit. Result: mild eye irritation, 24h.

Irritancy of product

This product may cause irritation to contaminated tissues.

Reproductive toxicity

This product is not reported to produce mutagenic, embryotoxic, teratogenic, or reproductive effects in humans.

Suspected cancer agent

No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by ACGIH, NTP, or OSHA.

Irritancy of product

This product may cause irritation to contaminated tissues.

Reproductive toxicity

This product is not reported to produce mutagenic, embryotoxic, teratogenic, or reproductive effects in humans.

Medical conditions aggravated by exposure

It is anticipated that mainly skin, eye, and respiratory disorders may be aggravated after over-exposure.

Additional information

RTECS: KW2975000

When ingested early symptoms mimic alcohol inebriation and are followed by nausea, vomiting, abdominal pain, weakness, muscle tenderness, respiratory failure, convulsions, cardiovascular collapse, pulmonary edema, hypocalcemic tetany, and severe metabolic acidosis. Without treatment, death may occur in 8 to 24 hours. Victims who survive the initial toxicity period usually develop renal failure along with brain and liver damage., Exposure to and/or consumption of alcohol may increase toxic effects.



Central nervous system - Irregularities - Based on Human Evidence

12. Ecological Information

12.1 Ecotoxicity (aquatic and terrestrial)

This product may be harmful to aquatic life if large quantities are released into bodies of water.

Ethylene glycol

Toxicity to fish: NOEC – Pimephales promelas (fathead minnow) – 39,140 mg/L, 96h.

NOEC – Pimephales promelas (fathead minnow) – 32,000 mg/L, 7d.

LC50 – Oncorhynchus mykiss (rainbow trout) – 18,500 mg/L, 96h.

LC50 – Leuciscus idus (golden orfe) – >10,000 mg/L, 48h.

Toxicity to invertebrates: NOEC – Daphnia – 24,000 mg/L, 48h.

EC50 – Daphnia magna (water flea) – 74,000 mg/L, 24h.

LC50 – Daphnia magna (water flea) – 41,000 mg/L, 48h.

12.2 Persistence and degradability

Ratio BOD/ThBOD: 0.78% Dynalene, Inc. Dynalene EG SDS Effective February 20, 2020

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12.3 Bioaccumulation potential

Bioaccumulation, other fish – 61d, 50 mg/L

Bioconcentration factor (BCF): 0.60

12.4 Mobility in soil

No data available.

12.5 Other adverse effects

None.

13. Disposal Considerations

13.1 Waste treatment methods

Waste disposal must be in accordance with appropriate Federal, State, and local regulations. This product, if unaltered by use, may be disposed of by treatment at a permitted facility or as advised by your local hazardous waste regulatory authority.

14. Transport Information

DOT



UN number: 3082 Class: 9 Packing group: III Proper shipping name:
Environmentally hazardous substance, liquid, n.o.s. (Ethylene glycol) Reportable
Quantity (RQ): 5000 lbs no Poison Inhalation Hazard: No

IMDG

Not dangerous goods.

IATA

Not dangerous goods.

15. Regulatory Information

SARA 302 No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302 (Extremely Hazardous Substances).

SARA 313 The following components are subject to reporting levels established by SARA Title III, Sections 313 (Toxics Release Inventory Chemicals).

Ethylene Glycol CAS-No. 107-21-1

SARA 311/312 (Emergency Planning and Community Right-to-Know Act)

Acute Health Hazard, Chronic Health Hazard

Massachusetts Right

To Know No components are subject to the Massachusetts Right to Know Act.

Pennsylvania Right

To Know Ethylene Glycol CAS-No. 107-21-1

New Jersey Right

To Know Ethylene Glycol CAS-No. 107-21-1

TSCA The following components are on the Toxic Substance Control Act Inventory: Ethylene Glycol CAS-No. 107-21-1

EINECS All of the components of this product are on the European Inventory of Existing Commercial Chemical Substances.

California Prop 65 This product contains ingredients that cause cancer or reproductive harm known to the state of California. Ethylene Glycol, CAS-No. 107-21-1 Dynalene, Inc. Dynalene EG SDS Effective February 20, 2020

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Canada DSL The following components of this product are on the Canadian Domestic Substance List: Ethylene Glycol, CAS-No. 107-21-1



16. Other Information

Revision Date

February 20, 2020

This SDS was prepared by Dynalene, Inc.

The information contained herein is based on data considered accurate. However, no warranty is expressed or implied regarding the accuracy of these data or the results to be obtained from the use thereof. Dynalene Heat Transfer Fluids assumes no responsibility for injury to the vendee or third persons proximately caused by the material if reasonable safety procedures are not adhered to as stipulated in the data sheet. Additionally, Dynalene Heat Transfer Fluids assumes no responsibility for injury to vendee or third persons proximately caused by abnormal use of the material even if reasonable safety procedures are followed. Furthermore, vendee assumes the risk in his use of the material.

**12.2 Material Safety Data Sheet****MATERIAL SAFETY DATA SHEET****ISOPROPYL ALCOHOL**

Ref: AK/002/IPA

CHEMICAL NAME

2- Propanol.

TYPICAL PROPERTIES

Density @ 20 °C		0.785 – 0.786 kg/l
Flash point, Abel		10°C
Auto-ignition temperature		425°C
Explosive Limits in Air,	Lower	2 % vol.
	Upper	12 % vol.
Vapour Density @ 0°C & 1 atm.		2.1
Threshold Limit Value,	TLV-TWA	400 ppm (985 mg/m ³)
ACGIH 1989-90)	TLV-STEL	500 ppm (1230 mg/m ³)

HAZARDOUS PROPERTIES

Isopropyl alcohol (IPA) is a highly flammable liquid and can form explosive mixture with air particularly in emptied uncleaned receptacles. The vapour is invisible, heavier than air and spreads along the ground; distant ignition is possible. Do not use compressed air for filling, discharging, or handling. IPA is irritating to the eyes and mucous membranes in liquid and vapour form. Generalized toxicity is similar to ethyl alcohol e.g. inebriation. Prolonged or repeated skin contact may cause dermatitis.



GENERAL PRECAUTIONS

Isolate from sources of heat, naked flames and sparks. Maintain adequate ventilation and avoid breathing the vapour. Avoid contact with the skin, eyes and clothing.

Wearing protective equipment as and when appropriate.

PROTECTIVE CLOTHING AND EQUIPMENT

Normal Operations.

Eye : Chemical goggles or full-face shield with forehead protection.

Hand : Wrist length gloves, nitrile rubber, PVC, neoprene.

Foot : Safety shoes or boots: chemical resistant.

Body : Standard issue work clothes.

Respiratory protection: Should be handled in a well-ventilated area.
Wear a full-face respirator with organic vapour cartridge if there is an inhalation risk. (Nominal Protection Factor NPF 20).

Spillage

Eye : Covered by respiratory protection.

Hand : Wrist length gloves: Nitrile rubber, PVC, neoprene.

Foot : Safety boots: knee length.

Body : Chemical resistant PVC one-piece suit.

Respiratory protection: Full-face piece respirator with organic vapour cartridge (Nominal protection factor NPF 20). If spillage occurs in a poorly ventilated area, wear compressed air line or self-contained breathing apparatus. (NPF 2000)

ACTION IN CASE OF SPILLAGE / LEAKAGE

Extinguish all naked lights and remove ignition sources.

Avoid sparks. Evacuate personnel from the area. Ventilate the contaminated area thoroughly.

Avoid inhaling the vapour. Wear appropriate protective clothing.

Small Spillage

Absorb the liquid with sand or earth.



Shovel up and place in a labelled sealable container for subsequent disposal.

Flush contaminated area with water.

Large Spillage

Prevent spillage from spreading by use of sand or earth.

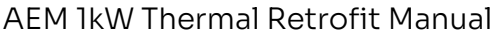
Transfer spillage to a salvage tank if possible; otherwise treat as for a small spillage.

Inform local authorities (eg. fire service) immediately if the spilt liquid enters the surface water drains since a potential explosive and toxic hazard will exist.

Leakage

Shut off leaks if without risk.

Put leaking drums in safe containers (overdrum). Otherwise treat as for appropriate spillage.

[illegible]